Advances in silicon carbides and their MEMS pressure sensors for high temperature and pressure applications

Authors

Renxing Wu[1], Hui Chen[1], Yichen Zhou[2], Yihao Guo[1], Zhangbin Ji[1], Long Li[1], Yuanfan Yang[1], Guoqiu Wang[3], Jian Zhou[1], Yongqing Fu[4*]*

Affiliations

1. *College of Mechanical and Vehicle Engineering, Hunan University, Changsha 410082, China*
2. *Hunan University of Technology and Business, International Business, Changsha 410205, China*
3. *Hunan Qitai Sensing Technology Co. Ltd, No.22, Dingsheng Road, Hi-Tech Industrial Development Zone, Liuyang, Hunan Province 410323, China*
4. *Faculty of Engineering and Environment, Northumbria University, Newcastle upon Tyne NE1 8ST, United Kingdom*

* Corresponding E-mail address: jianzhou@hnu.edu.cn; Richard.fu@northumbria.ac.uk

Abstract

High-temperature pressure sensors have recently attracted considerable interest for potential applications in the automotive, aerospace, and deep-well drilling industries, where they are required for monitoring gas or liquid pressures under extremely high temperatures and/or high pressures in harsh corrosive environments. Silicon carbide (SiC) is a third-generation semiconductor material with a wide bandgap, excellent high-temperature stability, and is regarded as a good candidate for overcoming the high-temperature intolerance of traditional pressure sensors. Currently, there are few reviews on recent advances in the synthesis, characterization, sensing mechanisms, design methodology, fabrication processes, operation, and application issues of SiC-based pressure sensors used in extreme application conditions. This review explores the following key topics, i.e., (i) key properties and special attributes of SiC materials; (ii) synthesis of SiC materials and thin films for high-temperature pressure sensor applications, and processing of SiC materials, including etching, ohmic contacts, and bonding;

(iii) recent development of SiC piezoresistive pressure sensors, including those based on silicon-on-insulator and all-SiC designs; (iv) recently reported SiC capacitive pressure sensors, including both 3C-SiC-based and all-SiC designs; and (v) advances in SiC-based fiber-optic pressure sensors. Finally, we highlight the key challenges and future prospects of next-generation SiC-based high-temperature pressure sensors.

Keywords: SiC, high temperature, pressure sensors, harsh environments, MEMS

Table of Contents

1. Introduction

Pressure sensors based on microelectromechanical systems (MEMS) are essential elements for measuring internal gas, fluid pressure, and monitoring equipment status, and are crucial for applications in automotive[1][2], aerospace[3][4][5][6], next-generation electronics/photonics, and deep-well drilling[7][8][9][10]. Conventional MEMS based high-temperature pressure sensors are mainly divided into two types based on their architectures, i.e., (1) silicon (Si) types [11][12][13][14], where silicon is served as both the sensing element and substrate, and (2) Si-on-insulator (SOI) types [15][16][17][18][19], which are consisted of layers of silicon, silicon dioxide, and silicon, as well as other relevant materials.

Nevertheless, owing to the relatively narrow bandgap of Si (~1.12 eV), Si-based high-temperature pressure sensors encounter major challenges. For example, the excessive charge carrier generation of Si at the elevated temperatures restricts its maximum operating temperature to typically below 300°C[20][21]. Whereas for the SOI type pressure sensors, the Si material is susceptible to creep at temperatures exceeding 300°C, leading to its diminished sensitivity and stability, despite their potential for being operated at temperatures up to 400°C[22][23][24][25][26].These limitations restrict their applicability in various demanding fields such as aerospace, automotive, and deep-well drilling, where operating temperatures often exceed 400°C.

To allow pressure sensors operated beyond 400°C[27][28][29], a variety of semiconductor materials with their excellent thermal stability and wide-bandgap characteristics have been investigated, including silicon carbide (SiC) [30][31], group III nitrides (e.g., AlN, GaN, and AlGaN), and diamond[32]. Among these materials, SiC is an excellent option for producing high-temperature MEMS pressure sensors because of its array of superior properties [33]. First, SiC has been reported to have fewer crystal dislocations than other wide-bandgap semiconductor materials such as GaN, AlN, diamond-like carbon (DLC), and zinc selenide (ZnSe) [34][35][36][37], which is beneficial for its superior high-temperature stability[38][39]. Additionally, SiC exhibits outstanding mechanical characteristics[40][41], with a Mohs hardness rating of approximately 9-9.5, which is marginally lower than that of diamond (10), but on par with gallium nitride[42]. The thermal conductivity of SiC (370 W/cm·K) is also significantly higher, for example[43] [44], twice that of gallium nitride (130-180 W/cm·K). SiC's superior properties enable their MEMS pressure sensors to be operated reliably at temperatures beyond 600°C, while maintaining high sensitivity and long-term stability[45][46]. Furthermore, SiC's high thermal conductivity and resistance to corrosion make it ideal for harsh environments, overcoming the limitations of traditional Si-based sensors and paving the way for next-generation high-performance pressure sensing technologies.

Recently, there have been several reviews on the advancement of high-temperature pressure sensors, with the majority focusing on operational mechanisms, such as capacitive[47][48][49], fiber-optic[50][51][52], and piezoresistive[53][54], for marine and aerospace [55]. For example, Chen et al. [56] reviewed advances in various types (piezoresistive, capacitive, piezoelectric, fiber-optic and friction electric) of high-temperature pressure sensors based on inorganic materials (such as silicon, silicon carbide, sapphire and ceramics) and high-temperature resistant polymers, with detailed discussions on material properties, sensing mechanisms and temperature compensation methods. Han et al. [57] provided a comprehensive overview of various types of MEMS pressure sensors, including the key aspects related to their design, manufacturing, and packaging. However, to date, no topic reviews have mainly focused on the latest advancements in SiC-based high-temperature pressure sensors, particularly in the areas of design methodologies, bulk and thin-film SiC preparation, fabrication, and system design/fabrication.

This review seeks to offer a thorough analysis of specific and potential topics related to SiC-based high-temperature pressure sensors by discussing existing SiC-based high-temperature pressure sensor technologies and exploring future trends with a focus on new materials and synthesis, key mechanisms, design methods, fabrication, and packaging applications. SiC high-temperature pressure sensors can be categorized into piezoresistive, capacitive, and optical types based on their structures and sensing mechanisms, and each offers distinct advantages in high-temperature applications, as depicted in **Figure 1**. Among the three categories of SiC high-temperature pressure sensors currently available, piezoresistive, capacitive, and fiber optical pressure sensors have the reported operation temperatures from 25°C up to 800°C, 500°C, and 1000°C, respectively.

In this review, we firstly summarize the key properties and characteristics of SiC materials and then introduce synthesis and fabrication/characterization methodologies for SiC materials and thin films. This is followed by a discussion on the processing of SiC materials for high-temperature pressure sensor applications (including etching, ohmic contact, and bonding). Subsequently, we focus on recent advancements in major SiC piezoresistive pressure sensors, including those utilizing SOI and all-SiC technology. Detailed discussions are focused on the recent development of SiC capacitive pressure sensors (including both 3C-SiC-based and all-SiC-based designs) and SiC fiber-optic pressure sensors. Finally, we highlight the key challenges in this field and propose future opportunities and perspectives for SiC-based high-temperature pressure sensors.

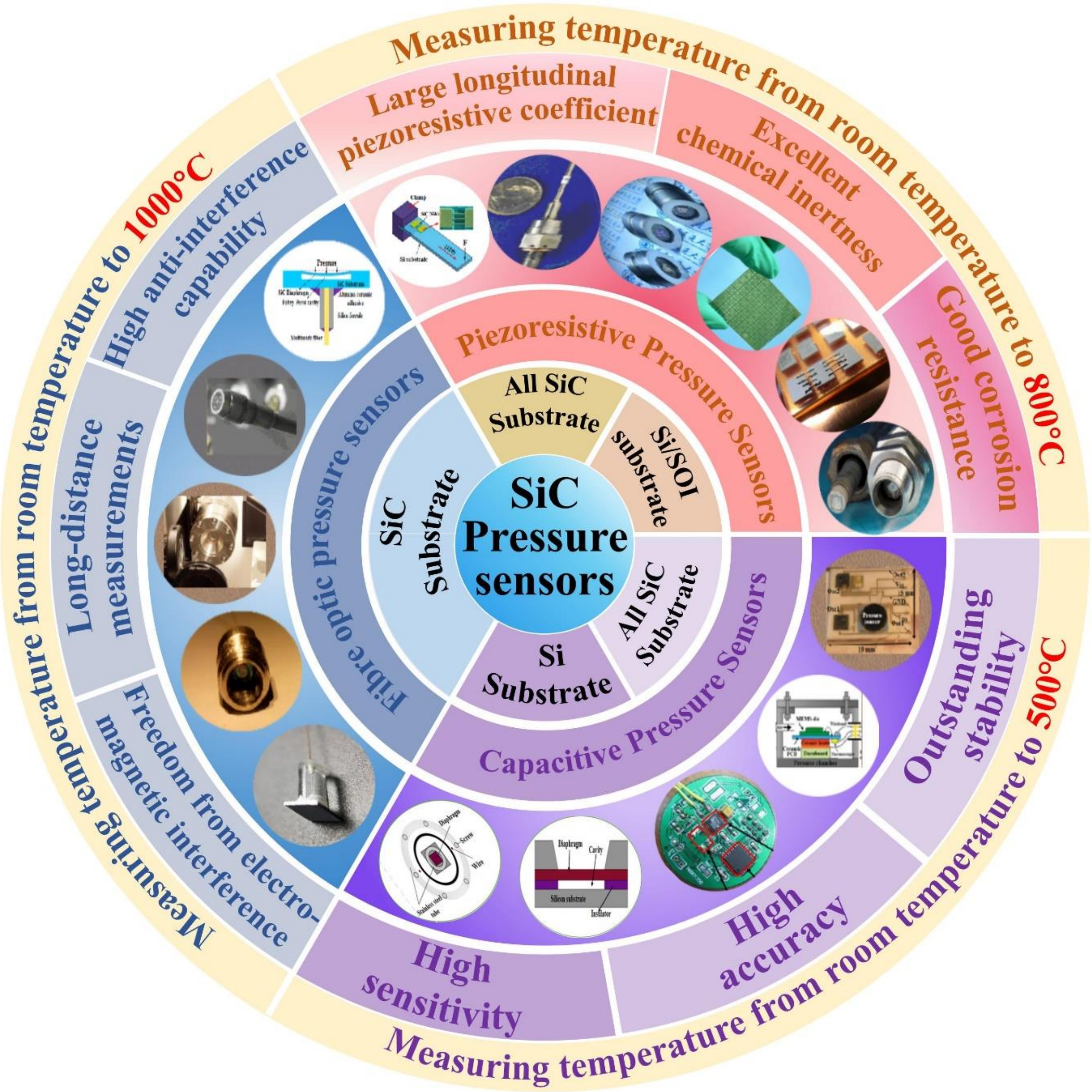


Figure 1. Overview on the structure and sensing mechanism of SiC high-temperature pressure sensors.

2. Silicon Carbide Materials

2.1. Properties of Silicon Carbide Materials

SiC is a prominent third-generation semiconductor material. It has extremely strong Si-C bonds with special crystal structures, a wide bandgap (~ 3.2 eV), superior thermal conductivity, high dielectric breakdown field strength, and numerous other outstanding physical and mechanical properties and chemical stability[58][59][60][61]. The following lists are some of its key properties:

(1) Mechanical properties. SiC materials possess excellent mechanical properties, such as high hardness, strength, rigidity, and resistance to creep, fatigue, and high temperatures. Therefore, they can be used as a coating in micromechanical devices to prevent erosion. With a Young's modulus 2–5 times higher than that of Si [62][63], and superior tensile and compressive strengths, SiC is well-suited for high-performance MEMS applications, including those in deep well drilling

(2) wide bandgap. The bandgaps of various polycrystalline SiC materials range from 2.2 eV for cubic structure to 3.3 eV for hexagonal structures. This wide range in bandgap enables high temperature operations of up to 1000°C[64]. Compared to other semiconductors (e.g., Si, GaN, GaAs), SiC's wide bandgap yields lower carrier concentration and defect density, enabling higher electron drift rates. This enhances charge transport in SiC-based pressure sensors, improving their response speed and power density.

(3) High-temperature stability. The unique crystal structure of SiC imparts excellent thermal stability and a significant potential barrier, thereby effectively mitigating the migration of electron-hole pairs across the bandgap at elevated temperatures. Consequently, SiC electronic devices and sensors demonstrate a remarkable stability at high temperatures[65]. Furthermore, it possesses an elevated melting point (exceeding 2,500°C) and a substantial thermal conductivity of up to ~5 $Wcm^{-1}K^{-1}$. It can preserve its structural integrity when subjected to rapid temperature fluctuations. It also demonstrates its exceptional stability in prolonged exposure to high temperatures owing to its minimal thermal expansion and superior resistance to thermal corrosion[66], making it ideal for pressure measurement in high-temperature and highly corrosive environments.

(4) Chemical inertness. Silicon (Si) and carbon (C) elements are bonded to form a hard carbide structure, which results in SiC with its excellent chemical corrosion resistance in various acids, alkalis, and other corrosive chemicals.

(5) Electrical properties. SiC materials have good electrical properties, including high breakdown field strength and low electron mobility, and are suitable for specific electronic devices and power electronics applications.

In summary, SiC effectively addresses the main challenges and limitations of current

MEMS pressure sensors, attributed to its exceptional properties. Its wide bandgap (~3.2 eV) and high thermal conductivity enable stable operations at temperatures up to 1000°C, while its low carrier concentration and low defect density enhance electron drift rates, improving sensor response speed and power density. The high hardness, rigidity, and good creep resistance of SiC make it suitable for high-stress environments, such as deep well drilling industry. Its chemical inertness and corrosion resistance ensure excellent performance in acidic, alkaline, and other corrosive media. Additionally, SiC's high breakdown field strength and thermal stability guarantee a high reliability under rapid temperature fluctuations and prolonged high-temperature exposure. These properties make SiC an ideal material for pressure sensing applications in high-temperature, high-pressure, and corrosive environments, with wide-ranging uses in fields such as oil/gas extraction and automotive engine monitoring.

2.2. Various types of silicon carbides

SiC has more than 200 polymorphic structures[67][68]. They can be uniquely separated into monocrystalline, polycrystalline, and amorphous phases, distinguished by the stacking order of each tetrahedrally bonded Si-C bilayer[69]. Crystalline SiC can also be classified as β-SiC and α-SiC. β-SiC is a cubic crystal system, often referred to as 3C-SiC, and is the most stable cubic crystal form of SiC [70]. Whereas α-SiC polycrystalline types include 2H-SiC, 4H-SiC, and 6H-SiC, all of which have a hexagonal lattice[71]. Currently, the most commonly used high-quality SiC crystalline structures are 3C-SiC, 4H-SiC and 6H-SiC. Among these, the 4H- and 6H-polytypes are the most favorable choices for piezoresistive devices [72].

Figure 2 compares piezoresistive effects of different semiconductor materials (including Si, Ge, GaN, SiC) and the performance, application scenarios, and operating temperature ranges of their corresponding pressure sensors. Results demonstrate that SiC exhibits exceptional performance of high-temperature adaptability, piezoresistive sensitivity, and applications in extreme environments (e.g., aerospace and nuclear energy). Notably, n-type 4H-SiC has been successfully developed for high-temperature pressure sensors, operated up to 800°C (as shown in **Figure 2 a**). In terms of applications, SiC fiber optic pressure sensors are recognized as having significant potentials for uses in extreme environments exceeding 600°C (as illustrated in **Figure 2c)**. **Table 1** also compares the key material parameters of three SiC polytypes (3C-SiC, 4H-SiC, 6H-SiC) with traditional materials (including Si, GaAs and diamond) [73][74][75]. SiC exhibits significant advantages over silicon-based materials in electrical properties (e.g., wider bandgap, higher breakdown electric field), thermal properties (e.g., higher thermal conductivity, lower thermal expansion coefficient), and mechanical properties (e.g., higher Young's modulus and hardness). Among the SiC polytypes, 4H-SiC demonstrates the most balanced performance (e.g., a bandgap of ~3.2 eV, a breakdown field of ~2.2 mV/cm, and the maximum operating temperature of ~1580°C), making it ideal for high-voltage and high-temperature devices. Although diamond outperforms SiC in bandgap (5.45 eV) and hardness (10,000 Kgf/mm²), its fabrication complexity limits practical applications. Future research should be focused on optimizing the doping processes of the SiC and developing heterogeneous integration technologies with Si-based devices to further enhance performance and reduce costs.

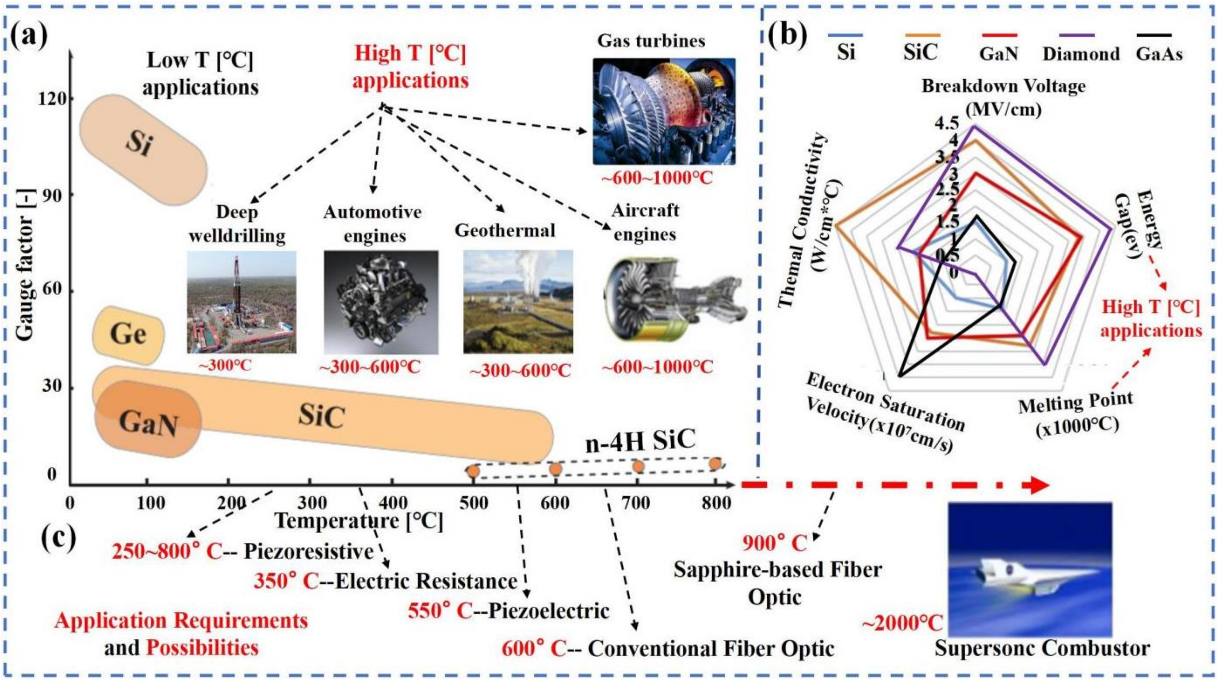


Figure 2 (a) The piezoresistive effect in common semiconductors and some applications; and (b) Performance comparison of popular semiconductor materials. Adapted with permission from ref [54]. (Copyright 2015 IEEE). (c) Pressure sensor categories and temperature ranges for applications in high temperature environments. Adapted with permission from ref [381]. (Copyright 2001 SPIE).

Table 1 Properties of different semiconductor materials.

Material parameters	unit	Si	GaAs	3C-SiC	4H-SiC	6H-SiC	Diamond
Bandwidth（Eg）	eV	1.12	1.43	2.4	3.2	3.0	5.45
Coefficient of thermal expansion(α)	10-6/k	2.6	5.9	4.7	4.8	4.8	1.1
densities(ρ)	g/cm^3	2.3	5.3	3.2	3.2	3.2	3.515
thermal conductivity(λ)	W/cm·K	1.5	0.5	5.0	5.0	5.0	2.0
melting point（T）	℃	1420	1240	2830	2830	2830	4000
Maximum working temperature(Tmax)	℃	600	760	1250	1580	1580	--
Breakthrough electric field	mV/cm	0.2	0.3	2.0	2.2	2.5	10
Young's modulus(E)	GPa	190	75	448	448	448	1220
durometer(HV)	Kgf/mm^2	1000	600	3980	2130	--	10000
electron mobility	cm^2/V s	1500	8800	800	1000	400	--
hole mobility	cm^2/V s	425	400	40	115	101	--

Based on information present in **Figure 2** and **Table 1**, it is clear that when selecting an appropriate SiC crystal structure, it is necessary to comprehensively consider manufacturing methods, application scenarios, cost factors, and device performance. Chemical vapor deposition (CVD) method is commonly used to synthesize 4H-SiC or 6H-SiC due to their excellent crystal quality and thermal stability; whereas the physical vapor transport (PVT) method is more suitable for 3C-SiC, as it can be grown at lower temperatures. In terms of application scenarios, 4H-SiC is the preferred choice for high-temperature electronic devices due to its high breakdown field strength and thermal conductivity; 6H-SiC is more suitable for high-frequency devices owing to its higher electron mobility; whereas 3C-SiC is suitable for optoelectronic devices due to its narrower bandgap. From a cost perspective, 3C-SiC is more economical due to its lower synthesis temperature, whereas 4H-SiC, despite its higher cost, often proves more valuable in high-performance applications due to its superior performance. Additionally, 4H-SiC is more suitable for high-power processing applications, whereas 6H-SiC performs better in scenarios requiring fast switching. By fully understanding their properties, suitable silicon carbide crystal structure can be selected for the specific manufacturing methods and application scenarios.

2.3. Advances in synthesis and processing of SiC materials

SiC possesses exceptional attributes including superior hardness, chemical and thermal durability, and outstanding resistance to corrosion. Since Acheson's first SiC production in 1892, SiC has emerged as a significant contender for high-power and high-temperature electronic devices, as well as for wear and cutting applications. [76]. This section provides a brief overview of the fabrication techniques, including Process Voltage Temperature (PVT), CVD, and sol-gel, used for synthesizing SiC materials (including SiC powder, bulk SiC, and SiC nanostructures). It also includes the synthesis methods of SiC thin films for high-temperature pressure sensors including CVD, Atomic Layer Deposition (ALD), and PVD, and also the key processing methods (such as etching process, ohmic contact, and bonding) of SiC.

2.3.1. Synthesis of SiC materials

Natural formed SiC, found as moissanite or in meteorites, is rarely existed, necessitating the synthetic production[77]. The Acheson process, a traditional carbothermal reduction method, synthesizes SiC by reacting silica sand and petroleum coke at temperatures exceeding 2500°C[78]. However, this method is energy-intensive and prone to oxygen contamination. It produces large-grained SiC with impurities like nitrogen (N) and aluminum (Al), which degrade material quality[78]. To address these limitations, modern techniques such as PVT, CVD, and sol-gel have been developed.

(1) Process Voltage Temperature (PVT)

PVT, or the seeded sublimation growth, has been widely used for producing large SiC single crystals[79][80]. While PVT-grown SiC exhibits good mass uniformity, its high-temperature (≈2500°C) growth in graphite crucibles complicates crystal size control and dopant concentration regulation**.** Innovations such as the gas tube integration (Straubinger et al.[81]) and modified PVT (M-PVT) with dopant gases (Wellmann et al.[82]) have enabled the growth of high-quality 4H and 6H SiC wafers up to 100 mm in diameter.

(2) Chemical Vapor Deposition (CVD)

CVD is a versatile method for producing SiC thin films, powders, and nanostructures[83]. By varying precursors and deposition parameters, CVD can yield SiC in diverse forms. For example, Kavcký et al.[84] synthesized fine SiC powders using a SiH4-C2H2 system, while Fu et al.[85] produced high-purity SiC nanowires using CH3SiCl3 (MTS) and H2 without metal catalysts. The deposition temperature significantly influences grain size and growth rate, with higher temperatures accelerating the crystallization of SiC.

(3) Sol-Gel

The sol-gel method offers advantages over the PVT and CVD, including lower synthesis temperatures, and high-purity/homogeneous SiC products[86] [87][88][89]. This technique involves hydrolysis and condensation of alkoxide precursors, followed by carbonization and carbothermal reduction[90][91]. Despite its high cost due to the usage of expensive precursors, sol-gel produces highly porous SiC with excellent microstructural homogeneity. For instance, Raman et al.[87] synthesized β-SiC with grain sizes of 9–53 nm, demonstrating stable function properties. Additives such as boric acid (H_3BO_3) enhanced carbothermal reduction efficiency[92], and crystalline β-SiC was formed at 1580°C using silica sol and sucrose.

For different application requirements, the synthesis methods for bulk SiC materials should be selectively optimized. The PVT method is preferred for large-sized single crystal growth, and the improved temperature field and gas convection control can enhance crystal quality and dimensions. CVD is more suitable for micro/nano structures such as nanowires/rods, enabling precisely morphological design through precursor ratio and deposition temperature adjustments. The sol-gel method excels in producing high-purity, porous nano-powders. When combined with boric acid catalysis, it significantly reduces the reaction temperatures while improving product uniformity.

2.3.2. Synthesis of SiC thin films

The commonly employed techniques for fabricating SiC thin films include CVD, ALD, and PVD.

(1) Chemical Vapor Deposition

CVD is a widely used method for depositing SiC thin films by decomposing organic or organometallic precursors in various reactor configurations. Common CVD variants include atmospheric-pressure CVD (APCVD) [93][94][95][96], low-pressure CVD (LPCVD), and plasma-enhanced CVD (PECVD)[97][98]. In a typical CVD process, gas precursors are introduced into the reaction chamber, where they become dissociated or reacted, and the resulting compounds migrate to the substrate surface to form the film [99].

APCVD enables high growth rates (up to a few micrometers per minute) [100][101] and is suitable for doping SiC films with n-type (e.g., nitrogen) [102] or p-type (e.g., boron)[103] materials. Methyl-trichloro-silane (MTS) is a common precursor for the APCVD due to its favorable Si/C stoichiometry and high film quality [104][105]. However, APCVD requires high temperatures (>1050°C), limiting its compatibility with certain substrates and integrated circuits (ICs).

LPCVD operates under vacuum, allowing for uniform temperature distribution and larger surface coverage [106]. It uses dual-source precursors (e.g., SiH_4 and C_3H_8) and produces films with consistent thickness and high purity[107][108][109].

PECVD utilizes plasma to decompose precursors, enabling film deposition at lower temperatures (200–600°C) [110][111]. Common precursors include methane (CH_4) and silane (SiH_4), as well as single-source precursors such as methylsilane and HMDS[112][113] [114]. PECVD is advantageous for applications requiring a low-temperature processing[115] [116].

For many applications, only the APCVD or LPCVD can generate the silicon carbide materials with the specifically required quality. The APCVD and LPCVD processes are often operated in a continuous gas feed, known as an "analogue-type mode." This sometimes results in difficulties in regulating the thickness and consistency of the films on different substrates, ultimately limiting the use of these technologies for applications that require extremely thin films (less than 100 nm).

(2) Atomic Layer Deposition (ALD)

ALD offers precise control over film thickness and composition by depositing atomic or molecular layers sequentially[117][118]. Unlike CVD, the ALD separates precursors to prevent gas-phase reactions, ensuring self-limiting surface reactions[119][120]. The common precursors include SiH_2Cl_2 or Si_2H_6 for silicon and C_2H_2 or C_2H_4 for carbon[121][122]. ALD produces highly conformal and homogeneous films, making it ideal for applications requiring ultra-thin (<100 nm) and uniform coatings[123][124][125].

(3) Physical Vapor Deposition (PVD)

PVD, particularly magnetron sputtering, is commonly used to deposit SiC films by physically ejecting atoms from a SiC target. This method offers advantages such as low deposition temperatures (e.g., at or lower than 400°C)[126][127], cost-effectiveness, and strong film adhesion[128][129]. Key parameters include target-substrate distance, sputtering power, and deposition pressure, and they influence film stress, density, and adhesion[130][131]. PVD is suitable for high-precision applications and can deposit SiC films on various substrates, including silicon, SiO_2, and Si_3N_4.

Suitable thin film deposition techniques should be chosen based on performance requirements. For example, APCVD supports high-speed growth and doping, making it ideal for thick films in power devices. LPCVD and PECVD overcome temperature limitations, achieving high-density films and low-temperature. ALD provides atomic-level thickness and composition control, meeting the demand for ultra-thin coatings in nano-devices. Finally, PVD process has its advantages such as low-temperature rapid deposition and strong adhesion.

2.3.3. Processing methods of SiC

(1) Etching of SiC

Unlike silicon materials, the high chemical bonding energy between Si-C makes SiC materials good chemical stability, making the conventional wet etching process difficult to etch SiC materials[132]. The commonly used SiC etching processes include photoelectrochemical etching (PEC), reactive ion etching (RIE), and femtosecond laser etching (FS).

Photoelectrochemical etching

PEC is a process that uses photoelectrochemical reactions to achieve the localized etching of a material surface[133]. It typically involves exposing a semiconductor material (e.g., Si or SiC) to a specific electrolyte solution and then using light to initiate an electrochemical reaction that results in a controlled etch or etching of specific areas of the material surface [133]. For example, Carrabba[134] used PEC to etch diffraction gratings into polycrystalline 6H-SiC and single-crystal 3C-SiC. They reported that the visible light utilized during the etching procedure led to a sluggish etching speed and coarse film surface [134]. To address these shortcomings, Shor et al.[135] used a focused laser beam and a diffuse incoherent light source for the PEC of n-type and p-type 6H-SiC, resulting in enhanced etch rates of up to 400 nm/min and 700 nm/min, respectively. They found that by controlling the potential in n- and p-type SiC in hydrofluoric acid (HF) solutions, only hole current injection occurred in n-type SiC, leading to selective etching based on the dopant [136]; in other words, the PEC selectively photoetched the n-type 6H-SiC on the surface of p-6H-SiC. Based on this research, 6H-SiC pressure sensors have been fabricated, capable of measuring with temperatures up to 500°C[137]. Additionally, the implementation of Ti/Ti/N/Pt metallised ohmic contacts has elevated the operational temperature of the 6H-SiC pressure sensor up to 600°C [138]. However, these PECs were reported to be poorly oriented, making it difficult to control the lateral dimensions (e.g., diaphragm size) of the pressure sensors [139].

Reactive ion etching (RIE)

Reactive ion etching (RIE) is an effective method for etching SiC[140][141] and has been used for the synthesis of SiC pressure sensors[142]. Commonly, plasma etching, inductively coupled plasma (ICP), and deep reactive ion etching (DRIE) can all be used for the etching of thin films[143], bulk substrates[144] and deep grooves[145] of SiC with a good etching direction and controlled size and groove depth of microstructures. The etching rate and roughness of the processed surfaces of RIE are related to the plasma density of the sources and the etching mask used in the process. Chabert et al. [146] employed the SF_6/25% O_2 plasma reaction and nickel as a masking material to create a through-hole on a 330 μm thick 4H-SiC substrate. The etching process took approximately 6 h, with an etching rate of 0.9 μm/min. An increased etching rate of 1.35 μm/min resulted in the production of a rough surface [146]. Cho et al.[147] utilized an ICP process with a mixed gas of 63% SF_6 and 37% O_2 to etch a 97μm deep hole in 4H-SiC. They achieved an etching rate of 0.32 μm/min and obtained a smooth surface, despite with a slow etching rate. Tanaka et al. [145] performed DRIE on silicon carbide (SiC) with an etched depth of 216 μm and an etching rate of 0.45 μm/min. They achieved a high-aspect-ratio wall within the diaphragm cavity and enhanced the uniformity of the sensing diaphragm.

However, groove effects commonly occur during the RIE processing of 4H-SiC diaphragms, leading to premature bursting and failure of the 4H-SiC sensor diaphragm under uneven pressure during operation [148]. The RIE process for large-area bulk SiC also showed problems of low etch rate (nm/s), poor etch selectivity, and inability to create high aspect ratio structures during etching processing[149]. Simulation clearly showed the occurrence of such a failure in a non-grooved diaphragm [148]. In addition, the etching rate of etching methods (including PEC, RIE, and DRIE) is commonly low, the etching accuracy is poor, and the masking process is complex[150][151][152]. There are also reports on using the polished bulk wafers to process SiC pressure sensor diaphragms. These obtained SiC layers are normally much thicker (>40 μm), with low spatial resolution and rough surfaces [153], which are inadequate for meeting the requirements of mass production and high-precision development.

Femtosecond laser etching

Femtosecond (FS) laser etching has emerged as a crucial technique for structural deposition of SiC [154]. Laser irradiation can modify the surface hardness, stress states, and semiconducting properties of SiC materials [155]. For example, Ahn et al.[156]fabricated periodic surface structures of 4H-SiC using a low fluence (<0.50J/cm^2) high spatial-frequency laser [156]. Furthermore, ultrafast laser irradiation can induce the surface oxidation and amorphization of SiC materials, in addition to the formation of periodic surface structures. Femtosecond lasers have very short pulse widths (typically in the femtosecond or 10^{-15} s class) and can achieve high precision, high efficiency, high cleanliness, and minimal collateral thermal damage[157][158].

One of the critical parameters of femtosecond laser ablation processing of SiC is the ablation threshold (J/cm^2), which refers to the minimum energy required to irreversibly damage the material by removing its monolayer [153]. The ablation threshold is closely associated with the number of laser pulses and processing medium. For example, Wu et al.[159] utilized a femtosecond laser with a central wavelength of 800 nm to ablate 4H-SiC in different environments, including air **(Figure 3.a),** water **(Figure 3.b),** and hydrofluoric acid (HF) **(Figure 3.c)**), and they obtained the ablation threshold of SiC **(Figure 3.d)**. The largest ablation threshold of 4.98 Jcm^2 was documented in an atmospheric atmosphere with a pulse count of

N=50, while the lowest ablation threshold of 0.53 Jcm^2 was recorded in an HF atmosphere with a pulse count of N=300.

The choice of the processing medium has been found to have a notable impact on the surface roughness and chemical bonding characteristics of SiC. In particular, the use of liquid processing medium has been shown to decrease surface roughness and prevent excessive oxidation [159]. For example, Iwatani et al. [160] conducted experiments on the underwater laser drilling of 4H-SiC using a nanosecond pulsed infrared (1064 nm) laser **(Figure 3.e)**. With a laser energy of less than 10 J/cm^2, clean through-holes were achieved without forming any debris, heat-affected zones, or cracks. Similarly, Wang et al. [157] examined the properties of femtosecond laser ablation of 4H-SiC using single and multi-pulse (1035 nm) and determined the ablation thresholds for material modifications and structural transformation to be 2.35 J/cm^2 and 4.97 J/cm^2, respectively. They indicated that an increase in the number of pulses and the accumulated impact resulted in a reduction in the ablation threshold, progressively, reaching a saturation point and attaining precisely manageable depth and minimal surface roughness.

At a specific ablation threshold value, the ablation rate of femtosecond laser processing of SiC may be influenced by the laser wavelength. For example, Molian et al. [153] utilized a wavelength of 800 nm and a 120 fs laser beam to micromachine 250 μm-thick 4H-SiC single-crystal wafers. The ablation rate at an ablation threshold of 2 J/cm^2 was approximately 140 nm per pulse and the drilling speed was as high as 25 μm/s. The surface roughness was 12 μm Ra and the etching depth was maintained within 1% control. The absence of thermal damage (such as recast layers and contamination) results in high aspect ratios and outstanding spatial resolution. Zhang et al·[161] explored the use of a 266 nm laser and a vacuum UV laser (133-184 nm) for multi-wavelength ablation and reported that the etch rate of 6H-SiC at 1 J/cm^2 was 35 nm per pulse. The femtosecond laser processing is also affected by laser pulse duration. For example, Zoppel et al·[162] examined the erosion speed of 3C-SiC at the erosion rate value of 2 j/cm^2 per pulse of 100 nm with a 400 FS pulsed laser (1026 nm), and the erosion speed values surpassed those obtained using a nanosecond pulsed excimer laser.

It is important to note that femtosecond laser ablation demonstrates high precision and efficiency in deep etching of SiC. For example, Zhao et al.[163] employed a femtosecond laser to fabricate opaque cavities measuring 1200 μm in diameter and 270 μm in depth in a 350 μm-thick 4H-SiC wafer, yielding an 80 μm-thick responsive membrane for a pressure sensor **(Figure 3.f)**. The dimensional errors for the diameter and depth are 2.02% and 0.38%, respectively, with an average roughness of 320 nm. However, as a result of the uneven dispersion of laser scanning energy and the lag in laser irradiation, the sensitive film being prepared exhibited elliptical shapes and grooves (over-etching phenomenon) **(Figure 3.g)**. The magnitude and breadth of the grooves expanded in tandem with the increase in power, pulse repetition rate, and scanning speed, thereby influencing the performance of the device [164]. Zhang et al.[165] used a femtosecond laser operating at a central wavelength of 800 nm and maximum repetition rate of 1 kHz to create grooves in 4H-SiC. By incrementing the number of repetitive scans, a surface roughness in the range of 0.05 μm of the grooves was obtained. However, the accumulation effect on the SiC subsurface during this process resulted in different energy-absorption accumulations, making the ablation zone discontinuous during femtosecond processing [165]. To solve this defect issue, Wang et al. [166] induced a morphology on the 4H-SiC surface by repeated irradiation with a femtosecond laser, and its femtosecond laser processing system is shown in **Figure 3.h.** The center wavelength was 1030 nm, and the near-threshold fluence was 1.1J/cm^2 **(Figure 3.i).** The discontinuous regions were ablated after femtosecond laser irradiation to obtain a smooth machined surface.

Femtosecond lasers can achieve high precision and etch rates when processing heterojunction blinds or through-holes. For example, Zehetner et al.[167][168] prepared epitaxial AlGaN/GaN heterostructured thin films using SiC as the substrate material and micromachined the films using a femtosecond laser (520 nm) ablation technique **(Figure 3. J& k)**. An array of 275 μm deep, 1000–3000 μm diameter blind vias were precisely etched on a 350μm thick 4H-SiC substrate without damaging the 2μm AlN layer on the backside [168]. They reported that this method eliminated the risk of compromising or ruining the functionality of the coated film and allowed a greater degree of design flexibility to be attained.

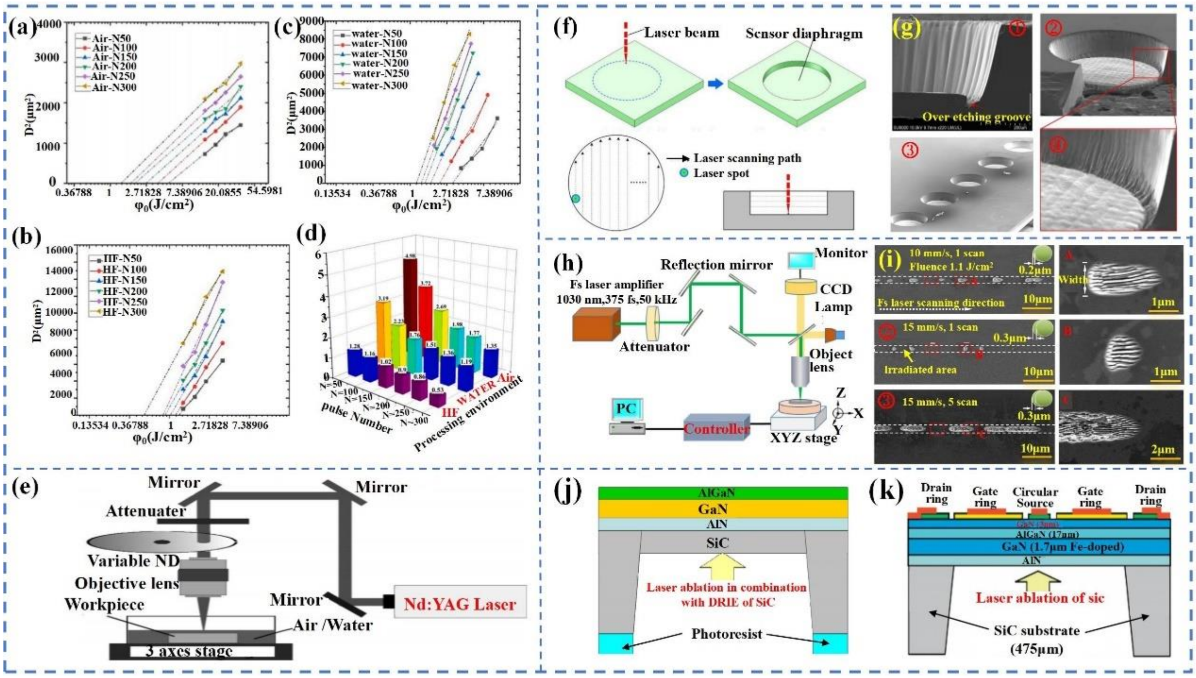


Figure 3 Fitting function curves: (a) in air; (b) in water; (c) in HF; (d) ablation threshold of SiC. Reproduced with permission from ref [159]. (Copyright 2021 Elsevier B.V.). (e) schematic of the laser workstation. Reproduced with permission from ref [160]. (Copyright 2013 Elsevier Ltd.). (f) Fs laser processing of circular diaphragms; and (g) Blind hole sidewall morphology. Reproduced with permission from ref [163]. (Copyright 2020 Elsevier B.V.). (h)Schematic diagram of femtosecond laser processing system; and (i)Fs laser-irradiated 4H-SiC surface at different scanning velocity and scanning times: ① 10 mm/s and 1 scan, ② 15 mm/s and 1 scan, ③ 15 mm/s and 5 scans. Reproduced with permission from ref [166]. (Copyright 2012 IPO Publishing.). (j) Design principle of SiC based pressure sensor. Reproduced with permission from ref[167]. (Copyright 2015 SPIE.). (k) Cross-section of a heterogeneous structural layer. Reproduced with permission from ref [168]. (Copyright 2016 J. Zehetner et al.).

(2) Ohmic contact annealing

For SiC piezoresistive pressure sensors, establishing a robust ohmic contact is crucial, and ensuring an effective connection between the force-sensitive resistor and metal lead is also essential[169]. At elevated temperatures, the metallized contacts are subject to degradation owing to enhanced microstructural processes, including interdiffusion among distinct layers, oxidation, and changes in composition and microstructure[170], These variations lead to instability at the metal/semiconductor interfaces, resulting in sensor's measurement output signals very noisy, with a large drift and a temperature dependence. Therefore, it is necessary to fabricate high-quality metal–SiC contacts that can withstand the operating temperature of the device. This, in turn, relies on various elements, such as the thermal stability, conductivity, and interface microstructure of the SiC pressure sensor [169][171]. Researchers have endeavored to diminish the contact resistance (or specific contact resistance ρ_c) and enhance the thermal durability of ohmic contacts through various metal layouts. The most common structures are Ni (nickel)/SiC, Cr(chromium)/SiC, and Ti(titanium)/SiC.

Ni/SiC

Ni forms excellent ohmic contacts on SiC and is widely employed as one of a primary metal for producing n-type SiC ohmic contacts, which necessitates elevated annealing temperatures (900-1000°C)[172][173]. The annealing of Ni/SiC led to the buildup of graphitic carbon at the interface as a result of the interplay between Ni and SiC, along with the formation of the Ni_2Si phase. [174][175]. Zhao et al.[176] used n-type 4H-SiC nickel-based ohmic contacts and obtained I-V curves for Ni in contact with n-type 4H-SiC after annealing at 1000°C. The metallic contact structure can be maintained stable in air at 300°C for 60 h. Li et al.[177] conducted a study on the ohmic contact properties of p-type SiC with Ni/Al/Ni/Au and found that a favorable ohmic contact of Ni/Al/Ni/Au was obtained under 950°C annealing condition. The synthesized sensor was encapsulated to test the piezoresistor in air from 25°C to 600°C, and a stable electrical connection was obtained at $\leqslant$600°C air environment.

It is worth mentioning that Ni on n-type 4H-SiC epitaxial silicon surfaces exhibited a reduced specific contact resistance (c) when subjected to rapid thermal annealing at ~1000°C ranged between $2.8E^{-3}$ and $1.1E^{-6}\Omega cm^2$ [178][179]. However, at approximately 1000°C, a distinct interdiffusion layer was developed between the Ni and Au layers during the annealing process. Additionally, the non-uniform structure of Ni contact after high-temperature annealing resulted in roughening of the surface morphology and the formation of interfacial voids, which in turn resulted in a decrease in the bonding strength at the interface of Ni/n-type SiC, ultimately causing contact with flakes, peels, or fractures during use [180][181]. These negatively affected the reliability, electrical properties, and adhesion properties of the sensors.

Cr/SiC

Cr demonstrates excellent anti-diffusion properties and suitable electrical characteristics, effectively inhibiting the inter-diffusion between Ni and other metals, as well as the SiC substrate, while maintaining its own electrical properties. Therefore, this results in a more stable metal-ohmic contact structure on SiC. For example, Patrick et al.[182] noted that the metal layers in the n-type 3C-SiC/Cr/Ni/Au structure exhibited significant interdiffusion during annealing at temperatures of 900°C and 1000°C. However, after interchanging the Cr and Ni layers, the metal-structured contacts exhibited limited interdiffusion of the metal (Ni:Au). In addition, Cr and NiCr alloys have been reported to be relatively ineffective as catalysts for graphite formation during annealing, and the tendency of Cr_3C_2 carbide to decompose into graphite is similar to that of Ni. [183]. Contacts with Ni/Cr/Au structures demonstrated a reduced thickness (10 nm) of the carbon layer formed during the reaction process while maintaining comparable electrical properties [184]. Furthermore, the inclusion of Cr in the contact layer decreased the level of interdiffusion between the Ni and Au layers during annealing at ~1000°C [183][184].

It is quite challenging for Ni- and Cr based ohmic contact annealing temperatures to reach 1000°C, leading to complexity and uncontrollable factors in processing. For example, the thermal cyclic stresses of Cr/Ni/Au metallization during annealing caused delamination between the respective materials[185] and the formation of irregular Ni/Au or Cr/Ni interfaces, which affected the structural stability. In addition, such high temperatures may induce creep and thermal mismatches in materials.

Ti/SiC

Encouragingly, Ti (titanium) could be a stable metallized interlayer when annealed at 700°C and form stable ohmic contacts[186]. Ti can serve as a barrier to prevent the formation of an undesirable NiSi phase by hindering the diffusion of Si and carbon, and can also react with carbon to produce TiC. The conversion of carbon into a graphite structure within NiTiSiC significantly decreases the sheet resistance and results in stable ohmic contact within the NiTiSiC structure [187]. More importantly, the Ti/$TaSi_2$/Pt metallization structure has been shown to eliminate the delamination problem caused by thermal cyclic stresses during high-temperature annealing of Cr/Ni/Au metallizations [188]. In addition, He et al.[189] examined the characteristics of n-type and p-type 4H-SiC/Ni/Ti/Al/W metal contacts and found that the specific contact resistances of the Ni/Ti/Al/W multi-metallic layers on n-type and p-type 4H-SiC were remarkably low at 8×10^{-4} Ωcm^2 and 4.1×10^{-5} Ωcm^2, respectively. Here the metal of

W was used as the top layer at an annealing temperature of 750°C. During the annealing process, there was a buildup of Ni and Al on the SiC side of the specimens, leading to the creation of Ni_2Si-rich and Al-rich areas at the interface, consequently improving the ohmic contact behavior.

(3) Bonding process

Wafer bonding is a procedure in which highly polished, smooth, and unsoiled wafers are initially contacted by van der Waals forces when they come into contact at a specific temperature and are subsequently brought together and joined or bonded by a process[190][191]. The bonding techniques for SiC/SiC wafers can be categorized into two approaches, i.e., direct and indirect [192]. As a crucial technology in the production of pressure sensors, the SiC bonding process significantly affects the reliability, stability, and sensitivity of these devices [193][194].

Indirect wafer bonding

Indirect wafer bonding (IWB) uses an interlayer for bonding, which is based on the interaction or fusion of the substrate with the interlayer at appropriate annealing temperatures and applied loads[195]. The common insulating interlayers used for SiC/SiC bonding include phosphor silicate glass, SiO_2[195][196][197], high-performance ceramic binders[198], sodium silicate solutions, and spin-coated glass [199], and conductive interlayers primarily consist of Ni[198] and Ti_3SiC_2[200]. These wafer-bonding techniques have been used to investigate potential SiC/SiC-bonded devices. However, there are certain limitations in these indirect wafer bonding processes. For example, a coefficient of thermal expansion (CTE) mismatch may occur when wafers are bonded indirectly with different adhesives as bonding layers and SiC, potentially resulting in device failure or performance degradation at elevated temperatures [195][196][201].

Direct wafer bonding

Direct wafer bonding (DWB) involves the bonding of two planar mirror-polished wafers in the ambient air at room temperature or under vacuum conditions without the use of an intermediate adhesive layer [193][202]. The DWB technology enables the creation of heterojunctions with various polymorphic combinations based on the characteristics of the individual SiC wafers, thus facilitating easy control of the electrical properties of the layers[203][204]. Yushin et al.[205] successfully achieved direct bonding of 6H-SiC/6H-SiC wafers in an ultrahigh vacuum environment with a pressure of 20 MPa and temperatures ranging from 800 to 1000°C. Although there were variations in the bonding performance for the different surface crystal orientations, they did not appear to significantly impact the process of direct bonding. Moreover, misalignment of the bonded wafers does not affect the bonding capability, but significantly influences the electrical characteristics of the fused heterojunction[206]. Low-resistance connections can be created through the high-temperature fusion of aligned 6H-SiC/6H-SiC wafers. However, such the UHV direct bonding requires a long period of time (2h) in a vacuum environment at a high temperature (800-1000°C), making the method low yield, costly, and difficult.

Bonding at room temperature is a good solution, but the bonding strength at the interface is low. Oxides on the surface of SiC can typically be eliminated through high-temperature annealing [193] or treatment with HF[207], which can increase the direct bonding energy. The experimental findings indicated a progressive increase in the bonding capacity of 3C-SiC wafers as the concentration of HF acid increased, reaching its maximum at a 2.0% concentration, and subsequently declined [196]. Liang et al. [195] fabricated 4H-SiC bonded structures containing cavities through a synergistic approach of room-temperature pre-bonding and subsequent high-temperature annealing, resulting in enhanced structural properties. The optimum tensile strength of 2 MPa was obtained [195]. However, the annealing temperature for this process usually needs to reach 1300 °C; therefore, thermal cyclic stresses after cooling down from annealing can cause delamination between the materials and the formation of irregular fusion interfaces, affecting the structural stability [185].

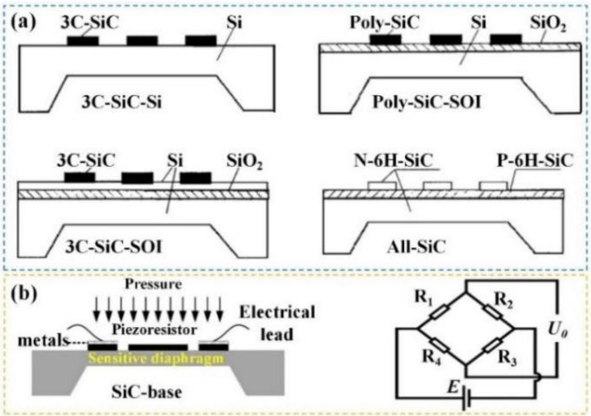
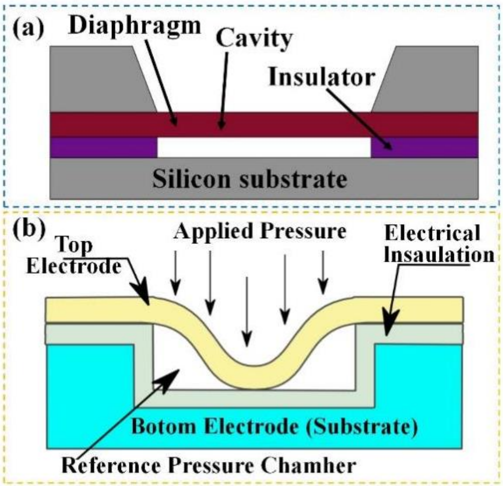
Notably, surface-activated bonding (SAB) is a technology that allows the direct bonding at room temperature, eliminating the need for high-temperature annealing treatments [208]. Subsequent studies have shown that the introduction of active excited ions, such as Ar, can further increase the bonding energy and yield more stable bonded structures. Mu et al.[209] utilized a Si-Ar ion beam modified SAB technique to achieve successful direct chip bonding of SiC. The argon ion beam effectively removed the oxide layer on the surface of SiC and increased the strength of the bonding interface by 30-60% compared with the traditional SAB process [208]. However, the use of argon ion beam bombardment introduces certain issues such as the increased complexity of the production process and the requirement for more expensive specialized equipment.

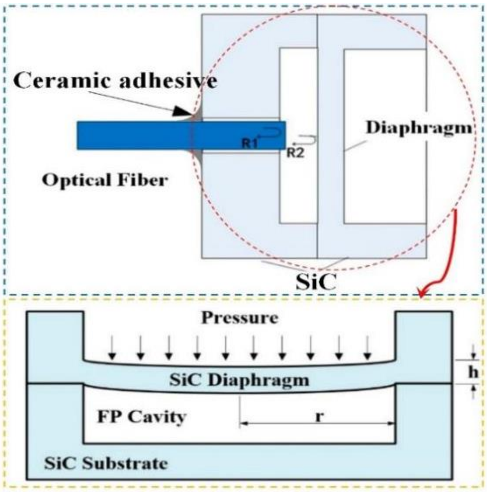
3. Advances in SiC Pressure Sensors

High-temperature pressure sensors based on SiC can be categorized into three main types according to their working principles and structural designs, i.e., piezoresistive, capacitive, and fiber-optic sensors[47][48][50][53][56][210]. Piezoresistive SiC pressure sensors detect pressure via resistance changes caused by the piezoresistive effect [54] . They feature a simple structure, large piezoresistive coefficient, high stability, and strong corrosion resistance[54][56] making them ideal for high-temperature, high-pressure, and corrosive environments such as chemical reactors and aerospace engines[55]. However, their sensitivity is limited and prone to temperature drift, often requiring compensation circuits. Capacitive SiC pressure sensors measure pressure through changes in capacitance[49], offering high sensitivity and precision suitable for biomedical, MEMS, and precision industrial applications[56]. Their use is limited by complex structure, high cost, and vulnerability to humidity and dust. Fiber-optic SiC pressure sensors detect pressure by monitoring variations in light signals (intensity, wavelength, frequency, or phase) [51][52]. They provide strong EMI resistance, thermal stability, and long-distance sensing, making them ideal for oilfields, mining, aerospace, and deep-sea exploration. However, their high cost, complex processing, and strict installation requirements hinder large-scale use.

In practical applications, sensor selection should consider measurement needs, environmental conditions, and technical constraints. For instance, piezoresistive sensors are favored in high-temperature, corrosive settings due to their simple design and strong stability. Capacitive sensors are better suited for detecting subtle pressure variations with high precision, while fiber-optic sensors excel in long-distance or EMI-prone environments, owing to their superior anti-interference and thermal stability. **Table 2** summarizes the structures, principles, advantages, limitations, and applications of these sensors. Understanding these distinctions helps guide the selection of suitable sensor types and mechanisms for accurate, efficient pressure measurement.

Table 2 SiC Pressure Sensor Categories and Brief Information

Type	Illustration of principle	Advantages	Disadvantages	Major applications areas
Piezoresistive	 (a) Four construction types of SiC piezoresistive pressure sensors. (b) Operating Principle of SiC Piezoresistive Pressure Sensors and Wheatstone Bridge.	Large longitudinal piezoresistive coefficient Good stability Excellent corrosion resistance	High influence of temperature changes Susceptible to mechanical vibration Limited accuracy	Automotive Aerospace Industrial Automation
Capacitive	 (a)Cross section through the MEMS capacitive pressure sensor. [337]. (b) Polycrystalline 3C-SiC thin film capacitive pressure sensors[340].	high precision, Strong anti-interference ability Good linearity	Complexity of manufacture High demands on environmental conditions High influence of parasitic capacitance	Aerospace Industrial Process Control Oil and Gas Extraction

Fibre-optic	 Schematic of SiC EFPI pressure sensor head and measurement scheme. Bottom left is the encapsulated sensor.	High temperature stability High corrosion resistance High precision and long life Remote monitoring	The process is complex High cost Dependent on ambient light	Aerospace Oil and gas extraction Chemical industry

3.1. SiC Piezoresistive Pressure Sensors

3.1.1. Sensing mechanisms

SiC piezoresistive pressure sensors utilize the piezoresistive effects of SiC, which are commonly designed and used to compensate for pressure readings in different directions [54]. In these piezoresistive pressure sensors, the output signal is the voltage reading; thus, the subsequent signal collection, processing, and conditioning circuit are quite simple. Currently, SiC is one of the most widely developed SiC pressure sensors.

In 1954, Smith discovered the piezoresistive effect[211]. Subsequently, researchers found that the piezoresistive effect plays a significant role in a wide range of essential semiconductor materials, resulting in significant changes in the resistance values when subjected to mechanical stress or strain [54]. The thickness, width, length, and resistivity of the resistor are denoted by t, ω, l and ρ, respectively.

$$R = \frac{l}{\omega t} \tag{3-1}$$

When strain is applied, the resistor's dimensions and resistivity are altered, leading to a change in the resistance.

$$\frac{\Delta R}{R} = \frac{\Delta\rho}{\rho} + \frac{\Delta l}{l} - \frac{\Delta\omega}{\omega} - \frac{\Delta t}{t} = \frac{\Delta\rho}{\rho} + (1 + 2\gamma)\varepsilon \tag{3-2}$$

where $\varepsilon = \Delta l / l$ represents the longitudinal strain of the material, and γ represents the Poisson's ratio of the material. The piezoresistive effect is commonly measured using the gauge factor (GF), which denotes the proportional change in resistance with respect to strain:

$$GF = \frac{\Delta R}{R\varepsilon} = \frac{\Delta\rho}{\rho\varepsilon} + (1 + 2\gamma) \tag{3-3}$$

where *R* represents resistance and “denotes strain.” GF is acausal, meaning it is positive when the resistance increases with increasing tensile or compressive strain, and negative when the resistance decreases with increasing tensile or compressive strain [212].**Table 3** lists the reported parameters of the piezoresistive effects in various SiC materials obtained from the literature.

Table 3 Lists of reported parameters of the piezoresistive effect of SiC materials

SiC Type	Doping	Growth	Carrier Concentration	Gauge Factor		Orientation
				Room Tem	High Tem	
Single 3C-SiC[214]	n/N	APCVD	10^{18}	-31.8	-18（450℃）	[100]
Single 3C-SiC[215]	n/N	APCVD	--	-18	-7（400℃）	[100]
Single 3C-SiC[217]	n/Al	LPCVD	--	5.8([100] longitudinal) 5.2([100] transverse) 30.3([110] longitudinal)	--	[110]
Single 3C-SiC[216]	p/Al	LPCVD	highly doped	28	25（300℃）	[110]
Single 3C-SiC[218]	p/Al	LPCVD	5×10^{18}	30.3(longitudinal) -25.1(transverse)	--	[110]
Single 3C-SiC[275]	n/N	--	--	-32	--	[100]
Single 3C-SiC[281]	n/N	LPCVD	--	-18	-10（200℃）	--
Single 3C-SiC[287]	n/N	HMCVD	10^{18}	-27	--	[100]
Poly **3C**-SiC[269]	n/N	LPCVD	--	18	7（400℃）	[100]
Single 4H-SiC[226]	n/N	--	1.5×10^{19}	20.8(longitudinal) -10(transverse)	--	--
Single 4H-SiC[228]	p/Al	--	1.12×10^{19}	26.7(longitudinal) -21.5(transverse)	--	[1100]
Single 4H-SiC[267]	p/Al	CVD	10^{18}	31.5(longitudinal) -27.3(transverse)	--	[1100]
Single 6H-SiC[230]	p/Al	LPCVD	2×10^{19}	27	12(250℃)	(0001)
Single 6H-SiC[137]	n/N	--	3.8×10^{18}	-29.4	-17(250℃)	(0001)
Amorphous SiC[290]	n/N	PECVD	--	49	--	--
Nanocrystalline SiC[244]	p/Al	LPCVD	2×10^{18}	14.5	--	--
nanobelts 3C-SiC[248]	n/N	LPCVD	_--	-61.7	--	[110]
Nanowires SiC[250]	n/N	--	--	33.6	--	（111）
nanowires 3C-SiC[253]	--	FIB	5×10^{18}	35	--	_--
nanowires 3C-SiC[256]	p/B	PSZ	--	620.5	--	[110]
Nanowires SiC[260]	N/P co-	--	--	-877.79	--	--

	doping					
nanowires 3C-SiC[264]	--	--	--	568.7	--	_--
nanobelts SiC [249]	B	PRE	--	-1875.1	--	[111]
3C-SiC/Si heterojunction[266]	--	--	--	58000	--	--

3.1.2. Piezoresistive effects in various SiC polytypes

For SiC polytypes (such as 6H- SiC, 4H- SiC, 3C-SiC, and SiC nanostructures), various dopants inside the SiC influence the gauge factor (GF). Doping elements of nitrogen (N), boron (B), and aluminum (Al) result in either N-type or P-type conductivities, and different doping concentration levels of doping elements have different effects on the piezoresistive properties of SiC[212]. In addition, crystal orientation affects the piezoresistive properties of SiC [54]. Two prevailing approaches for evaluating the piezoresistive phenomenon in SiC are the deforming diaphragm method [214][215][230] and the bending beam method[214][216][217][218] [228][213] [229][232]. In the deformed diaphragm method, a biaxial strain is induced in the piezoresistive layer of SiC resistors arranged on a diaphragm structure, and the strain is created by applying uniform pressure to the diaphragm[215]. In the bent beam method, the SiC resistor is either patterned or embedded in a substrate, which is then deflected by a load at the end of the beam, resulting in uniaxial strain in the SiC resistor[216][217].

(1) 3C-SiC

The piezoresistive characteristics of 3C-SiC materials are significantly influenced by the doping level of the dopant. For example, Shor et al. [214]explored the piezoresistive properties of N-doped n-type 3C-SiC films grown on Si substrates using APCVD at various doping concentrations. Unintentionally doped 3C-SiC films with N (doping concentration: Nd = 10^{16}-$10^{17}/cm^3$) exhibited a maximum GF of -31.8, whereas the films with non-uniform doping with N (doping concentration: Nd =$10^{20}/cm^3$) showed a lower GF of 12.7. However, post-annealing temperature for 3C-SiC films after the APCVD process was excessively high, exceeding 1000°C. In addition, owing to the thermal instability of the top silicon layer, the growth of high-quality SiC is extremely challenging. To address this challenge, Yasui et al. [215] utilized conventional CVD to epitaxially grow n-type 3C-SiC on SOI substrates, achieving an enhanced

GF of -27 by reducing the annealing temperature to 750°C. Additionally, stable electrical properties of highly B-doped p-type 3C-SiC were demonstrated at elevated temperatures. Phan et al.[216] investigated the piezoresistive characteristics of highly B-doped single-crystal 3C-SiC under high-temperature conditions by utilizing the Joule heating effect to heat up the piezoresistive resistance of 3C-SiC **(Figure 4. a & b)**. The GFs were approximately 25-28 in the temperature range of 300K~ 573 K. Phan et al. [217] further investigated the piezoresistive effect of p-type single-crystal 3C-SiC films and reported longitudinal and transverse gauge factors (GFs) of 30.3 and -25.1, respectively, for [110]-oriented p-type 3C-SiC at room temperature (**Figure 4.c & d**) [217]. Other investigations into p-type 3C-SiC films oriented at [100] (**Figure 4. e&f**) discovered that the transverse and longitudinal GFs were only 5.2 and 5.8, respectively [218]. Consequently, p-type 3C-SiC pressure sensors were engineered to exhibit greater sensitivity in the silicon (100) plane as opposed to the [110] direction **(Figure 4.g)** [218].

However, the deposition of 3C-SiC is limited to thin films on Si substrates using a heteroepitaxy approach[219][220][216], and the significant lattice mismatch and differences in the thermal expansion coefficients (TECs) between Si and 3C-SiC results in substantial intrinsic stresses in the SiC epitaxial layer. This results in heterojunction leakage and a mixed SiC/Si varistor[221]. This issue was partially overcome by choosing a suitable oxide isolation layer (i.e., forming a 3C-SiC-SiO_2-Si structure, also referred to as the SiC-OI structure) [222][223]. However, like those in the Si thin-film sensors, Si in the SiC-OI structure becomes easily crept when the temperature exceeded 400°C, which constraintly hindered the operational temperature span and overall efficiency of 3C-SiC pressure sensors. [229].

(2) 4H-SiC and 6H-SiC

The fabrication of SiC pressure sensors utilizing 4H-SiC and 6H-SiC wafers has been documented as a method to circumvent the challenges of high stress and creep issues [224]. Depositing 3C-SiC films on Si substrates could be a favorable option. By manipulating the doping concentrations of doped materials (such as N, P, and Al), a consistent and reliable piezoresistive effect can be achieved[225][226][227], as well as the dimensions of the piezoresistor (length and width) [228]. For example, Joseph et al.[229] investigated n-type 6H-SiC piezoresistors that were doped with N at levels of $1.8\times10^{17}/cm^3$ and $3\times10^{18}/cm^3$, The longitudinal GFs of the 6H-SiC piezoresistors were 35 and 29.4 for the two doping concentrations. Similarly, Okojie et al.[230] studied the piezoresistive characteristics and electrical properties of heavily doped ($2\times10^{19}/cm^3$) p-type (B-element doping) and n-type (N-element doping) 6H-SiC piezoresistors within the integrated cantilever beam sensors. They reported that the GFs of the p-type and n-type 6H-SiC were 27 and 22, respectively. They also developed a 6H-SiC pressure sensor designed to function at 600°C[231] and found that at this temperature, the maximum output of these sensors decreased by approximately 50-65% compared to their values at room temperature. These findings suggest that these devices, which exhibit subpar performance at high temperatures, are unsuitable for operation at elevated temperatures [231].

Nguyen et al.[232] investigated the piezoresistive properties of p-type 4H-SiC piezoresistors doped with aluminum (Al) at a doping level of $10^{18}cm^{-3}$ in both the longitudinal and transverse directions **(Figure 4. h).** They reported that the transverse and longitudinal GFs of the p-type 4H-SiC were -27.3 and 31.5°, respectively **(Figure 4. i)**. They further reported **(Figure 4. j)** that the transverse and longitudinal GFs of heavily doped p-type 4H-SiC along the [1100] crystallographic direction were 26.7 and -21.5, respectively **(Figure 4. k)** [233]. In addition, Meng et al. [234] reported that the piezoresistive curves of 4H-SiC with different doping concentrations showed significant nonlinear fluctuations, but increasing the doping concentration of 4H-SiC effectively suppressed the nonlinear piezoresistive effect caused by such the intrinsic excitation.

The GFs of piezoresistors can be increased by doping different elements simultaneously, which enhances the sensitivity of 6H-SiC and 4H-SiC based pressure sensors. For instance, Tomasz et al. [212] conducted research on the piezoresistive properties of 4H-SiC and 6H-SiC doped with various elements (such as Sc, B, N, and Al) and discovered that the absolute value of the gauge factors for the N-doped samples was decreased within the range of $1–5x10^{18}/cm^3$. The conductivity of the piezoresistive materials was increased upon doping with more elements. The GF of co-doped 6H-SiC was reported to have the highest value of 25-35, much higher than that of singly doped 6H-SiC (GF=22). The GF values were reported to remain constant values at 300°C.

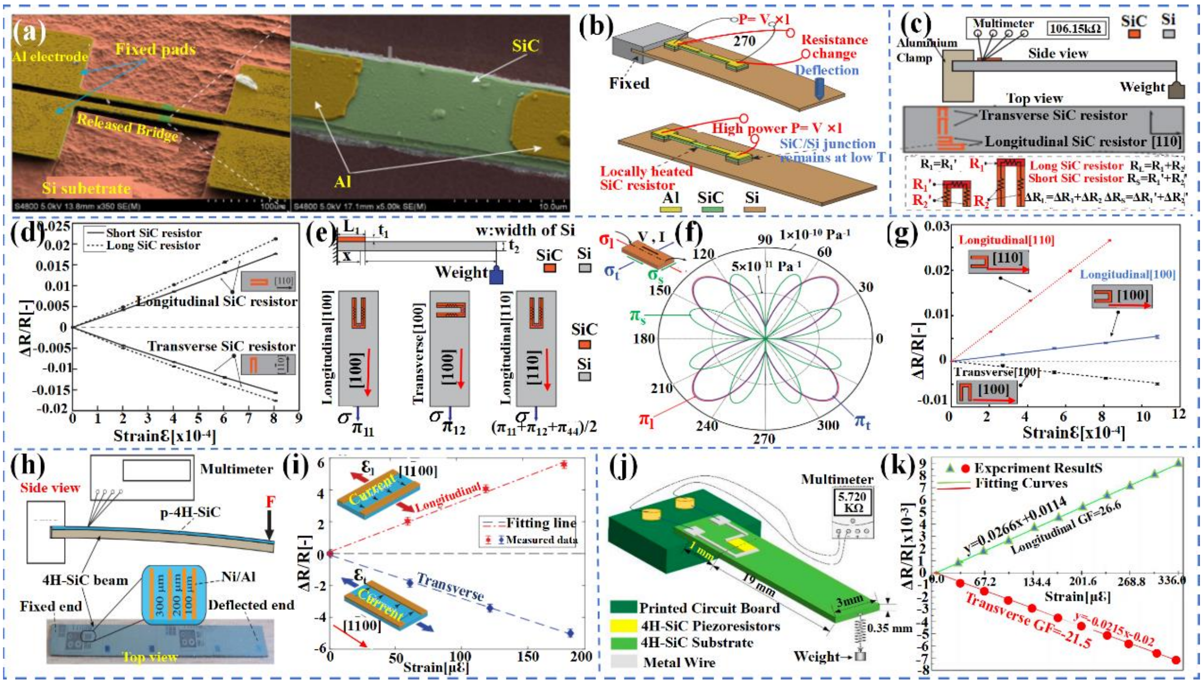


Figure 4 (a) SiC resistors on Si substrate; (b) The proposed in situ measurement. Reproduced with permission from ref [216]. (Copyright 2016 Hoang-Phuong Phan.). (c) Experimental setup. (d) Longitudinal/transverse fractional resistance variation of 3C-SiC resistors. Reproduced with permission from ref[217]. (Copyright 2014 IEEE.). (e) Schematic diagram of a bending experiment to measure GFs; and (f) Piezoresistive coefficients πl, πt, πs for p-3C-SiC (100); and (g) Relationship between ΔR/R and strain for SiC resistors with different orientations. Reproduced with permission from ref [218]. (Copyright 2014 AIP Publishing.). (h) Schematic diagram of 4H-SiC piezoresistor uniaxial strain experiment. (i) The linear relationship between ΔR/R and strain Reproduced with permission from ref [232]. (Copyright 2017 IEEE.). (j) Test structure of the p-type 4H-SiC piezoresistive effect. (k) Relative change in resistance of beams as a function of strain. Reproduced with permission from ref [233]. (Copyright 2021 Yongwei Li.).

(3) SiC nanostructures

New strategies could be applied to develop nanoscale SiC structures (such as nanosheets, nanocrystalline phases, nanobelts, or nanowires) as active components, attributed to their unique structures and morphological features and excellent mechanical and electrical properties[235][236][237]. These enable fast responses to external pressure changes, high sensitivity, and high accuracy for high-temperature pressure-sensor applications [237][251]. In the subsequent sections, we will discuss application advantages, piezoresistive properties, and influencing factors of the three structures of nanocrystals, nanoribbons, and nanowires.

SiC Nanocrystalline structures

Compared to single-crystalline SiC, SiC with a nanocrystalline structure and submicron grains can be synthesized at significantly lower temperatures on a range of substrates (such as silicon nitride, silicon dioxide, and silicon) using MEMS fabrication techniques. [238][239][240][241][242][243]. Phan et al. [244] reported **(Figure 5. a& b)** that the electrical conductivity of p-type nanocrystalline SiC is significantly affected by mechanical strain. They investigated the piezoresistive effect and GF value of p-type SiC nanocrystals by examining the changes in resistivity at nanograins and grain boundaries, as well as the random orientation of SiC nanocrystals [244]. However, to date, few studies have been conducted in this direction.

SiC Nanobelts

SiC nanobelts have recently attracted considerable interest because of their distinctive aspect ratios and extensive surface areas[245][246][247]. The piezoresistive effect of SiC nanobelts can be significantly affected by their dopants (including B, Al, and N) and doping thickness (often referred to as size effect). For instance, Li et al.[248] examined the lateral piezoresistive behavior of N-type 3C-SiC nanobelt N doping **(Figure 5.c&d&e)** and found a corresponding GF of 61.7. However, higher GFs were obtained when B dopants were introduced, which did not show synergized size effect of the SiC nanoribbons. Li et al. [249] explored the piezoresistive effect of individual 3C-SiC(111) nanobelts synergized with the size effect of B-doping, and their GF values were as high as -1875.1. They also performed controlled trials using SiC nanobelts with thicknesses of 90 and 200 nm under identical stress conditions. The 90 nm thick SiC nanobelts exhibited a GF of 796.1, surpassing that of 200 nm nanoribbons (GF = 373.1) [249].

SiC Nanowires

SiC nanowires (SiC NWs) have a one-dimensional structure with superior electron transport characteristics, a significant specific surface area, and mechanical properties in the axial direction[250][251]. These properties allow the SiC NWs to have high reaction rates and sensitivity, and these NWs commonly exhibit strong piezoresistive responses to compressive strains. Gao et al. [252]investigated the piezoresistive properties of p-type 6H-SiC nanowires **(Figure 5.f&g&h)** and observed a decrease in resistance with increasing compressive strain. The measured transverse piezoresistive coefficients π [010] of the nanowire-structured p-type

6H-SiC were varied between 51.2 and 159.5 10^{-11} Pa when subjected to loading forces ranging from 43.8 to 140.2 nN. In addition, Phan et al.[253] studied the piezoresistive effects of nanowire-structured p-type 3C-SiC **(Figure 5.i)** and verified the excellent electrical conductivity of the uniaxially strained arrays of SiC NWs **(Figure 5.j&k)**. A GF value of 35 was obtained for the SiC NWs prepared using the top-down approach, which was highly compatible with the mass production of the IC process.

The use of nanoscale SiCs with different B, N, and P elements is an effective approach to improve their piezoresistive properties [254]. N-doped n-type 3C-SiC nanowires displayed a remarkably weak current feedback response to the applied pressure at the nN scale, reflecting their extremely high sensitivity[255]. The introduction of B doping in SiC nanowires has been found to alter the acceptor states within their bandgap to a lower impurity energy level, resulting in the improved piezoresistive behavior and enhanced sensitivity to the piezoresistive effect [256][257] . Furthermore, Li et al. [256] reported an increase in the resistance of B-doped 3C-SiC nanowires when subjected to compressive stress, providing further evidence of the transverse piezoresistive behavior of SiC nanowires. When applying a force ranging from 51.7 to 181.0 nN, the piezoresistive coefficient $\pi[110]$ was ranged from 8.83 to103.42x$10^{-11}$$Pa^{-1}$ and resulted in a significantly high GF value of -620.5[256]. Moreover, doping with different elements can yield synergistic effects for increasing the GF of the SiC nanowires. For example, B doping has been reported to effectively alter the carrier density and mobility under tensile stress[258], whereas doping with P and N doping increases the lattice constants and electrical activation[259]. Cheng et al. [260] demonstrated that the lateral piezoresistive coefficient $\pi[110]$ of N and P co-doped SiC nanowires was increased from 5.07 to $-146.30 \times 10^{-11}Pa^{-1}$ as the applied forces were varied from 24.95 nN to 130.51 nN, resulting in a GF of about -877.79.

However, it should be noted that contamination and damage could occur when doping SiC NWs, leading to large discrepancies between the obtained results for deposited SiC NWs and the original properties of SiC NWs[261][262][263]. To address this challenge, Cui et al.[264] fixed SiC NWs with goat hair and conductive silver (Ag) epoxy resin for experiments and succeeded in eliminating the contamination and damage caused by the process. They discovered that the piezoresistive coefficient of SiC NW is 82 times greater than that of naturally formed SiC NW, with a measurement of -94.78×10^{-11} Pa^{-1}. Its GF was 568.7, representing a 17-fold increase compared with that of normal bulk SiC [212].

Integrating piezoresistivity with other physical phenomena (e.g., piezoelectricity) has been recognized as a viable approach to enhance the piezoresistive characteristics of SiC NWs[265]. Nguyen et al.[266] modulated tensile and compressive strains (i.e., the photoelectric coupling effect) by coupling the photoexcitation of charge carriers, strain and carrier mobility modification, use of electric field modulation of carrier energy, modulation of current, and coupling of transverse photovoltaic voltage. The GF of the 3C-SiC/Si heterojunction reached 58,000 under visible-light irradiation, representing a 30,000-fold increase compared to commercial metallic strain gauges and over a 2,000-fold increase compared to 3C-SiC under dark conditions [266]. It was also demonstrated that photocoupling substantially improved the efficiency of micromechanical heterojunction pressure sensors[267][268]. The utilization of photocoupling led to an increase in pressure sensitivity from 4.7×10^{-6} kPa^{-1} to 0.87 kPa^{-1}, which is about 185,000 times higher compared to the pressure sensitivity without the application of photocoupling. Replacing the irradiation light with UV irradiation has also been reported to enhance the piezoresistive properties of SiC NWs [269].

In general, SiC nanostructures exhibit a much larger surface-area-to-volume ratio than those of SiC thin films or bulk SiC. This characteristic renders them more responsive to external pressure, and their surface conductivity undergoes significant alterations upon the application of stress or adsorption of molecules on the surface of SiC nanowires[270]. Compared to the conventional SiC thin films or bulk SiC, pressure sensors prepared from SiC nanostructures have commonly shown higher sensitivity and faster response time[252][253][255][267].

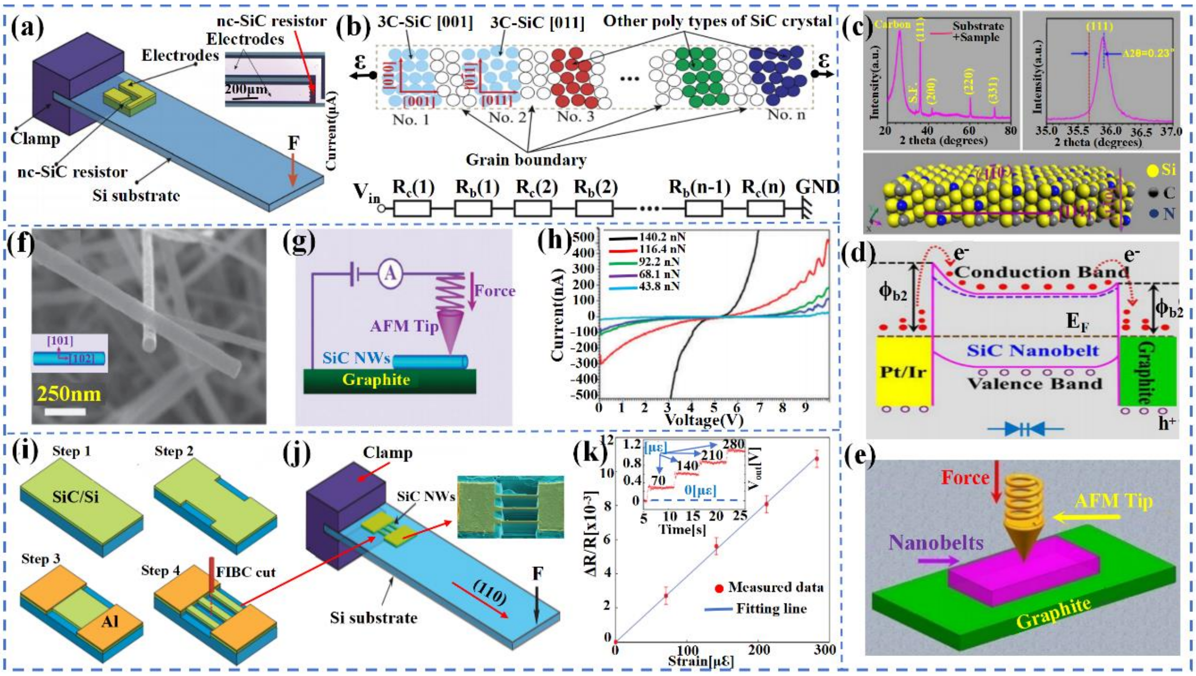


Figure 5 (a) Bending experiment; and (b)An one dimensional (1-D) model of the crystalline structure of nc-SiC. [244]. (Copyright 2012 Royal Society of Chemistry). (c) XRD patterns of SiC nanoparticles + substrate; and (d) Schematic energy bands of Pt/Ir-SiC structure; and (e) Measurement of piezoresistive effects in a single SiC nanotape. Reproduced with permission from ref [248]. (Copyright 2018 Elsevier Ltd.). (f) SEM images of p-6H-SiC nanowires; and (g)Schematic diagram of the electromechanical characterization measurements; and (h) I-V curves of p-6H-SiC nanowires. Reproduced with permission from ref [252]. (Copyright 2011 Royal Society of Chemistry.). (i) SEM image of the as-fabricated SiC NWs; and (j) Schematic sketch of the bending experiment; and (k) Linear relationship between the relative resistance change (ΔR/R) and strain for SiC nanowire arrays. Reproduced with permission from ref [253]. (Copyright 2016, IEEE.).

Overall, SiC materials with high GF values (such as 4H-SiC, 6H-SiC, and certain nanostructures) exhibit high sensitivity, excellent temperature stability, and a wide temperature range in high-temperature pressure sensors, making them suitable for high-temperature, high-precision applications. In contrast, SiC materials with low GF values (such as heavily doped 3C-SiC) show lower sensitivity and poorer high-temperature performance, making them more suitable for low-temperature or room-temperature environments where performance requirements are less demanding. Clearly the GF coefficient of SiC materials is significantly influenced by factors such as crystal structure, doping type and concentration, crystal orientation, and nanostructure. In terms of crystal structure, 4H-SiC and 6H-SiC generally exhibit higher GF values. The GF value of 3C-SiC is more affected by doping concentration, with higher GF values at low doping levels (-31.8) and significantly lower values at high doping levels (12.7). Regarding doping type and concentration, elements such as N, B, and Al, along with their concentrations, directly influence the GF values. For example, N-doped 6H-SiC showed a GF value of 35 at low doping concentrations, which was decreased to 29.4 at high concentrations. Co-doping (e.g., with B, N, and Al) can further enhance the GF value to 25-35. Crystal orientation also significantly affects the GF value. For instance, [110]-oriented p-type 3C-SiC showed a GF value of 30.3, while the [100] orientation one yielded only a value of 5.8. Nanostructures (such as nanobelts and nanowires) exhibit extremely high GF values due to their high surface-area-to-volume ratio and unique electronic properties. Additionally, temperature has a notable impact. For example, the 4H-SiC and 6H-SiC maintain stable GF values at high temperatures (>500°C), whereas 3C-SiC shows a significant decline in GF value above 200°C. By optimizing material selection and doping processes, the performance of SiC pressure sensors can be significantly improved, particularly in high-temperature and high-sensitivity applications.

The application of SiC materials in piezoresistive pressure sensors has been extensively developed recently. **Table 4** summarizes the crucial information on the processing, application temperature, and operating sensitivity of various piezoresistive pressure sensors.

Table 4 SiC Piezoresistive Pressure Sensor Development History（R.T=Room Temperature）

SiC Type	Substructure	Temperature	Growth	Etching	Sensitivity at work
Single Crystalline 3C-SiC[275][272]	SOI	R.T --573k	--	--	20.2μV/V/Kpa(R)
Single Crystalline 3C-SiC[281]	SOI	25--200°C	LPCVD	PECE	3.5mV/Vbar(R) 2.1m V//Vbar(200°C)
Single Crystalline 3C-SiC[282]	SOI	25--300°C	PECVD	RIE	1.9 mV/MPa(R) 1.2 mV/MPa (300°C)
Single Crystalline 3C-SiC[283]	SOI	R.T --385°C	APCVD	KOH	101.5 μ V /V psi(R) 53.4μ V /V psi (85°C)
polycrystalline 3C-SiC[272]	SOI	R.T --573k	CVD	RIE	2.0 mV /V/bar(R)
polycrystalline 3C-SiC[284]	SOI	R.T --400°C	APCVD	RIE	177.6 mv /V psi (25°C) 63.1 mv /V psi (400°C)
3C-SiC[214]	SOI	25--800°C	APCVD	--	8.7μV/V/Kpa (25°C) 4.4μV/V/Kpa(350°C)
3C -SiC[271]	SOI	--	PECVD	KOH	3.90 mV/psi(R)

3C-SiC[290]	SOI	R.T --400℃	PECVD	KOH	--
N-6H-SiC[293]	All-SiC	20°C--500°C	APCVD	PEC	0.019 mV/V-ppm
N-6H-SiC[138]	All-SiC	20°C--600°C	--	PEC	0.013mV/V/psi
N-6H-SiC[294]	All-SiC	25°C--600°C	--	DREI	32.5 nV/V/psi(R)
4H-SiC[295]	All-SiC	23°C--800°C	CVD	RIE	18.8 µV/V/psi
4H-SiC[310]	All-SiC	23°C--225°C	--	RIE	859 µ V/bar (25℃)
4H-SiC[302]	All-SiC	- 50◦C--300◦C	CVD	FS laser	132.4 µV/V/MPa (25℃) 144.4 µV/V/MPa (-50℃) 76.0 µV/V/MPa (300℃)
n-4H-SiC[151]	All-SiC	−50°C--600°C	CVD	FS laser+ICP	--
n-4H-SiC[305]	All-SiC	30°C--500°C	CVD	ICP-RIE	3.4mv/MPa (500℃)
n-4H-SiC[303]	All-SiC	R.T --600℃	APCVD	RIE	--
n-4H-SiC[309]	All-SiC	30°C--350°C	--	ICP-RIE	13.2mV/V/kPa
N-4H-SiC[207]	All-SiC	30°C--250°C	--	CMP	1.56 mV/V/MPa (30℃) 0.775mV/V/MPa (250℃)
P-6H-SiC[319]	All-SiC	< 600°C	--	RIE	--

3.1.3. SOI-based 3C-SiC pressure sensor

Structure and performance

During the initial stages of research on SiC piezoresistive pressure sensors, the primary focus of researchers was on producing SiC pressure sensors by epitaxially depositing 3C-SiC films on SOI or Si substrates, owing to the significant limitations posed by high-quality SiC crystal growth and SiC processing procedures **(Figure 6. a&b&c)** [271]. However, the performance of pressure sensors fabricated through the direct epitaxial growth of 3C-SiC on Si was often unsatisfactory because of the significant lattice mismatch between Si and 3C-SiC, as well as the substantial disparity in their thermal expansion coefficients, resulting in substantial inherent stresses in the SiC epitaxial layer. This resulted in heterojunction leakage and the formation of SiC/Si hybrid varistors[272]. SiO_2 was then applied as the oxide isolation interlayer (i.e., the 3C-SiC-on-SOI structure, also known as the SiC-OI structure), which has been reported to overcome the above problems[273]. It has been shown that a high-quality oxide layer not only broadens the temperature range of buried oxides as insulators, but also obtains large wafers with effective electrical isolation between the silicon block and the SiC film[274].

Ziermann et al.[275] demonstrated a 3C-SiC piezoresistive pressure sensor using an SOI substrate **(Figure 6.d)**, where the SiO_2 isolation layer effectively prevented the reverse leakage current from the piezoresistor to the Si substrate within the temperature range of room temperature to 573 K (or 300°C). Reichert et al. [273][276] characterized the doping properties of SiC-OI structures and revealed that P atoms caused notable lattice distortion [276]. The diffusion coefficient of SiC was enhanced by N doping, and the electron mobility and electrical activation were increased with the increasing injection temperature [273].

3C-SiC can be divided into single-crystal and polycrystalline forms, based on the continuity of the crystal structure[277],[278]. Single-crystal 3C-SiC exhibits a continuous structure with excellent crystal quality and uniformity, while polycrystalline 3C-SiC is composed of numerous grains with grain boundaries within the crystal structure. [279][280]. Eickhoff et al.[281] utilized LPCVD process to selectively deposit single-crystal 3C-SiC on SOI and fabricate pressure sensors. The GF of the fabricated piezoresistor was found to decrease with increasing temperature, with values of -18 and -10 at room temperature and 200°C, respectively. When employing a deep dry etching process for the 3C-SiC film, a significantly enhanced operational sensitivity of 3.5 mV/Vbar and 2.1 mV/Vbar at room temperature and 200°C respectively was achieved, indicating improved stability. Berg et al.[282] designed a piezoresistive pressure sensor with an SOI structure **(Figure 6.e)** to monitor the combustion cycle and cylinder pressure within the internal combustion engines **(Figure 6. f)**. As the operating temperature was increased from room temperature to 300°C, the sensitivity of the sensor was decreased from 1.9 mV/MPa to around 1.2 mV/MPa **(Figure 6. g).** It was also noted that the sensor sensitivity was decreased markedly as the temperature was increased, thereby severely restricting the pressure sensor's operating temperature range(<300°C)[281][282]. To enhance their performance at higher temperatures, it is essential to account for the temperature sensitivity and out-of-phase voltage and make the necessary compensations[272]. However, single-crystal 3C-SiC varistors do not commonly show such an improvement.

Polycrystalline 3C-SiC films have good attributes such as low-temperature drift characteristics and relatively small changes in electrical characteristics at different temperatures

[283]. Additionally, the polycrystalline 3C-SiC pressure sensor exhibited a good temperature dependence, enabling it to offset the bias voltage effect and temperature sensitivity across a broad temperature range. Consequently, it functions effectively at a temperature of 673 K(400°C) [272]. Furthermore, polycrystalline 3C-SiC technology offers the potential to eliminate the requirement for the additional wafer-bonding process, which were often used in single-crystal 3C-SiC technology. **(Figure 6. h)** [284][273].

The key issue with polycrystalline 3C-SiC is that the strain factor of the corresponding thin films is rather poor; thus, its sensitivity is often quite low. To solve this issue, Tong et al·[285] created a SiC piezoresistive pressure sensor that exhibited exceptional ultra-sensitivity and stress amplification **(Figure 6. i)**. This technology has increased the maximum stress of the device by 750% when compared to conventional sensors of similar dimensions, while still maintaining a very high sensitivity of 0.276 mV/VkPa [285].

Disadvantages

However, 3C-SiC piezoresistive pressure sensors utilizing the SOI structures have encountered significant issues. Specifically, the underlying Si in the SOI configuration is susceptible to deformation at temperatures exceeding 400°C, resulting in the failure of the sensor structures[286]. This restricts the high-temperature application of SOI-based 3C-SiC pressure sensors. In addition, the production of high-quality and extensive 3C-SiC films is a significant obstacle. Owing to constraints in the fabrication process, the thin silicon top layer of the SOI substrate was usually incompletely converted into a 3C-SiC film while structuring the SiC piezoresistor pattern in the SiO_2 mask layer [287]. The generated Si/3C-SiC heterojunction hybrid structures affected the sensitivity stability of the pressure sensor. Wu et al. [287] utilized wafer bonding and bulk micromachining to develop a single-crystal 3C-SiC/ SiO_2/Si piezoresistive pressure sensor **(Figure 6. j)**. However, due to the difficulty in achieving a reliable doping standard during the deposition of single-crystal 3C-SiC, the sensitivity of the sensor was decreased as the temperature was increased. In addition, it was difficult to control the surface roughness of the processed 3C-SiC diaphragm. A higher concentration of KOH was used, but the back-etching speed of the SOI-based 3C-SiC wafers has decreased. It was reported that the addition of isopropyl alcohol (IPA) during the etching of 3C-SiC wafers in a KOH solution improved the surface smoothness of the wafers [288]. Marsi et al. [289]investigated the effects of combined KOH and isopropyl alcohol (IPA) at different modified concentrations and temperatures on the back-etching speed and smoothness of silicon-based silicon carbide 3C SiC wafers. They showed that the inclusion of 10% IPA led to a significant reduction in the surface roughness of 3C-SiC diaphragms.

Fraga et al·[290] investigated the characteristics of 4H-SiC films deposited on Si substrates using PECVD and RF magnetron sputtering techniques. This study revealed that the sensor sensitivity was superior to that of sensors using single-crystal and polycrystalline 3C-SiC films. An analysis of the thermal properties indicated that 4H-SiC films are more suitable for high-temperature piezoresistive sensor applications[290]. In addition, owing to the diverse pulsed pressure stimulations, the dynamic sensitivity of this silicon carbide sensor remained highly stable, exhibiting significantly reduced hysteresis errors, nonlinearities, and maximum hysteresis compared with silicon-based sensors. When the pulse amplitude and duration were varied, the SiC-based sensors showed a better resistance to shock disturbances than the Si-based sensors[291]. Fang et al·[292] also developed a 4H-SiC electrically isolated piezoresistive pressure sensor of the n-p-n type, in contrast to Si-based pressure sensors. The sensor exhibited an exceptional stability and high corrosion resistance and radiation resistance, which are far superior to those of Si-based 3C-SiC pressure sensors.

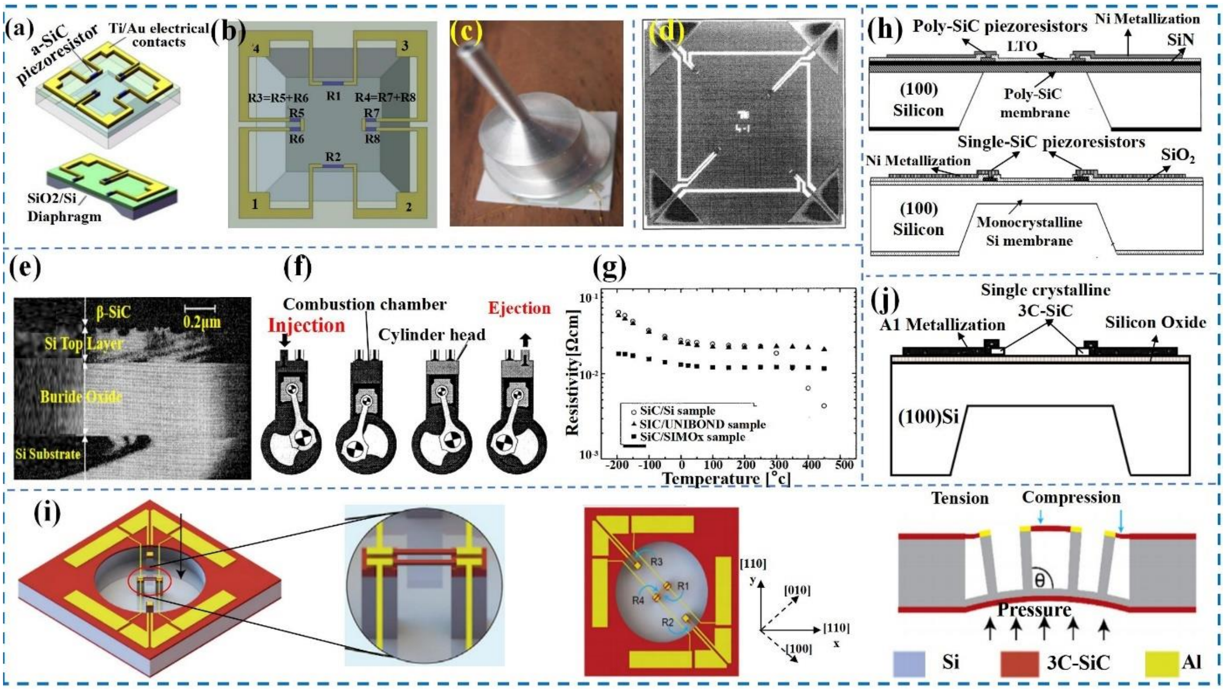


Figure 6 (a)Layout of 3C-SiC/SiO_2/Si piezoresistive pressure sensor; and (b) Distribution of 3C-SiC piezoresistor; and (c) packaged sensor. Reproduced with permission from ref [271]. (Copyright 2010 Elsevier Ltd.). (d)Photograph of the fabricated sensor chip. Reproduced with permission from ref [275]. (Copyright 1997 IEEE.). (e)XTEM photograph of the SiC-OI layer structure; and (f)Pressure and temperature behaviour in engine combustion chambers; and (g) Relationship between resistivity and temperature. Reproduced with permission from ref [282]. (Copyright 1998 IEEE.). (h) Cross-sectional view of the pressure sensor. Reproduced with permission from ref [284]. (Copyright 2006 IEEE.). (i) 3D schematic of the sensor. Reproduced with permission from ref [285]. (Copyright 2022 Braiden Tong et al.). (j) Cross-sectional view of the pressure sensor. Reproduced with permission from ref [287]. (Copyright 2001 C-H Wu et al.).

3.1.4. All-SiC Structure Piezoresistive Pressure Sensors

The advancement of pressure sensors utilizing all-SiC structures relies on the processing of SiC-sensitive diaphragms and MEMS fabrication techniques, which influence the performance of SiC pressure sensors. This section provides an overview of recent advancements and breakthroughs in all-SiC piezoresistive pressure sensors, including processing and MEMS fabrication techniques, zero drift, packaging applications, and optimized design.

(1) Integrated SiC processing and MEMS

The performance of SiC pressure sensors can be enhanced by integrating the SiC processing and MEMS fabrication techniques. The stability and temperature range of SiC piezoresistive sensors have been improved by co-etching by PEC or ICP with RIE etching. For instance, Okojie et al.[293] developed piezoresistive pressure sensors by producing 6H-SiC sensitive diaphragms using a combination of PEC and RIE etching. The sensor had a double-epitaxial (n-p-n) 6H-SiC structure. Electrochemical etching controls the shapes of the diaphragm and smoothens the sharp edges, and RIE process selectively etches the top layer of n-type SiC to form a piezoresistor with a maximum process temperature of 350°C. Okojie et al.[294] utilized a combination of inductively coupled plasma and deep reactive ion etching (ICP-DRIE) to collectively produce a batch of pressure sensors from SiC wafers, which were then subjected to testing at temperatures up to 600°C. However, SiC pressure sensors fabricated using ICP-DRIE etching showed a premature failure due to cracking of the diaphragm before reaching the predicted pressure handling capacity [148]. Okojie et al·[295] utilized the RIE to etch a 4H-SiC diaphragm and fabricate 4H-SiC piezoresistive sensors. Their pressure sensitivity was found to recover at operating temperatures above 400°C-800°C. As the temperature was increased, the full-scale output (FSO) voltage gradually diminished and then exhibited upward fluctuations as the temperature was increased further. Their results showed that the measured sensitivity values at 700°C were nearly equivalent to or greater than those acquired at the ambient temperature[296], and consistency and repeatability of FSO voltage were demonstrated to achieve in multiple devices. Pass temperature, pressure, and fuel/air ratio tests were also performed, and the results of the FSO and sensitivity showed excellent agreement [297].

For many brittle materials with femtosecond laser processing, the accumulation of microscopic damage in the SiC diaphragm after laser processing due to large stress is quite common [298][299]. However, other MEMS etching processes, such as Inductively Coupled Plasma (ICP), can minimize the harm inflicted on the diaphragm by the FS. This can lead to the production of superior diaphragms, thereby enhancing the precision of pressure sensors [150][151]. Wu Chen et al reported that the femtosecond laser-assisted SiC dry etching integrated processing method can achieve high surface quality and high rate etching of SiC microstructures[300]. This means that integrating MEMS processes with femtosecond laser ablation could yield SiC pressure sensors with high accuracy and low roughness.

Wang et al. [150] further utilized femtosecond laser micromachining in combination with ICP etching to process the membrane structure of a 4H-SiC pressure sensor **(Figure 7. a)**, and subsequently fabricated and packaged the sensor **(Figure 7. b)**. ICP etching can significantly diminish the surface roughness of the 4H-SiC diaphragm and improve the surface morphology of the membrane substrate **(Figure 7.c)**. Zhao et al. [151] used a combination of femtosecond laser deep etching, high-temperature rapid annealing, and ICP-based dry etching techniques to fabricate a 4H-SiC sensor **(Figure 7.d&e),** which were subsequently sealed using a gold wire bonding technique. The resulting SiC pressure sensors demonstrated an accuracy of 0.33% FS and were operated effectively across a temperature range of -50 to 600°C. In 2021, Wang et al.[301] combined the MEMS and femtosecond laser processing technologies to fabricate piezoresistive pressure sensors based on 4H-SiC sensitive diaphragms, as shown in **Figure 7.f**. The blind holes exhibited a surface roughness of 153 nm and a depth error within 2%. The resulting varistors demonstrated favorable voltage characteristic curves **(Figure 7. g)**. Furthermore, Wang et al.[302] employed a MEMS technique in conjunction with a femtosecond laser to produce 128 SiC-sensitive diaphragms on 4H-SiC wafers **(Figure 7. h)**, as shown in **Figure 7. i**. The output voltage results were obtained using a high-temperature test chamber **(Figure 7. j)** to compare the simulated output voltage of the ideal pressure sensor model with the actual values **(Figure 7. k)**. Tests conducted determined that repeatability and nonlinearity errors of the sensor were 1.49% FS and 0.13% FS, respectively, within a pressure range of 0-5 MPa at room temperature [302]. The impressive accuracy and stability of the measurements highlighted the potential of batch production of SiC piezoresistive pressure sensors on 4H-SiC wafers through the integration of MEMS processes and femtosecond laser processing technology.

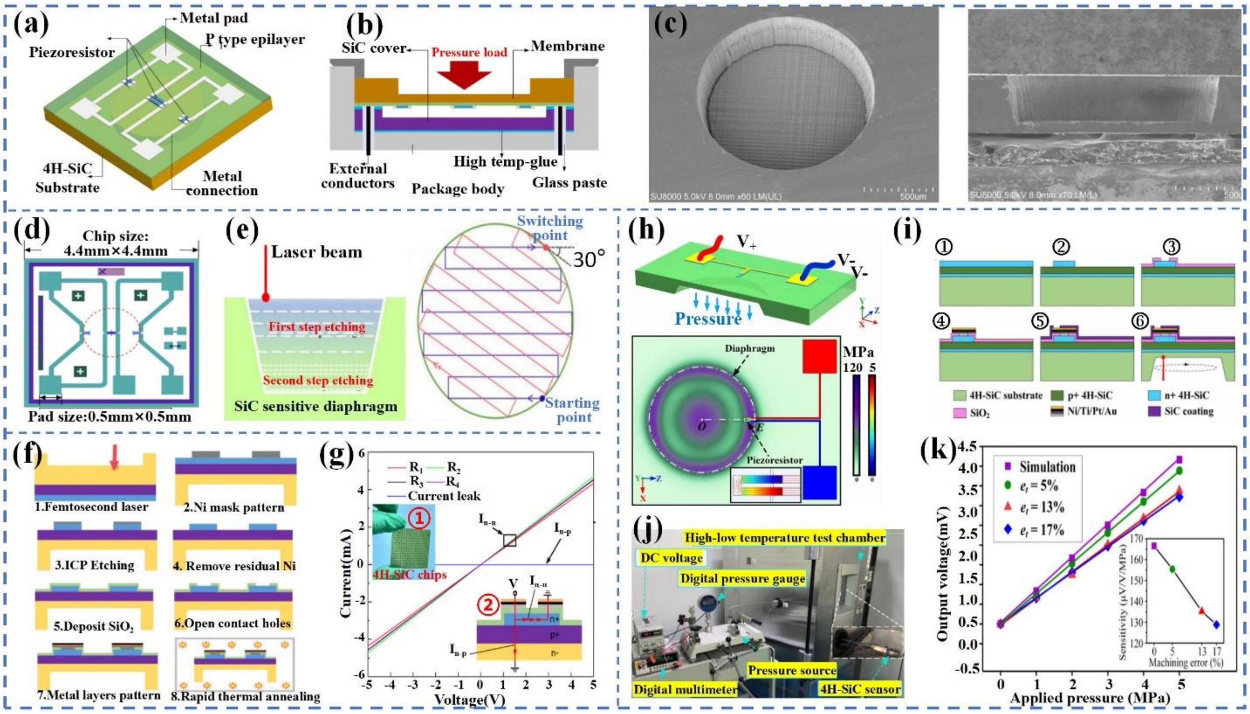


Figure 7 (a) 4H–SiC pressure sensor chip; and (b) Cross-sectional view of the sensor package structure; (c) 4H-SiC thin film diagram. Reproduced with permission from ref [150]. (Copyright © 2020 Elsevier Ltd and Techna Group S.r.l.). (d) Alignment of SiC varistors; and (e) Schematic of laser etching. Reproduced with permission from ref [151]. (Copyright 2023 You Zhao et al.). (f) Manufacturing process of the chip; and (g) Forward Voltage Characteristics of Varistors. Inset: ① Photograph of the 4H-SiC pressure sensor chips array;② Schematic diagram of sensor current measurement. Reproduced with permission from ref [301]. (Copyright 2021 Lukang Wang et al.). (h) 4H-SiC pressure sensor cross-section and diaphragm finite element analysis; and (i) Fabrication process of 4H-SiC diaphragms; and (j)Experimental setup of the pressure sensor measurement; and (k) Comparison of output results. Reproduced with permission from ref [302]. (Copyright 2022, IEEE.).

(2) Zero-point drift

The reliability of the pressure sensor relies heavily on the consistency of the reference zero-output voltage Voz post-calibration and during testing, with the level of zero drift during measurement influencing the accuracy[303]. A good metal ohmic structure has a significant effect on suppressing zero drift. For example, Hoogerwerf et al.[304] prepared a SiC piezoresistive pressure sensor with the ability to function at temperatures up to 600°C **(Figure 8.a)**, using Ta-Pt as the metal connection. However, the sensor's recorded output signal exhibited significant noise as well as substantial drift and temperature dependency. Fang et al.[305] fabricated pressure sensors based on 4H-SiC with ohmic connections in an alloy scheme that was functioned well at temperatures exceeding 500C and demonstrate outstanding consistency and precision across a temperature span from 30C to 500°C **(Figure 8.b)**. Okojie et al.[306] also fabricated a 4H-SiC piezoresistive pressure sensor with $Ti/TaSi_2/Pt$ as the metallization structure **(Figure 8.c)**. By optimizing the ohmic contact metal to suppress the time-zero drift of the output signal at high temperatures, the ohmic contact metal structure was selected as $Ti/TaSi_2/TaSi_2/Pt/Ti/Au$ **(Figure 8.d)**, and the relative drift rate of the output voltage of the sensor was suppressed to ±0.5 mV/h at 600 °C. This resulted in the formation of a thicker and more compact Pt-Si diffusion barrier, which effectively inhibited the diffusion of AuSi eutectic couples and prevented the infiltration of Au into the SiC interface **(Figure 8.d)**. Enhancements in regulating intermetallic diffusion and managing microstructural phase shifts within the metallization could diminish the offset and bolster the stability of the metallization ohmic point[307].Similarly, Fang et al.[308] utilized Al_2O_3 as a transition layer and fabricated a Pt thin-film resistance temperature detector (RTD) on a SiC substrate using the micro-nano machining technology. The Al_2O_3/Pt bilayer exhibited excellent interfacial stability at 950°C, enabling the Pt RTD to maintain good linearity and high-temperature electrical performance even above 850°C. This lays the groundwork for the integration of SiC pressure sensors.

Taking advantage of the low annealing temperature of Ti to form ohmic contacts, Okojie et al. [170] investigated monolithic 6H-SiC pressure sensor prototypes with $Ti/TaSi_2/Pt$ and Ti/TiN/Pt multilayer metallization strategies. After subjecting the n-type 6H-SiC coating to heat treatment in air at 600°C for over 20 h, its contact resistivity was remained stable, and the 6H-SiC pressure sensor was operated up to 600°C. However, at room temperature, the 6H-SiC piezoresistor GF values were small, with transverse and longitudinal GFs of 19 and 22, respectively, limiting the sensitivity and application temperature range of the 6H-SiC sensors. Whereas the 4H-SiC showed higher GFs[228][267], and the prepared pressure sensors showed recovery of sensitivity at temperatures above 600°C[295]. In addition, the stable Ti multilayer metallized structure has promoted the development of pressure sensors for multi-environment monitoring. Fang et al.[309] fabricated an n-type 4H-SiC piezoresistive pressure sensor with $Ti/TaSi_2/Pt$ metal ohmic contacts designed to function under hydraulic and pneumatic pressures at 350°C **(Figure 8.e)**. The sensors were operated under the hydraulic loads at 220°C and pneumatic loads at 350°C with high linearity and repeatability **(Figure 8.f)**. The preparation process is illustrated in **Figure 8.g**.

In addition, ohmic contact bonding has been adopted as a viable approach to improve the performance of SiC pressure sensors. For instance, Mackowiak et al.[310] prepared a novel 4H-SiC piezoresistive pressure sensor **(Figure 8.h)** by employing RIE to etch the Ti/Ti/WN metallization system, achieving an etching rate of 4 μm/min, and demonstrated the ability of the metallization system to withstand high temperatures up to 600°C.

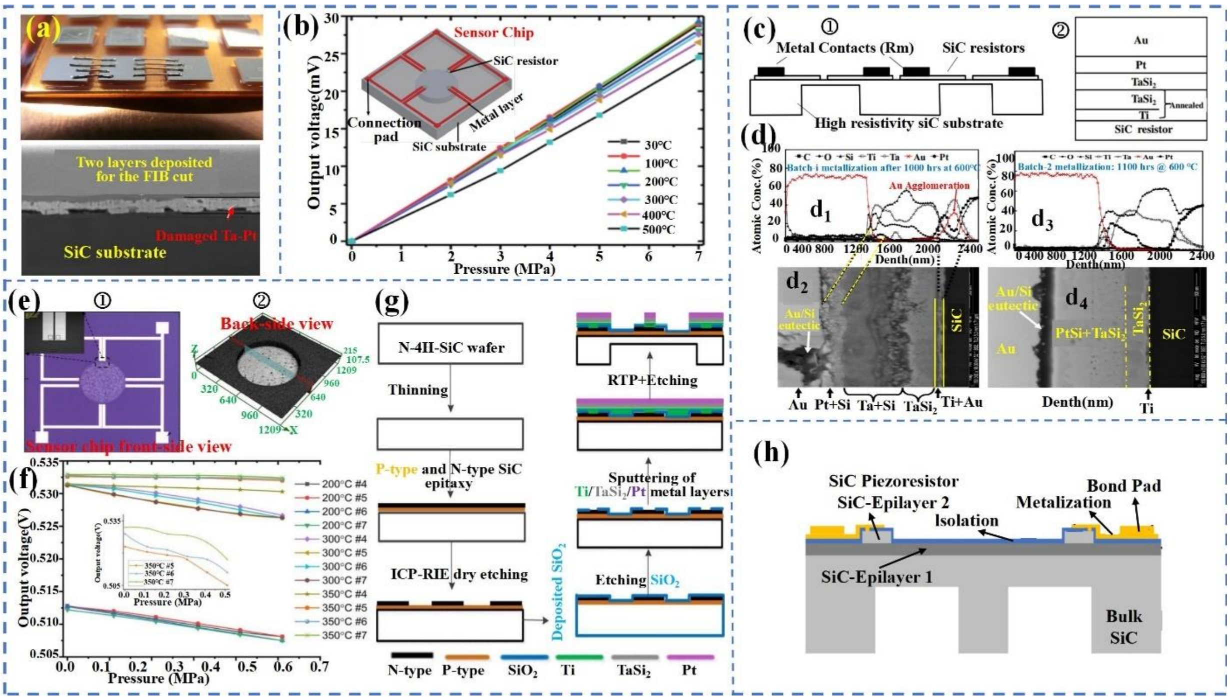


Figure 8 (a) Platinum wire bonded test chip and right: FIB cross-section of the Ta-Pt layer on the contact area. Reproduced with permission from ref [304]. (Copyright 2019 IEEE.). (b)Sensor test data from 30°C to 500°C with pressure range 0-7MPa; illustrate: Schematic of the sensor Chip. Reproduced with permission from ref [305]. (Copyright 2019 IEEE.). (c) ①Cross-section of a 4H-SiC pressure sensor; ② Contact metallization stacks; and (d) AES profiles and cross-section FESEM images. Reproduced with permission from ref [306]. (Copyright 2011 IEEE.). (e) All-SiC Sensor Chips; and (f)The static performance of the sensor; and (g)Fabrication process of the all-SiC pressure sensor chip. Reproduced with permission from ref [309]. (Copyright 2012 Institute of Physics.). (h)Schematical of the designed piezoresistive pressure sensor. Reproduced with permission from ref [310]. (Copyright 2021 IEEE.).

(3) Packaging process

In high-temperature environment applications, the packaging process is a key factor that affects the reliability and stability of SiC pressure sensors. The packaging technologies for SiC high-temperature pressure sensors are mainly lead wire bonding packaging and leadless packaging technology (direct chip attachment (MEMS-DCA) packaging).

Wire Bonding Package

Leads in wire-bonded packages are commonly connected to the encapsulated substrate using the bonding technique, and the sensor detection signals are transmitted through the leads. Burla et al.[311] used nickel wire bonded packages for the 3C-SiC pressure sensors capable of detecting signals up to 550°C. Wang et al.[312] achieved high-temperature lead-free packaging for N-type 4H-SiC high-doped pressure sensor chips by utilizing a nickel-based metallization system combined with glass frit and nanoparticle conductive silver paste sintering technology. Tests conducted at 600°C for 10 hours demonstrated that the sensor exhibited excellent temperature drift performance and high-temperature stability. Nevertheless, the stability of nickel wire leads is compromised under vibrations or prolonged exposure to environments exceeding 400°C. Zhao et al. [151] used Au wire bonding to encapsulate SiC pressure sensors. A schematic illustrating the wire bonding encapsulation process, as well as the encapsulated sensors, is shown in **Figure 9.a**. The input pressures and output voltages of the sensors maintained good linearity at different temperatures **(Figure 9.b)**.

Leadless Packages

Leadless encapsulation effectively eliminates failure mechanisms in lead-bonded packages at high temperatures [313][314]. This technology reduces the thermal resistance between the external environment and the sensor, enhancing the sensor's durability and dependability by encasing the sensor element directly in a material with high-temperature resistance. Commonly used high-temperature resistant materials include A1N and glass. SiC exhibits a coefficient of thermal expansion (CTE) comparable to that of A1N and glass; thus, A1N is frequently employed as an interlayer to reduce the CTE disparity within the subsystem [315]. The thermomechanical stresses transferred to the sensor device can be minimized at high temperatures[316]. Masheeb et al.[317][318] used a metal-glass frits mixture as the electrical interconnection between a 6H-SiC pressure sensor and sensor connector pins without bonding wires for sensor encapsulation. The encapsulation structure is shown in **Figure 9.c**, which demonstrates a mixture of metal-glass frits and acts as a hermetic protective layer. This layer enables the measurement of pressure sensitivity at pressures and temperatures of 1000 psi and

500°C, respectively [318]. Furthermore, Okojie et al.[316] utilized MEMS-based direct chip attachment (MEMS-DCA) technology to package a 6H-SiC pressure sensor **(Figure 9.d)**, The SiC sensor was affixed to the AlN header using DCA, eliminating the need for wire bonding. This allowed the chip to be packaged at a higher density; thus, only the circular diaphragm of the sensor was freely moved. The sealing glass was intended for use solely within the confined space (<10 μm) between the SiC sensor and the inner surface of the AlN, serving as a pressure reference for detection within the cavity. The net output voltage of the SiC pressure sensor was evaluated within the temperature range of 25 to 600°C, exhibited outstanding stability and linearity following a three-hour test at 600°C [319]. The MEMS-DCA technology allows for the matching of CTEs and good connections of the sensor to the package connector, which significantly reduces adhesive failures [320]. However, leadless packaging and MEMS-DCA technology are limited to indirect pressure detection and measurement using liquid-filled sensing techniques. The pressure deformations of the liquid medium and small temperature changes can affect the accuracy of the encapsulated sensor. Wieczorek et al·[321] developed a new pressure sensor prototype based on a fully floating encapsulated SiC mold **(Figure 9. e)**, which solves the aforementioned problems. The package structure is illustrated in **Figure 9.f**. The sensor chip was mounted in a ceramic holder, which provided an electrical connection. The pressure was transferred from the membrane to the chip using a pusher, and the housing was free of fluid. A steel membrane was used to separate the medium. The sensor provided the direct and accurate measurement of pressure and temperature, overcoming the drawbacks of indirect pressure detection based on fluid-filled measurements. It can be operated up to 400°C with a high fatigue resistance.

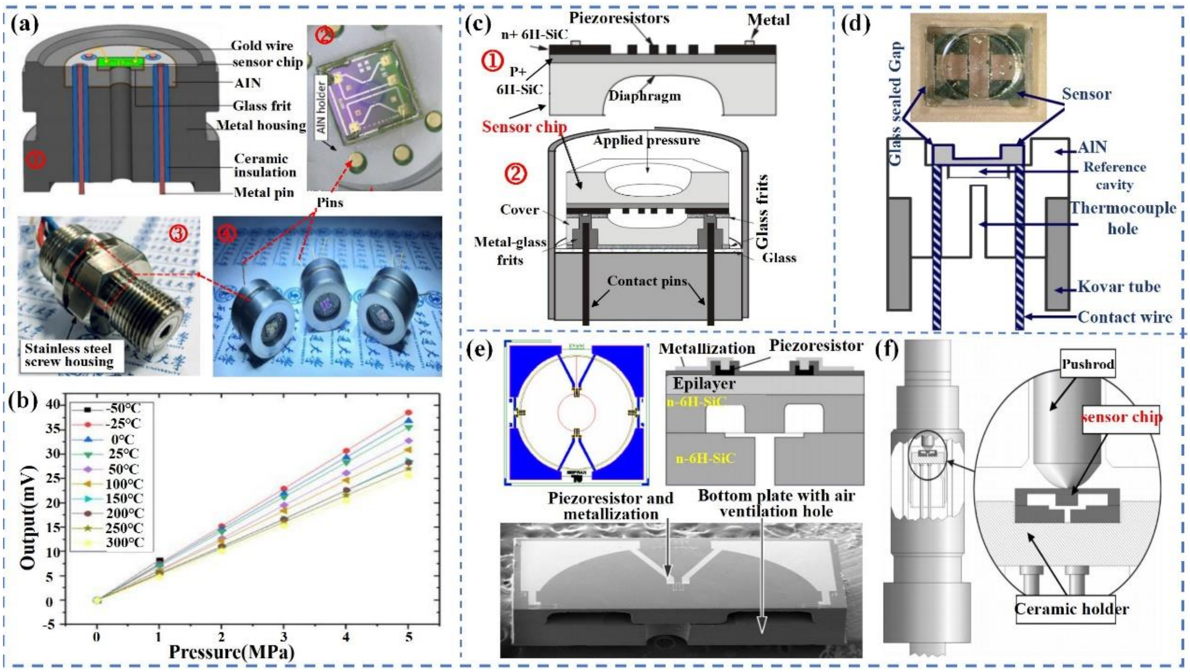


Figure 9 (a)①:Gold wire bonding package of the SiC pressure sensor;②Leadless sensor link pins;③④:Pictures of packaged sensors; and (b)Static calibration output curve at different temperature. Reproduced with permission from ref [151]. (Copyright 2023 You Zhao.). (c) ①Cross-section and top view of 6H-SiC sensor chip; ②Schematic Drawing of the Packaged Sensor Structure. Reproduced with permission from ref [317]. (Copyright 2002 IEEE.). (d) Cross-sectional view of the MEMS-DCA packaging concept. Reproduced with permission from ref [316]. (Copyright 2004 IEEE.). (e) Sensor chip layout (①: top view, ②: cross section) and ③:Cross-sectional view of SiC chip; and (f)Drawing of sensor housing. Reproduced with permission from ref [321]. (Copyright 2007 IEEE.).

(4) Simulation of optimized designs

To optimize SiC pressure sensors, it is effective to use various simulation software. For example, Belwanshi et al.[322][323] designed a pressure sensor with an edge-clamped configuration on a rectangular diaphragm using finite element analysis and theoretically investigated the linearity and sensitivity of the sensor by employing wide-bandgap materials, such as diamond and SiC. Tian et al.[324] designed SiC piezoresistive pressure sensors using circular diaphragms and analyzed the effects of various parameters on the nonlinearity and sensitivity of the pressure sensors, such as the shape, location, dimension, and connection method, on the metal wire. Wu et al.[325] developed a prototype SiC pressure sensor and conducted an in-depth analysis of its diaphragm structure, material composition, piezoresistive arrangement, and dimensions. Kanekal et al.[326] introduced a MEMS SiC pressure sensor with a free support on a rectangular diaphragm. Simulations were conducted using MATLAB and

COMSOL to analyze the deflection, stress, strain, and sensitivity of the piezoresistive pressure sensors based on the free-edge conditions of the rectangular SiC diaphragm [326]. However, these piezoresistive pressure sensors often exhibit a temperature dependence. The temperature change causes an electrical resistance change, which is difficult to distinguish from the measured pressure induced by the piezoresistive change. This affected the accuracy of the sensor.

3.2. SiC Capacitive Pressure Sensors

Capacitive pressure sensors typically use a pair of parallel plates as capacitors. SiC capacitive pressure sensors generally consist of edge-clamped SiC diaphragms (commonly 4H-SiC, 6H-SiC, and 3C-SiC films) suspended over a sealed cavity on a substrate (commonly Si substrate)[327]. Compared to SiC piezoresistive pressure sensors, these SiC capacitive pressure sensors have shown advantages of high accuracy and temperature insensitivity because they do not depend directly on contact resistance, but they measure pressure based on changes in capacitance[328]. Wireless sensing schemes[329][330] can be easily implemented, thus eliminating the possible degradation of the sensor's stability and accuracy due to parasitic capacitance in the line.

Table 5 lists the reported SiC capacitive pressure sensors, including that based on Si-3C-SiC structure and all-SiC capacitive ones. In the following, we will comprehensively overview the progress of capacitive pressure sensors, covering application temperature, manufacturing processes, and operational sensitivity.

Table 5 The development history and processing of SiC capacitive pressure sensors (R.T=Room Temperature)

Sensor structure	Temperature	Growth	Etching	Sensitivity at work
3C-SiC-Si[327][339]	R.T--400℃	APCVD	CMP+H_2	7.7 fF/torr
3C-SiC-Si[338]	300 K-- 800 K	APCVD + LPCVD	Plasma Etching	--
3C-SiC-Si[343]	R.T--500℃	LPCVD	KOH +IPA	0.00962 pF/MPa(27℃) 0.0127 pF/MPa（300℃）
poly-3C-SiC-Si[341]	R.T--400℃	LPCVD	RIE+CMP	~0.3%/psi（21℃-100℃） ~0.1%/psi（200℃-400℃）
SiC-Si[342]	R.T--400℃	LPCVD	CMP	--
poly-SiC-Si[340]	R.T--400℃	LPCVD	CMP+TMAH	0.62pF/psi（400℃） 0.53pF/psi（500℃）

poly-SiC-Si[344]	R.T--500℃	--	--	6.1x10-4fF/Pa
All SiC[344]	R.T--500℃	--	--	6.5x10-4fF/Pa
All SiC[351]	R.T--600℃	LPCVD	ICP	272 μ V/psi（200-450 psi） 251μV/psi（300℃） 7.2 fF/psi（574℃）
All SiC[352]	R.T--600℃	LPCVD	ICP	3. 95 mV/MPa（R.T） 3. 64 mV/MPa（300℃） 104. 43 mV/MPa（574℃）
All SiC[354]	--	PECVD	DRIE	0.13 pF/bar(R.T)
All SiC[355]	--	PECVD	DRIE	0.09494 pF/bar(R.T)
All SiC[356]	--	PECVD	DRIE	0.021 pF/bar(5×5 array) 0.023 pF/bar(after KOH etching.)
All SiC[370]	RT--540°C	--	--	6.8 x 10-2 MHz/psi
poly-SiC[358]	20--180℃	LPCVD	RIE	1.03 fF/kPa（20℃） 0.98 fF/kPa（100℃） 0.92 fF/kPa（180℃）
poly-SiC[359]	20--180℃	APCVD+ LPCVD	TCP	13.7 fF/kPa（20℃） 13.4 fF/kPa (180℃）
6H-SiC-AlN-SiC[366]	R.T--500℃	MBE+CVD	RIE	--
6H-SiC-AlN-SiC[367]	R.T--500℃	CVD	RIE+TMAH	3.5pF/MPa(R.T)
4H-SiC[371]	RT--500°C	--	--	0.113 kHz/psi(25°C 50 psi) 0.026 kHz/psi(500°C 50 psi)

The capacitance response to the external load is primarily influenced by the deflection of the diaphragm, which bends towards the substrate when subjected to increasing external pressure. This results in an increase in the device capacitance between the substrate and diaphragm [327]. When the diaphragm is compressed against the substrate, the sensor capacitance increases nearly proportionally with increasing pressure as a result of the linear expansion of the contact area[331]. The variation in capacitance is a distinct indication of the sensitivity to pressure, and the sensitive film of SiC capacitive pressure sensors adheres to the principles of both large and small deformations when subjected to deformation[332][333]. The plate/diaphragm is defined as having small deflections if the deflection is less than 1/5th of the diaphragm thickness. Conversely, a large deflection is defined as deflection up to three times the thickness of the diaphragm [333].

Generally, pressure can be determined by evaluating the capacitance C of a parallel-plate capacitor, as expressed by the following equation:

$$C = \frac{\varepsilon A}{d} \quad (3\text{-}4)$$

where *d, A* and *ε* represent the dielectric constants of the spacing of the plates, plate area, and gap, respectively. In the case of clamped circular plates with minimal deflection (i.e., less than half the diaphragm thickness), the deflection has the following form [333]:

$$\omega(r) = \frac{Pa^4}{64D}[1 - (\frac{r}{a})^2]^2 \quad (3\text{-}5)$$

where *r; P, a,* and ω represent the radial distance from the diaphragm center, applied pressure, diaphragm radius, and deflection, respectively. *D* denotes the flexural stiffness, which is computed as. [334]:

$$D = \frac{Eh^3}{12(1-v^2)} \quad (3\text{-}6)$$

where *v*, *h* and *E* represent Poisson's ratio of the diaphragm, thickness, and Young's modulus, respectively. It is evident from the above equation that the deflection varies directly with the applied pressure [334][335]. In the flexural state of a circular diaphragm sensor, capacitance (C) can be expressed as [334]:

$$C = \iint \frac{\varepsilon}{d-\omega(r)} r\, dr\, d\theta \quad (3\text{-}7)$$

3.2.1. 3C-SiC Capacitive Pressure Sensor Based on Si Substrate

The 3C-SiC films exhibit significantly better mechanical properties (including chemical stability, thermal conductivity, and strength) than the Si films. In terms of electrical properties, they can achieve more than 50% energy savings compared to Si when used as a sensitive diaphragm in pressure sensors[336]. Therefore, 3C-SiC films are often used (instead of Si films) to fabricate high-temperature capacitive pressure sensors **(Figure 10. a)**[337]. They can be grown epitaxially on Si or deposited onto a sacrificial layer using surface micromachining techniques to create capacitive pressure sensors [338]. According to the lattice structure of 3C-SiC, capacitive pressure sensors can be classified into single-crystal and polycrystalline 3C-SiC types.

Capacitive pressure sensors utilizing single-crystal 3C-SiC diaphragms on Si substrates have demonstrated excellent sensing performances at temperatures up to 400°C[339]. However, there are noticeable limitations, such as the constrained thickness of the SiC film in single-crystal 3C-SiC capacitive pressure sensors. When the epitaxial film thickness surpasses 0.5 μm, excessive residual stress (>200 MPa) significantly hinders the successful manufacturing of the pressure sensor [327]. Conversely, film thicknesses below 0.5 μm fall short of the requirements for pressure monitoring in larger-scale applications. To address this issue, Du et al·[340]fabricated low-stress structures with sealed vacuum cavities using nitrogen-doped polycrystalline 3C-SiC (poly-SiC) films **(Figure 10.b)** instead of epitaxial single-crystalline 3C-SiC films, allowing the utilization of thicker diaphragms. In addition, thin-film diaphragms made of polycrystalline 3C-SiC were coated with polysilicate films to create capacitive pressure-sensing elements **(Figure 10.c)** [341]. This has achieved a large geometrical aspect ratio as large as 800:1, a maximum flexural load deflection of eight or more times of their thickness, as well as a test temperature up to 500°C. However, limited by the process of deposition and etching during thin-film processing, the deposition rate and roughness of polysilicon are inferior to those of single-crystal 3C-SiC [340].

In addition, encapsulation materials and encapsulation processes should be selected appropriately to enhance the high-temperature resistance and corrosion of 3C-SiC capacitive pressure sensors, ensure stable signal transmission, and protect the internal structure of the sensor. For example, Marsi et al·[342] developed a MEMS prototype by encapsulating high-temperature SiC capacitive pressure sensors **(Figure 10.d)** in a set of stainless-steel O-ring vacuum chambers. The stable operating temperature was increased up over 500°C. Marsi et al·[343] further packaged a prototype 3C-SiC capacitive pressure sensor (CPS) **(Figure 10.f)**. The sensitivity of the developed sensor to pressures ranging from 1.0-5.0 MPa was 0.00962 pF/MPa at room temperature. At 300ºC, the sensitivity was increased to 0.0127 pF MPa, showing an enhanced sensitivity of 0.0031pF/MPa. However, as mentioned above, there is a thermal expansion mismatch between Si and 3C-SiC materials. Si undergoes juxtaposition above 500°C, leading to failure of sensor. An all-SiC capacitive type can overcome this issue. For example, Sheng et al·[344] designed two sets of SiC diaphragm capacitive pressure sensors using identical dimensions and fabrication processes on polycrystalline SiC and Si wafers to investigate the impact of thermal expansion mismatch **(Figure 10.g).** Their tests conducted in both ambient and 500°C environments revealed that the sensitivity of the sensor on Si substrate was inferior to that on SiC substrate, and the stability of SiC pressure sensors is quite favorable. Similarly, Marsi et al.[345] compared the performance of Si and 3C-SiC in the fabricating of capacitive pressure sensors. Both these materials showed good linear properties at temperatures up to 500°C. However the sensitivity of SiC pressure sensors was higher in contrast. Furthermore, at detection pressures of 700 °C **(Figure 10.h)** and 1000 °C **(Figure 10.i)**, the Si pressure sensors began to malfunction at 50 MPa, whereas the SiC pressure sensors continued to exhibit superior linearity and sensitivity.

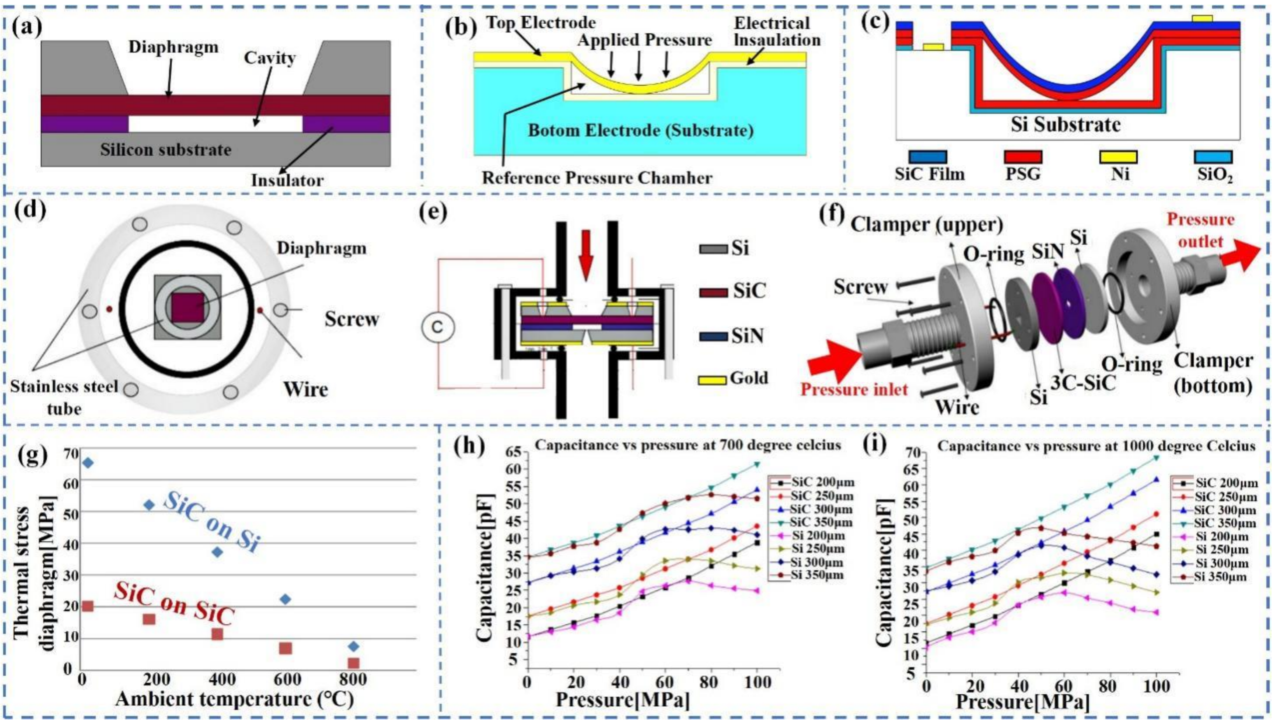


Figure 10 (a)Cross section through the MEMS capacitive pressure sensor. Reproduced with permission from ref [337]. (Copyright 2014 N. Marsi et al.). (b) Polycrystalline 3C-SiC thin film capacitive pressure sensors. Reproduced with permission from ref [340]. (Copyright 2013 The Materials Research Society.). (c) Cross-section of a miniature SiC capacitive pressure sensing structure operated in touch mode Reproduced with permission from ref [341]. (Copyright 2005 IEEE.). (d)The new design concept of packaged MEMS capacitive pressure sensor a top view (upper); and (e)side view (bottom); and (f)The MEMS capacitive pressure sensor with a reliable stainless steel O-ring packaging concept. Reproduced with permission from ref [343]. (Copyright 2015 Marsi et al.). (g)Thermal expansion mismatch stresses in SiC films based on SiC and Si substrates. Reproduced with permission from ref [344]. (Copyright 2011 IEEE.). (h)Capacitance versus pressure of 3C-SiC and Si at 700°C; and (i)at 1000°C. Reproduced with permission from ref [345]. (Copyright 2013 Marsi et al.).

3.2.2. All-SiC Capacitive Pressure Sensors

Compared to 3C-SiC capacitive pressure sensors, all-SiC capacitive pressure sensors have better high-temperature stability, corrosion resistance, and mechanical strength because all components are made of SiC. There is no significant issue of mismatch in the CTEs among the different materials. The main types of all-SiC-based capacitive pressure sensors include touch capacitive pressure sensors (TMCPS) and touchless capacitive pressure sensors [346][347]. Both units were composed of two opposing plates serving as the top and bottom electrodes, separated by a sealed chamber. The bottom electrode is stationary and insulated, whereas the top electrode is consisted of a flexible SiC diaphragm[348]. In the touchless operation mode of the capacitive pressure sensor, the sensitive diaphragm is isolated from the substrate to prevent physical contact[349]. The TMCPS allows the diaphragm to make contact with the substrate, which achieves a greater capacitance per unit area[350].

(1) Touchless SiC capacitive pressure sensors

Touchless SiC capacitive pressure sensors have the advantage of avoiding friction and wear, which reduces the risk of contamination because they do not require physical contact between the diaphragm and substrate. Therefore, they are suitable for pressure-measurement applications that require high accuracy, long-term stability, and reliability. Chen et al.[351][352] prepared all-SiC capacitive high-temperature pressure sensors by using a surface micromachining process **(Figure 11.a)**. The sensor encapsulated by high-temperature ceramics was tested in a single-cylinder internal combustion engine, and the measured pressures were as high as 700 psi, capable of operating at 574°C [352]. However, the temperature stability of SiC films prepared using surface micromachining was poor, and there were problems with contaminants or incomplete etching occur during processing, which led to restrictions in sensor performance owing to the horizontal and vertical machining dimensions during the process of surface micromachining of the sensitive diaphragm[353]. In addition, the deposited SiC films generally had low conductivity; therefore, complementary metals were often needed as electrodes to enhance the efficiency and stability of the sensor for transmitting information. For example, Tang et al.[354] prepared capacitive pressure sensors structured as SiC-W-SiC using the bulk micromachining method. Using tungsten(W) as an electrode to increase the electrical conductivity. They were fabricated using a bonding technique to circumvent adhesion issues related to surface micromachining. This resulted in a sensitivity of 0.13 pF/bar, which was double that of a device produced through surface micro-machining [353]. However, the compressive strain of W was still high, which affected the yield of SiC capacitive pressure sensors.

Another study showed that using gold (Au) instead of W as the electrodes improved the yield of the sensors[355]. Combining anodic bonding techniques with the PECVD method of depositing SiC films at low temperatures made the entire fabrication process more CMOS-compatible. The strong connection between the gold base electrode and the silicon carbide layer resulted in the formation of a SiC-Au-SiC structure, and an absolute capacitive pressure sensor made of SiC was created using the anodic bonding method **(Figure 11.b)**[355]. This anodic bonding process overcomes the fabrication limitations of surface micromachining geometry. The increase in the sensing film size significantly improved the pressure sensitivity, although there were apparent losses in the pressure linear operating range of the sensor. Subsequent research has indicated that this issue can be addressed by designing arrays of small sensing membranes. For example, Meng et al· [356] fabricated SiC capacitive pressure sensors with arrayed sensing membranes **(Figure 11.c)**. The sensitive membrane was formed by depositing multiple layers of SiC-Au-SiC structures, and the design of the small-sized arrayed sensing membrane effectively addressed the trade-off between sensitivity and linear response range, resulting in an exceptional sensitivity across a wide range of applied pressures.

The response of capacitive pressure sensors is non-linear in relation to input changes, and there are many parasitic capacitances (stray capacitance effects) that cannot be ignored, both of which affect the sensitivity in the near-linear region. In addition, MEMS capacitive sensors generally measure small variations in capacitance from fF to pF, and the parasitic capacitance is similar to the capacitance of capacitive sensors[357]. Therefore, background noise can significantly disrupt the measurement of the sensor capacitance, which in turn affects the sensitivity characteristics of the sensor. Combining conditioning circuits and sealing shields is an effective way to eliminate parasitic capacitance. For example, Zhang et al. [357] developed a unified conditioning circuit that integrated a tightly sealed shield and an integrated capacitance measurement chip to reduce circuit noise and enhance the overall sensitivity of the sensor. They successfully reduced the test noise and parasitic capacitance and improved the sensor sensitivity. Furthermore, Beker et al.[358] [359] prepared a sensor with concentrically matched differential capacitance output (DCO) using a concentric matched capacitor (CMC) design **(Figure 11.d).** This design is a differential capacitance scheme implemented using concentric rings and circular diaphragms to eliminate common-mode noise. A temperature-controlled pressure chamber was constructed to simulate a low-temperature geothermal well for the experiments **(Figure 11.e)**. The research team also fabricated circular capacitive pressure sensor arrays to achieve higher capacitance output[359]. These sensor arrays were phase-responsive **(Figure 11.f)** and the sensitivities in the 20C and 180C tests were 13.7 fF/kPa and 13.4 fF/kPa, respectively. We must acknowledge that the aforementioned designs increase the process complexity and exhibit a significantly narrower temperature range compared to SiC capacitive pressure sensors.

(2) Touch-mode capacitive pressure sensors

Touch-mode capacitive pressure sensors (TMCPS) are operated by bringing the SiC diaphragm into contact with the substrate when the SiC diaphragm makes contact with the substrate in a near-linear operation. In contrast, the near-linear operation in the non-touch mode yielded a significantly lower sensitivity. Therefore, the effects of parasitic capacitance can be negligible[360]. In addition, these TMCPS significantly extended the dynamic range and immunity to the interference of the measured temperature. Generally, TMCPS are classified as single-sided touch capacitive pressure sensors (STMCPS) and double-sided touch capacitive pressure sensors (DTMCPS).

The traditional STMCPS structure consists of a diaphragm and a substrate serving as the top and bottom poles of the capacitive plate arrangement. The bottom electrode is crossed with an insulating layer and the interlayer is sealed in a capacitive chamber. The diaphragm and substrate are in contact, with the advantages of near-linear output characteristics and large pressure ranges [350]. Hao et al.[361] [362] discovered that etching a stepped cavity at the bottom of a conventional STMCPS **(Figure 11.g)** resulted in a new STMCPS that showed enhanced sensitivity, stability, and linearity. This was a SiC-AlN-SiC three-layered structure with AlN as the capacitive insulating layer. The sensor exhibited excellent measurement accuracy of 800 K and a sensitivity of 3.5 pF/MPa within the pressure range of 0.9-1.4MPa [361], which was 1/3 higher than that of the traditional STMCPS.

For the upper diaphragm of the STMCPS, the existing stepped cavity can be designed to contact both the first- and second-stepped surfaces during operation, forming a DTMCPS[363] [364]. This can further improve the linearity range and sensitivity of STMCPS [365]. For example, Hao et al.[366] [367] designed a DTMCPS with a SiC-AlN-SiC sandwich structure **(Figure 11.h)**. AlN has a crystal structure similar to that of 6H-SiC, which can be used as an electrically insulating material. The DTMCPS is capable of functioning at elevated temperatures of 500°C, exhibiting a nearly linear operational range with minimal temperature drift, negligible parasitic capacitance, large overload protection, and higher sensitivity[362][366]. In addition, the sensor encapsulation could endure temperatures exceeding 500°C **(Figure 11.i)**.

(3) Integrated System

It is of great interest to integrate power supplies, wireless circuits, electronics, and high-temperature sensors into a single high-temperature-resistant system, which is crucial for the development of high-temperature wireless pressure sensors that can effectively address practical challenges in high-temperature and high-pressure environments. For example, Meredith et al.[368] constructed a cohesive system of SiC capacitive pressure sensors and SiC ring oscillator circuits that can be operated at 500°C. These sensors were utilized to modulate the frequency of an electronic oscillator, generating a signal indicating pressure, which was then wirelessly transmitted to a receiver situated in an environment with ambient temperature and atmospheric pressure [368]. However, constructing an oscillator requires a significant number of environmentally responsive and exceedingly precise passive elements, and ensuring long-term system reliability presents a challenging issue. In addition, Cardelletti et al.[369] developed a capacitive pressure sensor system that combined a clap-type oscillator and 6H-SiC MESFET-based electronics **(Figure 11.j)**. It incorporates a SiCN capacitive pressure sensor and SiC MESFET-based electronics into an oscillator circuit that includes resistors, inductors, capacitors, sensors, and transistors [369]. and was successfully underwent vibration, pressure, and temperature tests at 10.6Grms, 300 psi, and 500C, respectively. Furthermore, it demonstrated the precise tracking and monitoring of a flightworthy turbofan engine for up to 17 h, with a pressure sensitivity of 68kHz/psi across a pressure range of 0–350 psi [370]. However, the sensitivity of the system is affected by temperature and decreases significantly at 500°C and temperature compensation is required to enhance the sensitivity of the system[371].

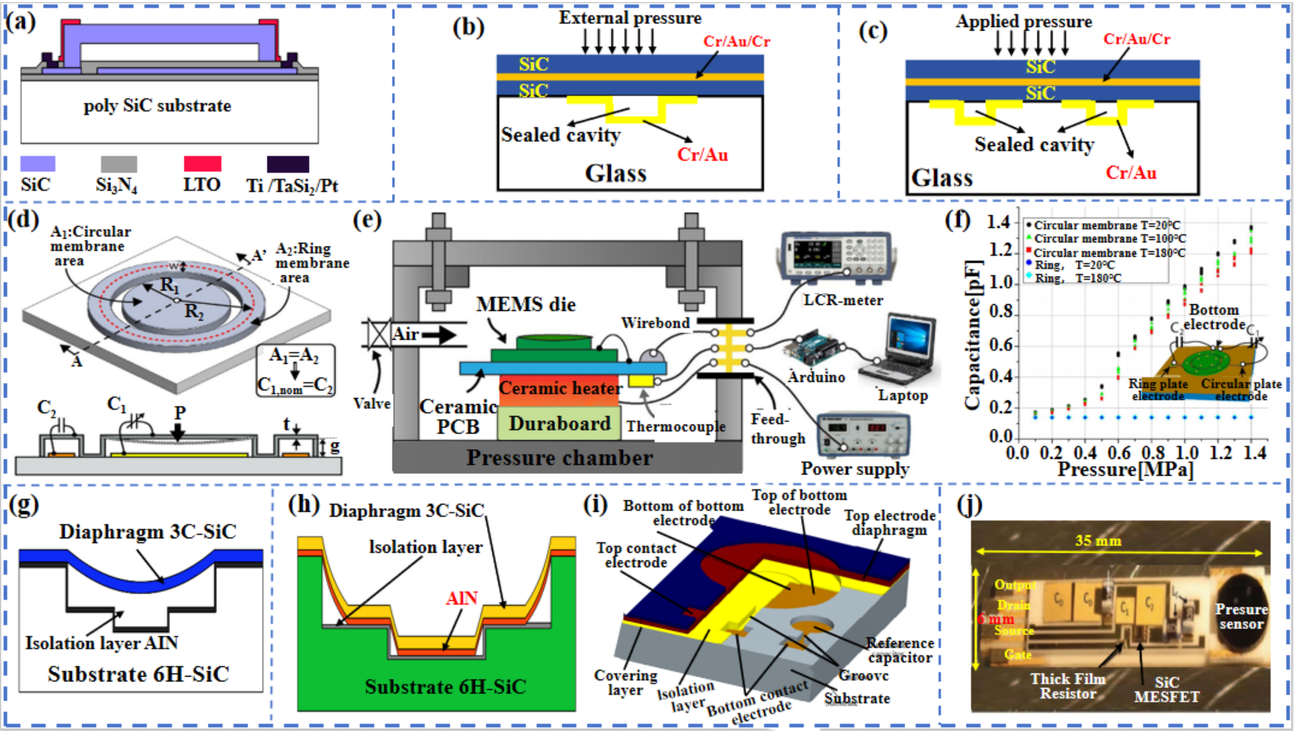


Figure 11 (a) All-SiC capacitive pressure sensors. Reproduced with permission from ref[351]. (Copyright © 2007, IEEE.). (b) SiC/Au/SiC Structure capacitive pressure sensor. Reproduced with permission from ref [355]. (Copyright 2012, IEEE.). (c) SiC/Au/SiC Array Structure Capacitive Pressure Sensor. Reproduced with permission from ref [356]. (Copyright 2013 IEEE.). (d) Cross-sectional view of a CMC pressure sensor; and (e) Temperature controlled pressure chamber test structure; and (f) Pressure/Capacitance response. Reproduced with permission from ref [359]. (Copyright 2018 Elsevier B.V.). (g) Single-touch capacitive pressure sensor (STMCPS) with SiC-AlN-SiC structure. Reproduced with permission from ref [362]. (Copyright 2008, IEEE.). (h) SiC-AlN based Dual Touch Mode Capacitive Pressure Sensor (DTMCPS); and (i) Three-dimensional structure profile of the sensor. Reproduced with permission from ref[367]. (Copyright 2013 Trans Tech Publications, Ltd.). (j) Photograph of the pressure sensor system prior to packaging. Reproduced with permission from ref[369]. (Copyright 2016 IEEE.).

3.3. SiC Fiber Optic Pressure Sensor

The primary structure of the SiC fiber-optic pressure sensor is a SiC diaphragm-based external Fabry-Perot interferometer (EFPI) [372], as shown in **Figure 12 a** [373]. This innovative sensor employs a new technique to measure changes in gas pressure. The F-P cavity is composed of two components, i.e., the end surface of the optical fiber (R_2) and the inner side of the diaphragm(R_1). As light travels through the fiber, part of it is reflected at the air interface (R1) (R_1)[373]. The other part entered the two reflectors positioned on either side of the fiber. The remaining light passes through the air gap between the reflecting surface and optical fiber and is eventually reflected back into the optical fiber (R_2). External pressure changes are converted into changes in the F-P cavity lengths using a sensitive diaphragm. The equation for two-beam interference is expressed as follows [373]:

$$I_R(\lambda) = I_1 + I_2 + 2\sqrt{I_1 I_2}\cos(\Delta\varphi + \varphi_0) \tag{3-8}$$

$$\Delta\varphi = \frac{4\pi n L}{\lambda} \tag{3-9}$$

where I_1 and I_2 represent the reflected intensities of R_1 and R_2, respectively; φ_0 denotes the initial phase; n represents the refractive index of the medium within the cavity; L denotes the actual length of the cavity; and $\Delta\varphi$ describes the phase dissimilarity between two light beams reflected from different interfaces.

Based on the theory of elastic deformation of thin films, the application of an external pressure to a sensitive diaphragm causes deformation that can alter the length of the F-P cavity, subsequently impacting the interference spectra of reflecting surfaces 1 and 2 (R_1 and R_2). The relationship between the cavity length and pressure variation can be expressed by the following equation [374]:

$$\Delta L = \frac{3(1-v^2)R^4}{16Ed^3}\Delta p \tag{3-10}$$

where E and v denote the Young's modulus and Poisson's ratio of the material sensitive to the cavity, Δp represents the variation in ambient pressure, and R and d represent the diameter and depth of the sensitive membrane, respectively.

Compared to piezoresistive and capacitive pressure sensors made of SiC, SiC fiber-optic pressure sensors demonstrate exceptional performance in withstanding extreme temperatures, electromagnetic interference, and electrical inactivity. This makes them superior for monitoring the environmental dynamics[375]. The sensors were tested to operate in the range of room temperature to 1000C, enduring pressures up to 50 bar, and exhibiting equivalent sensitivity at the nanometer scale [376].

3.3.1. Influence factors

The precision of micromachining on the SiC diaphragm and the effectiveness of the sensor cavity hermetic bonding technology are crucial in determining the performance of SiC fiber-optic pressure sensors. Pulliam et al.[377] introduced an innovative all-SiC thin-film pressure sensor utilizing SiC diaphragms deposited on a single-crystal SiC substrate **(Figure 12.b).** At the central opening of the sensing element, they creatively utilized epoxy resin to secure the optical fiber to a glass tube fiber spacer, resulting in a distinctive structure **(Figure 12.c)**. The key feature of this structure is the creation of a fine optical gap between the diaphragm and the end face of the optical fiber, precisely forming the F-P cavity, which enables accurate pressure sensing through the light interference phenomenon caused by pressure variations [377]. This design not only enhances the sensor's response speed and accuracy but also greatly improves its resistance to electromagnetic interference, marking a groundbreaking advancement in high-precision pressure measurement. For example, Jiang et al.[378] fabricated a single-crystal SiC diaphragm fiber optic pressure sensor (EFPI) by integrating the ultrasonic vibratory milling method (UVMG) with the nickel diffusion bonding technique **(Figure 12.d)**, This sensor exhibited exceptional linearity characteristics during tests across a pressure range of 0.1-0.9 MPa. The sensor achieved a resolution of 0.27%FS (percent of full scale) at room temperature, showcasing a high level of precision [378]. Furthermore, it delivered an exceptional performance in extreme temperature settings, such as 600°C, with an impressive sensitivity of 104.42nmMPa, indicating its stability and dependability in high-temperature conditions.

SiC-SiC direct bonding has been demonstrated to be a superior technique for bonding SiC. Because direct bonding technology bonds the chip directly to the substrate, it omits the wires and connecting materials needed in conventional packages, thus enabling more compact and highly integrated package structures [195]. For example, Sheng et al.[379] successfully integrated a fiber-optic pressure sensor into an all-SiC structure through a combination of ultrasonic vibratory milling (HF) and direct bonding techniques. The critical factor in this process is the utilization of HF to efficiently eliminate the oxide interlayer at the SiC bonding interface, thus achieving a high-strength SiC sensing diaphragm bonded to a SiC substrate with exceptional airtightness [207]. The sensor has the ability to precisely measure pressure within the range of 0 to 4 MPa, even at temperatures as high as 600°C, showing outstanding linearity. In addition, Liang et al.[380] developed an all-SiC fiber-optic pressure sensor using hydrophilic direct-bonding technology **(Figure 12.e)**, which demonstrates outstanding performance within the 800kpa pressure range and can reliably operate at extreme temperatures of up to 400°C, significantly broadening its potential application areas and enhancing its reliability. The direct bonding process results in a seamless connection at the interface, rendering conventional bonding interfaces virtually imperceptible, and creating a highly enclosed chamber with superior performance.

3.3.2. Multi-parameter measurements

It is important to note that by employing rational design, SiC fiber optic pressure sensors can not only achieve precise pressure measurements but also accurately detect temperature fluctuations simultaneously. The integration of these two crucial measurement functions not only results in substantial cost and physical space savings but also significantly simplifies the system design and maintenance[381][382]. This offers an efficient, compact, and versatile solution for modern precision-measurement technologies. For example, Riza et al. proposed a pressure measurement mechanism with a hybrid temperature and pressure sensor[383] and prepared a wireless integrated hybrid temperature and pressure sensor **(Figure 12.f)** using a remotely thick single-crystal SiC chip encapsulated in a pressurized adhesive. The designed encapsulated sensor is shown in **Figure 12.g** with a packaged structure **(Figure 12.h)**. The SiC chip serves as a crucial optical interface within the pressure capsule package and plays a vital role in global positioning. Through precise observation of both the local and overall spatial optical characteristics of the free-space laser beam, an efficient and accurate wireless pressure sensing function can be achieved [384]. Temperature sensing ranging from room temperature to 1000 °C has been achieved[385].

However, after integrating the temperature and pressure sensing functions, the overall structure of the hybrid temperature and pressure SiC fiber optic sensors is more complex, and many influencing factors need to be further explored. For example, Chakravarty et al. investigated the variations in the refractive index with pressure and temperature for interferometric optical sensors fabricated from 4H-SiC[386] and 6H-SiC[387], and the study was based on the principle that the refractive indices of SiC optical sensors vary with temperature and pressure. The refractive indices at the wafer-gas interface were determined by analyzing the interferograms, and the corresponding temperature and pressure gradients were calculated. These gradients and reflectance measurement results were used to decouple temperature and pressure. Zhang et al.[388] introduced a cutting-edge system featuring a silicon carbide film as the primary sensitive component, which was combined with a flexible laser reflector to create this advanced optical sensing apparatus. Optical measurements demonstrate the greatest sensitivity in the area outside the center of the SiC film, particularly in large-scale measurement applications, with the optical measurement displaying outstanding linear operational characteristics.

Compared to piezoresistive and capacitive SiC pressure sensors, the ability of SiC fiber-optic pressure sensors to perform multi-parameter measurements stems from their unique optical sensing mechanisms and structural design. These sensors detect external pressure by modulating optical signals (e.g., intensity, wavelength, phase), enabling simultaneous measurement of multiple physical quantities such as pressure, temperature, and strain. Specifically, wavelength modulation leverages the differential responses of optical signals at different wavelengths to distinguish between pressure and temperature. Distributed sensing arranges multiple sensing points along a single optical fiber, enabling spatially resolved multi-parameter measurements. Multi-modal sensing integrates structures like fiber Bragg gratings (FBG) and Fabry-Pérot interferometers (FPI) to concurrently monitor pressure, temperature, vibration, and more. In contrast, piezoresistive and capacitive sensors typically rely on single electrical signals (resistance or capacitance changes), limiting their sensing dimensions. While additional circuitry can extend their functions, it also increases system complexity and cost. Thus, the intrinsic multi-parameter sensing capability of SiC fiber-optic sensors offers a distinct advantage, particularly in harsh environments—such as aerospace, energy, and industrial monitoring—where high temperature, high pressure, and strong electromagnetic interference are present.

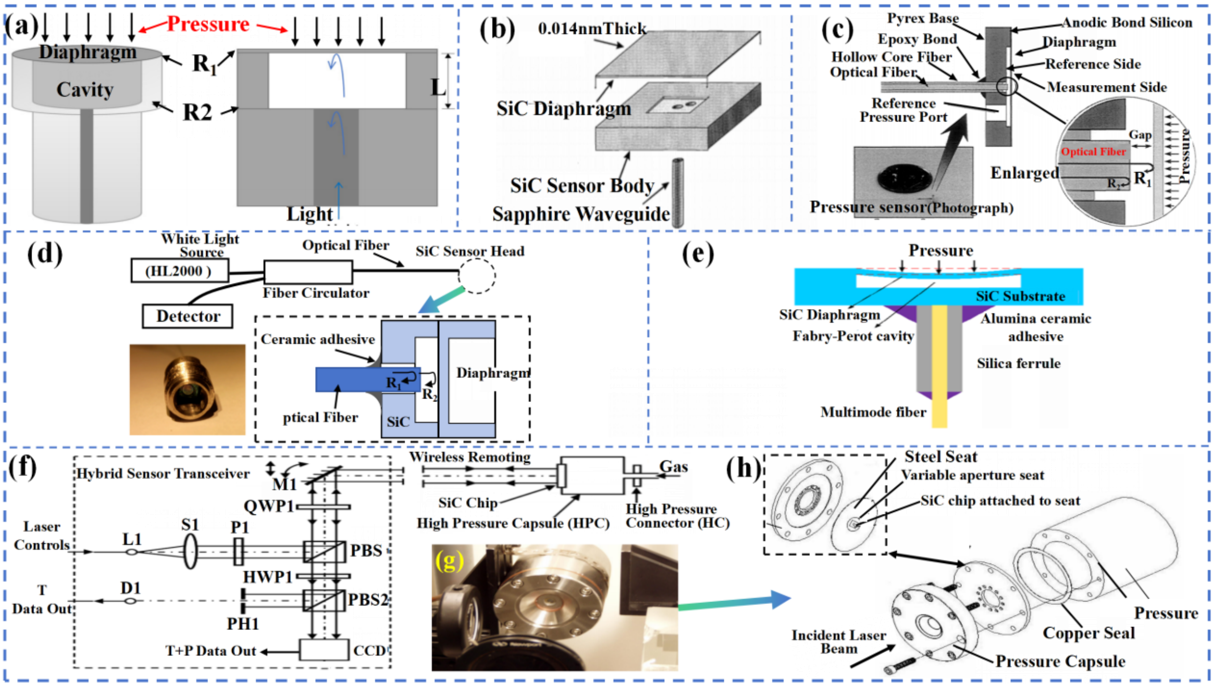


Figure 12 (a) Schematic of MEMS fiber-optic FP pressure sensors. Reproduced with permission from ref [373]. (Copyright 2024 Chen, Ye, et al.). (b) Schematic of an all-SiC fiber-optic pressure sensor; and (c) Assembled SiC fiber-optic (Fabry-Perot cavity) pressure sensor. Reproduced with permission from ref [377]. (Copyright 2000 SPIE.). (d) Schematic of SiC EFPI pressure sensor head and measurement scheme. Bottom left is the encapsulated sensor. Reproduced with permission from ref [378]. (Copyright 2016 Y.G. Jiang, et al.). (e)schematic of the SiC fiber-optic pressure sensor. Reproduced with permission from ref [380]. (Copyright 2021 Ting, et al.). (f)Proposed Wireless Hybrid Optical Sensor for Simultaneous Temperature and Pressure Measurements; and (g)Close-up front-view of the experimental SiC chip mounted in the high pressure capsule; and (h)Experimental design used for seating the SiC chip in the high-pressure capsule. Reproduced with permission from ref [383]. (Copyright 2006 Riza et al.).

4. Conclusions and future prospects

4.1. Conclusions

SiC can be used to prepare piezoresistors with high GF values or sensitive diaphragms owing to its extraordinary properties (e.g., wide bandwidth, high thermal conductivity, high hardness, high breakdown strength, and good chemical stability). Therefore, they can be used to prepare piezoresistors with high GF values and sensitive diaphragms. The SiC pressure sensors can be widely applied in automobiles, aerospace engineering, and deep-well drilling. We thoroughly discuss their respective advantages and macro designs for three main types of SiC-based pressure sensors (i.e., piezoresistive, capacitive, and optical fiber).

(i) Firstly, we highlight the key properties and characteristics of SiC materials and then introduce the synthesis and fabrication/characterization methodologies of SiC materials and films for pressure sensors.

(ii) Subsequently, we discuss key processing techniques (including etching, ohmic contact, and bonding) used to fabricate high-temperature pressure sensors using SiC materials and examine the variables related to these processes that impact the performance parameters of the pressure sensors.

(iii) We further comprehensively summarize the sensing mechanisms (i.e., the piezoresistive effect) of SiC materials. We discuss the piezoresistive properties of several SiC polytypes (including single crystal/polycrystalline 3C-SiC, n/p-type 4H-SiC, n/p-type 6H-SiC, and SiC nanowires). Using strategies such as doping, orientation dependence, and opto-mechanical coupling, the GF values of various SiC polytypes can be enhanced. SiC nanostructures with large piezoresistive effects also show great potential for the preparation of piezoresistive pressure sensors. We have emphasized recent developments in piezoresistive pressure sensors, encompassing SOI-based and all-SiC structures.

(iv) We then focused on detailed discussions of the recent development of SiC capacitive pressure sensors (including both 3C-SiC-based and all-SiC-based designs). Based on the sensitive membrane deformation principle of the small and large deflection theory, SiC capacitive pressure sensors have advantages such as high sensitivity, high accuracy, and temperature invariance. Conventional solution-based processes (e.g., doping, surface micromachining, film bonding, and encapsulation) have been widely applied for the preparation of SiC capacitive pressure sensors. In particular, TMCPS are one to two orders of magnitude more sensitive when operating in the near-linear range than when operating in normal mode, demonstrating significant progress in performance optimization, application expansion, etc.

(v) In addition, we focused on detailed discussions of the recent development of SiC fiber-optic pressure sensors. Such the SiC fiber-optic pressure sensors with hybrid temperature pressure measurement mechanisms have shown excellent characteristics (e.g., resistance to electromagnetic interference, electrical passivity, long-term stability, and resistance to extreme temperatures). These are crucial for targeted applications in high-temperature and high-pressure environments.

(vi) Finally, we highlight the key challenges in this field and propose future opportunities and perspectives in the field of SiC-based high-temperature pressure sensors.

4.2. Key issues identified

Although various advanced technologies have been developed to improve the performance of SiC high-temperature pressure sensors, many problems still exist and need to be solved urgently before they can be practically applied in high-temperature, high-pressure, and strongly corrosive environments to solve real engineering problems. Further exploration and research on sensing mechanisms and structural fabrication processes are required to realize superior sensors with high sensitivity, high stability, and a wide temperature range. In addition, exploring applications in the field of SiC nanostructures requires extensive research on their material and structural properties (e.g., orientation dependence and growth mode of SiC nanowires) to exploit their excellent material properties. Additionally, the anti-interference strength, linearity, and other auxiliary properties of the pressure sensor should be considered. Finally, integrating pressure sensors with detection circuits in miniaturized integrated structures remains a significant challenge. The following sections highlight the key challenges for SiC high-temperature pressure sensors in terms of overall performance improvement and related processing breakthroughs.

4.2.1. Overall performance improvement

Silicon carbide (SiC) high-temperature pressure sensors demonstrate significant potentials in high-temperature, high-pressure, and highly corrosive environments. However, several critical challenges must be addressed before they can be practically applied. Current research is primarily focused on optimizing individual performance metrics, such as sensitivity or temperature range, while improvement of overall performance has not been addressed significantly. To address this, interdisciplinary research should be conducted to develop sensors with a wide temperature range, high sensitivity, high stability, and high selectivity. For example, the B-doped 3C-SiC nanowires excited by photoelectric coupling can be developed to achieve extremely high GF values, suitable for highly sensitive pressure sensors[256][260][261][266]. Additionally, using Ti/TaSi2/Pt as metallized ohmic contacts can effectively suppress structural delamination during high-temperature annealing and reduce piezoresistive temperature drift[187][188]. Combining temperature compensation techniques with hermetic packaging can further enhance sensor stability and accuracy [357]

Secondly, current research is predominantly focused on small stress ranges (from a few MPa to a few hundred MPa), while sensing tests under large stress ranges (GPa) remain insufficiently performed. To address this, the preparation process of SiC-sensitive membranes must be optimized by studying the impact of membrane thickness and surface roughness on stress limits. Improved processing techniques, such as femtosecond laser machining, can enhance the uniformity and flatness of sensitive membranes to meet the requirements of high-stress environments. Furthermore, application of multiple materials (e.g., polycrystalline SiC, B, N-doped materials, and protective materials like AlN, glass, and ceramics) often leads to mismatched coefficients of thermal expansion and stress variations. To mitigate this, novel composite materials and multilayer structures can be developed to optimize the thermomechanical compatibility of materials. Accelerated Stress Testing (AST) protocols[316] can be employed to evaluate the long-term reliability of sensors and establish effective calibration schemes[320]. Additionally, exploring wireless sensing networks and leadless packaging structures can reduce thermal mismatch issues in high-temperature environments.

Lastly, the etching process of SiC materials is complex and time-consuming, and surface roughness adversely affects device performance. To address this, standardized etching processes with high speed and precision, such as improved reactive ion etching (RIE) techniques or the introduction of ultrasonic drilling and femtosecond laser machining, need to be developed. Research on metallized ohmic contact structures can also suppress metal diffusion at high temperatures, improving device stability and reliability. In terms of packaging technology, thermal mismatch and parasitic capacitance of packaging materials in high-temperature environments can easily degrade sensor’s performance. This can be mitigated by designing high-temperature-compatible packaging materials (e.g., ceramics or metal-based composites) and incorporating conditioning circuits and hermetic shielding to eliminate the effects of parasitic capacitance. In addition, recent studies have found that flake silver powder improved silver paste density and resistance to electromigration, significantly improving high-temperature stability and conductivity. This can be used to study the process impact of silicon carbide high-temperature pressure sensor packaging and electrical performance[389].

By optimizing material selection, improving manufacturing processes, and developing advanced packaging technologies, the overall performance of SiC-based high-temperature pressure sensors can be significantly enhanced, promoting their widespread application in high-temperature and high-pressure environments. Future research should be focused on interdisciplinary innovation to unlock the potential of these sensors in aerospace, energy, and industrial monitoring fields.

4.2.2. Process breakthroughs

The advancement of manufacturing techniques for high-temperature SiC-based sensors is still in its infancy and further advancements in this area are crucial. High-temperature silicon carbide (SiC)-based sensors face critical challenges in manufacturing processes, including difficulties in etching, degradation of metallized contact performance, and thermal mismatch in packaging materials.

To address etching challenges, novel etching technologies such as plasma-enhanced chemical vapor deposition (PECVD) or laser-assisted etching can be developed, and parameters of existing reactive ion etching (RIE) processes can be optimized to improve etching rates and surface quality. Additionally, the introduction of femtosecond laser machining or ultrasonic drilling techniques, combined with post-processing methods such as the chemical mechanical polishing (CMP), can significantly reduce surface roughness, mitigating stress concentration and non-uniform membrane thickness issues.

For metallized ohmic contacts, research into new high-temperature-stable contact materials (e.g., $TaSi_2$ and TiN) and their multilayer structures, along with the incorporation of diffusion barrier layers (e.g., AlN or Si_3N_4), can effectively suppress metal diffusion at high temperatures. Furthermore, optimizing annealing process parameters or exploring non-metallic contact materials (e.g., graphene or carbon nanotubes) can enhance the stability of contact performance.

In terms of packaging technology, the development of high-temperature-compatible ceramic-based or metal-based composite materials, combined with leadless packaging structures and wireless sensing networks, can reduce the impact of thermal mismatch and parasitic capacitance. Meanwhile, self-calibrating packaging systems integrated with temperature compensation and signal conditioning circuits can correct signal drift and noise in real-time under high temperatures, significantly improving sensor accuracy and reliability. By comprehensively optimizing etching processes, metallized contacts, and packaging technologies, the performance of high-temperature SiC-based sensors can be significantly enhanced, promoting their widespread application in extreme environments.

4.3. Future aspects

With the progress of industrial technology and the expansion of application scenarios, many applications must be operated at high temperatures, high pressures, or corrosive environments. SiC high-temperature pressure sensors can be operated stably under these extreme conditions to ensure the safety and stability of the equipment, and therefore have important application value in the energy, chemical, aerospace, and other industries.

4.3.1. Material optimization and new structures

Material optimization and novel structural design are among the core directions driving the performance enhancement of SiC (silicon carbide) high-temperature pressure sensors. As a wide-bandgap semiconductor material, SiC exhibits excellent high-temperature stability, chemical inertness, and mechanical strength. However, its application in high-temperature pressure sensors still faces varius challenges, such as material defects, optimization of the piezoresistive effect, and innovation in structural design. Therefore, future research should be focused on the growth and defect control of high-quality SiC materials, optimization of doping processes, and novel structural designs.

Firstly, the quality of SiC materials directly impacts the performance and reliability of sensors. Currently, CVD and epitaxial growth are the primary techniques for preparing SiC thin films. Future research needs to further optimize growth processes to reduce crystal defects (e.g., dislocations and stacking faults), thereby improving material uniformity and electrical properties. For instance, by controlling growth temperature, gas flow, and substrate pretreatment, defect density can be minimized, enhancing sensor sensitivity and long-term stability. Additionally, exploring new substrate materials (e.g., graphene or SiC-on-insulator) may provide new pathways for growing high-quality SiC thin films.

Secondly, doping is a critical method for tuning the electrical properties of SiC. By precisely controlling the concentration and distribution of dopants (e.g., nitrogen, aluminum, and boron), the piezoresistive effect of SiC can be optimized, thereby improving sensor sensitivity. For example, nitrogen doping significantly enhances the conductivity of n-type SiC, while aluminum doping is suitable for p-type SiC. Future research should explore more efficient doping techniques, such as ion implantation combined with high-temperature annealing, to achieve uniform distribution and activation of dopants. Moreover, novel co-doping or multi-element doping strategies may further enhance the piezoresistive properties of SiC.

In terms of novel structural design, traditional SiC pressure sensors predominantly employ planar structures, which exhibit limited performance under high-temperature and high-stress conditions. As a result, researchers are exploring innovative structural designs to improve the mechanical performance and response characteristics of sensors. For instance, the fabrication of SiC nanowires or ultrathin films can significantly increase the specific surface area and sensitivity of sensors. Nanowire structures also possess excellent mechanical strength and thermal shock resistance, making them suitable for extreme environments. Porous SiC structures, with their high specific surface area and low density, can enhance the pressure response speed and sensitivity of sensors while effectively reducing thermal stress and improving temperature stability. Furthermore, the use of MEMS technology to design complex three-dimensional structures (e.g., cantilevers and diaphragm arrays) can achieve higher sensitivity and a wider measurement range. Three-dimensional structures can also amplify the piezoresistive response through stress concentration effects. Lastly, combining SiC with other materials to form heterojunctions or composite materials represents another important direction for enhancing sensor performance. For example, SiC-graphene heterojunctions can significantly improve sensor response speed and sensitivity, while SiC-ceramic composites can enhance thermal stability and corrosion resistance while reducing manufacturing costs. These innovative designs provide new possibilities for the performance improvement and widespread application of SiC high-temperature pressure sensors.

4.3.2. Novel Sensing Mechanisms

Novel sensing mechanisms represent a pivotal direction in advancing the technological innovation of SiC (silicon carbide) high-temperature pressure sensors. Traditional piezoresistive and capacitive sensing mechanisms face limitations in high-temperature and extreme environments, such as temperature drift, reduced sensitivity, and insufficient long-term stability. To overcome these bottlenecks, researchers are exploring new sensing mechanisms based on SiC to achieve higher-performance sensor designs. These innovative mechanisms not only enhance sensor sensitivity, response speed, and environmental adaptability, but also expand their applicability in complex operating conditions. Among these, optical and quantum sensing mechanisms are currently the focus of intensive research.

Optical sensing mechanisms leverage the interaction between light and matter to achieve pressure measurement, offering advantages such as non-contact operation, immunity to electromagnetic interference, and high sensitivity. SiC-based optical pressure sensors can be realized through methods such as the photoelastic effect, photonic crystal structures, and fluorescence sensing. For instance, the refractive index of SiC changes under stress, enabling the design of optical interferometers or fiber-optic sensors that infer pressure values by measuring light intensity variations. This approach demonstrates exceptional stability in high-temperature and corrosive environments. Additionally, the bandgap structure of SiC-based photonic crystals alters under external pressure, modifying transmission or reflection spectra to enable high-precision pressure measurement. Fluorescence sensing, on the other hand, utilizes changes in the fluorescence properties of defect states (e.g., silicon vacancy centers) in SiC under stress, achieving high-sensitivity pressure measurement, particularly suited for high-temperature and radiation environments.

Quantum sensing mechanisms exploit the sensitivity of quantum states to achieve ultra-high-precision measurements, representing a cutting-edge frontier in sensing technology. SiC-based quantum sensing mechanisms primarily include color center sensing, quantum dot sensing, and spin-based quantum sensing. For example, silicon vacancy centers (e.g., VSi defects) in SiC possess stable spin states and optical properties. Under the external pressure, their energy level structures change, enabling nanoscale-resolution pressure measurement through techniques such as optically detected magnetic resonance (ODMR). Quantum dot sensing utilizes the quantum confinement effect in SiC quantum dots, achieving high-sensitivity pressure measurement by analyzing their luminescence spectra or electrical properties. Spin-based quantum sensing leverages the coherence of spin states in SiC, enabling ultra-high-sensitivity pressure measurement even in extreme environments, such as strong magnetic fields or high temperatures. Furthermore, SAW sensing mechanisms also show significant potential. Although SiC itself is not piezoelectric, combining it with piezoelectric materials (e.g., AlN or ZnO) allows the design of high-performance SAW sensors. For instance, depositing piezoelectric thin films on SiC substrates leverages SiC's high thermal conductivity and mechanical strength, along with the acoustic properties of piezoelectric materials, to enable pressure sensing in high-temperature environments. External pressure alters the propagation speed and frequency of surface acoustic waves, and measuring these frequency shifts enables high-precision pressure measurement with rapid responses and strong anti-interference capabilities. These novel sensing mechanisms open new possibilities for enhancing the performance and expanding the applications of SiC high-temperature pressure sensors.

4.3.3. Expanding Application Areas

SiC high-temperature pressure sensors have demonstrated significant application potentials in extreme environments due to their exceptional thermal stability, chemical inertness, and mechanical strength. With advancements in materials science and sensing technologies, their applications could be expanded from traditional aerospace and energy sectors to a broader range of fields, including industrial, automotive, medical, and scientific research. In the aerospace sector, SiC sensors are critical components for high-precision monitoring in aircraft engines, rocket propulsion systems, and supersonic vehicles. They can withstand extreme temperatures and pressures, enabling real-time monitoring of combustion chamber pressure, turbine temperature, and airflow conditions, thereby providing essential data for flight safety and performance optimization. Additionally, in space explorations, SiC sensors are used to monitor pressure changes inside and outside spacecraft, ensuring stable operation in the harsh conditions of space. In the energy sector, SiC sensors are applied across nuclear power, geothermal energy, clean energy, and fossil fuels. In the nuclear reactors, they can endure high temperatures, high pressures, and intense radiation, providing real-time monitoring of internal pressure and temperature to ensure the safe operation of nuclear power plants. In the geothermal energy development, they can monitor the state of high-temperature and high-pressure underground fluids, optimizing the design of geothermal wells. In the hydrogen energy and fuel cell systems, they can track the state of high-temperature and high-pressure hydrogen, improving energy conversion efficiency and safety. In the oil and gas exploration, they can be used for downhole pressure monitoring, optimizing extraction processes and reducing environmental risks.

As the automotive industry shifts toward electrification and intellectualize, the application of SiC sensors in both new energy or renewable energy vehicles and traditional internal combustion engine vehicles is becoming increasingly widespread. In new or renewable energy vehicles, they are used to monitor internal pressure and temperature within battery packs, enhancing battery safety and lifespan, while also playing a crucial role in motor control systems. In traditional internal combustion engine vehicles, they enable real-time monitoring of cylinder pressure and exhaust temperature, optimizing combustion efficiency and reducing emissions to meet stringent environmental standards.

Furthermore, the SiC sensors show great potentials in industrial manufacturing, medical and life sciences, and scientific research. For example, in high-temperature furnace monitoring, gas turbine monitoring, and chemical process control, they can withstand extreme environments and provide high-precision data. In the medical field, they are used for high-temperature sterilization and bioreactor monitoring, ensuring equipment safety and process optimization. In scientific research, they can be served as reliable tools for high-temperature and high-pressure experiments, as well as space and deep-sea exploration in extreme environments. With ongoing technological advancements, the application scope of SiC high-temperature pressure sensors will continue to expand, injecting new momentum into technological innovation and industrial upgrading across various fields.

4.3.4. Integration Concepts

As the demand for performance in applications is increasing, breakthroughs in sensor materials, design, manufacturing, and packaging methods have been achieved. The miniaturization and integration of SiC pressure sensors holds great promise for the future prospects of integration with functionality through front-end sensor readout[390]. The complexity of the fabrication techniques is a major obstacle, especially for smaller sizes and higher integration of sensors. These require high-precision machining and nanoscale process control, which tends to increase the cost of fabrication, complexity of technology development, variability, and uncontrollable factors in the manufacturing process. Integrating multiple functions and modules may increase the complexity and cost of the design, including difficulties in circuit design, signal processing, and layout optimization. There is a need to ensure interoperability between different modules (e.g., self-sensing conceptual design and circuit-integrated signal processing) and overall system reliability.

In addition, in high-temperature or high-pressure environments, integrated SiC pressure sensor systems must overcome compatible thermal management and stability issues. Although the SiC material itself has excellent thermal conductivity and high-temperature resistance, high temperatures may lead to thermal expansion of the internal components of the sensor in systems with multiple integrated material individual modules, which affects its performance and accuracy. High pressures may increase the mechanical stresses between integrated sensor components, which may lead to material fatigue or damage. All these problems can be solved through structural design optimization and material selection. However, there are no good solutions for component compatibility to stabilize integrated pressure sensors for long-term applications in high-temperature and high-pressure environments.

When sensors are used as arrays, the integrated combination design allows for both sensitivity and linear range and makes it possible to explore and experiment over large stress ranges. However, sensors integrated into complex arrays require complex circuits for operation. The tolerance of inter-sensor variability decreased significantly as the number of sensors in the array increased. As the size of the sensors decreases and the arrays become denser, the effects of ambient noise and circuitry components make variable operation quite difficult. Therefore, much more effort is needed to explore a set of compatible, yet stable, efficient, stable, and efficient interactive and integrated pressure sensor platforms.

4.3.5. Nanostructures Applications

As mentioned earlier, SiC nanostructures (including nanowires and nanoribbons) possess significant potential for pressure sensor applications, they also face multiple challenges. For example, achieving the controlled synthesis of SiC nanostructures is one of the key challenges, and during the preparation process, it is necessary to ensure that the size, morphology, and crystalline quality of the nanostructures meet the requirements. Techniques such as CVD and PVD are commonly used, and fine process control and characterization techniques to ensure that nanostructures are consistent in terms of size and morphology are critical to their performance. This is attributed to the large piezoresistive effect and high sensitivity response of SiC nanostructures and therefore needs to be optimized continuously to improve the yield and consistency. It is worth noting that SiC nanowires have GF or even mega-GF values of a higher order of magnitude than bulk or thin film SiC, and thus the corresponding sensors could achieve much higher sensitivity. This can be achieved using the local fabrication of nanowires on separate structures of sensors with a high strain concentration[391], which amplifies the strain induced in the nanoscale sensing element at small strain states through the strain concentration phenomenon. Maximizing the mechanical stimulus applied to the nanowire increased the sensitivity of the sensor. However, none of the current studies have explored the carrier concentration, orientation dependence of SiC nanowires, and the effect of the growth method on the characteristics of SiC nanowire pressure sensors, and there is a great deal of room for development.

In addition, there are challenges in the preparation of SiC nanostructures into sensors in terms of their mechanical strength and stability. Despite the inherently high hardness and corrosion resistance of SiC materials, nanoscale SiC structures can lead to issues of reduced strength and increased brittleness. When designing pressure sensors, it is important to consider the effects of pressure changes on nanostructures to ensure that the sensors do not deform or become damaged during long-term use. In addition, SiC nanostructures need to be accurately integrated into microelectromechanical systems (MEMS), which involve complex micronanofabrication techniques and require precision machining at the nanoscale to ensure that the SiC nanostructures can work stably. For example, the compatibility of SiC nanostructures with MEMS substrate materials (e.g., silicon, glass, and polymers) includes thermal expansion coefficient matching, chemical stability, and mechanical bond strength. The design of the interfacial bonding and buffer layers (e.g. Si_3N_4, SiO_2) can be optimized to enhance the adhesion of SiC to the substrate material and to alleviate stress concentration owing to the difference in the coefficients of thermal expansion. However, the stability and compatibility of the overall integration have not yet been emphasized and studied.

4.3.6. Digital Twins

The application of digital twin technology in the field of SiC high-temperature pressure sensors has great potential. Digital twin technology offers a new approach to sensor design, optimization, and maintenance through a combination of real-time data monitoring and virtual simulation models. First, it helps engineers accurately design sensors and optimize their construction and material selection, thereby improving their stability and accuracy under extreme operating conditions. Second, through real-time monitoring and comparative analysis with the digital twin model, possible malfunctions or performance degradation of the sensors can be detected and diagnosed in a timely manner, which improves the reliability and uptime of the sensors. In addition, the digital twin technology supports the customized design of sensors according to specific application requirements. This is beneficial for meeting the practical needs of various high-temperature and high-pressure force environments, such as petrochemical and aerospace fields.

4.3.7. Artificial Intelligence (AI) applications

AI-based simulation methods play an important role in the analysis and optimization of SiC high-temperature pressure sensors. An AI-driven simulation approach uses big data analysis and machine learning algorithms to analyze the behavior of SiC materials in high-temperature and high-pressure environments. This analytical capability goes beyond the limitations of traditional simulations to more accurately predict the performance, stability, and long-term reliability of the material under different operating conditions. Second, AI simulations can help optimize the structural design of SiC pressure sensors. By simulating different SiC pressure sensor structures and layouts, AI can rapidly evaluate the issues of various designs and suggest the best design solution to meet specific high-temperature and high-pressure force applications. In addition, AI simulations support real-time monitoring and analysis of sensor runtime data, which can be used to identify and predict potential problems in a timely manner by comparison with the simulation model to improve the sensor's operational efficiency and reliability. In addition, AI simulation helps accelerate the development cycle of new materials and technologies, reduce the cost and time of actual testing through virtual testing and optimization of design solutions, and promote the rapid progress and application expansion of SiC high-temperature pressure sensor technology.

Acknowledgements:

This work was supported by the Natural Science Foundation of Hunan Province (2025JJ20044), the Science and Technology Innovation Program of Hunan Province (2023RC3099) and the National Natural Science Foundation of China (No. 22090053). The Authors also thank the Center for international cooperation and disciplinary innovation of Micro-Nano Electronic Devices and Integration Technology.

References

[1] Johnson, R. Wayne, et al. "The changing automotive environment: high-temperature electronics." IEEE transactions on electronics packaging manufacturing 27.3 (2004): 164-176.

[2] Kononchuk, Oleg, and B-Y. Nguyen. Silicon-on-insulator (soi) technology: Manufacture and applications. Elsevier, 2014.

[3] Gastineau, Zane D., and US Air Force. "High-temperature smart actuator development for aircraft turbine engines." Proc. 5th Int. High-Temperature Electronics Conf. 2000.

[4] Javed, Yaser, Mohtashim Mansoor, and Irtiza Ali Shah. "A review of principles of MEMS pressure sensing with its aerospace applications." Sensor Review 39.5 (2019): 652-664.

[5] Reinhardt, Kitt C., and Michael A. Marciniak. "Wide-bandgap power electronics for the More Electric Aircraft." IECEC 96. Proceedings of the 31st Intersociety Energy Conversion Engineering Conference. Vol. 1. IEEE, 1996.

[6]Senesky, Debbie G., et al. "Harsh environment silicon carbide sensors for health and performance monitoring of aerospace systems: A review." IEEE Sensors Journal 9.11 (2009): 1472-1478.

[7] Werner, Matthias Ralf, and Wolfgang R. Fahrner. "Review on materials, microsensors, systems and devices for high-temperature and harsh-environment applications." IEEE Transactions on Industrial Electronics 48.2 (2001): 249-257.

[8] Hunter, Gary W., et al. "An overview of high-temperature electronics and sensor development at NASA Glenn Research Center." J. Turbomach. 125.4 (2003): 658-664.

[9] Carlin, Christopher M., and J. K. Ray. "The requirements for high-temperature electronics in a future high speed civil transport (HSCT)." Proc. 2nd Int. High-Temperature Electronics Conf. 1994.

[10] Neudeck, Philip G., Robert S. Okojie, and Liang-Yu Chen. "High-temperature electronics-a role for wide bandgap semiconductors." Proceedings of the IEEE90.6 (2002): 1065-1076.

[11] Li, Xin, et al. "A Novel Packaging Structure and Process for High Temperature Silicon Piezoresistive Pressure Sensor." Journal of Engineering Science & Technology Review 14.2 (2021): 90-97.

[12] Li, Xin, et al. "High-temperature piezoresistive pressure sensor based on implantation of oxygen into silicon wafer." Sensors and Actuators A: Physical 179 (2012): 277-282.

[13] Sheeparamatti, B. G., and Kirankumar B. Balavalad. "Fabrication and characterization of polysilicon-on-insulator (Poly-SOI) and a-SOI based micro piezoresistive pressure sensor for harsh environment applications." Microsystem Technologies 25.11 (2019): 4119-4133.

[14] Yuan, Weizheng, et al. "A review of silicon micromachined resonant pressure sensor." J. Mech. Eng 49.20 (2013): 2-9.

[15] Li, Sainan, et al. "A novel SOI pressure sensor for high temperature application." Journal of Semiconductors 36.1 (2015): 014014.

[16] Mosser, V., et al. "Piezoresistive pressure sensors based on polycrystalline silicon." Sensors and Actuators A: Physical 28.2 (1991): 113-132.

[17] Kroetz, Gerhard H., Martin H. Eickhoff, and Helmut Moeller. "Silicon compatible materials for harsh environment sensors." Sensors and Actuators A: Physical 74.1-3 (1999): 182-189.

[18] Liu, G. D., et al. "Silicon on insulator pressure sensor based on a thermostable electrode for high temperature applications." Micro & Nano Letters 10.10 (2015): 496-499.

[19] Broden, David Andrew, et al. "Characterization of process pressure sensor." U.S. Patent No. 6,935,156. 30 Aug. 2005.

[20]Corvasce, C., et al. "Measurement of the silicon resistivity at very high temperature with junction isolated van der Pauw structures." Proceedings of the 21st IEEE Instrumentation and Measurement Technology Conference (IEEE Cat. No. 04CH37510). Vol. 1. IEEE, (2004): 133-138.

[21] Reggiani, Susanna, et al. "Electron and hole mobility in silicon at large operating temperatures. I. Bulk mobility." IEEE Transactions on Electron devices 49.3 (2002): 490-499.

[22] Guo, Shuwen, et al. "High temperature smart-cut SOI pressure sensor." Sensors and Actuators A: Physical 154.2 (2009): 255-260.

[23] Ngo, Ha-Duong, et al. "Liquid-free, piezoresistive, SOI-based pressure sensor for high temperature measurements up to 400 C." SENSORS, 2012 IEEE. IEEE, (2012): 1-4.

[24] Ngo, Ha-Duong, et al. "Piezoresistive pressure sensors for applications in harsh environments—A roadmap." Modern Sensing Technologies (2019): 231-251.

[25] Neudeck, Philip G., Robert S. Okojie, and Liang-Yu Chen. "High-temperature electronics-a role for wide bandgap semiconductors." Proceedings of the IEEE 90.6 (2002): 1065-1076.

[26] Fahrner, Wolfgang R., R. Job, and M. Werner. "Sensors and smart electronics in harsh environment applications." Microsystem technologies 7 (2001): 138-144.

[27] Mills, David A., et al. "Development of a sapphire optical pressure sensor for high-temperature applications." Sensors for Extreme Harsh Environments. Vol. 9113. SPIE, 2014.

[28] Wang, Z., et al. "Sapphire Fabry–Perot interferometer for high-temperature pressure sensing." Applied Optics 59.17 (2020): 5189-5196.

[29] Chapin, Caitlin A., et al. "In AlN/GaN high electron mobility micro-pressure sensors for high-temperature environments." Sensors and Actuators A: Physical 263 (2017): 216-223.

[30] Marsi, Noraini, et al. "A Review: Properties of Silicon Carbide Materials in MEMS Application." International Journal of Nanoelectronics & Materials 13 (2020) : 113-128.

[31] Wright, N. G., and A. B. Horsfall. "SiC sensors: a review." Journal of Physics D: Applied Physics 40.20 (2007): 6345.

[32] Werner, Matthias, et al. "High—Temperature sensors based on SiC and diamond technology." Sensors Update 5.1 (1999): 141-190.

[33] Lloyd, A., et al. "Current status of silicon carbide based high-temperature gas sensors." IEEE Transactions on Electron Devices 46.3 (1999): 561-566.

[34] Casady, Jeffrey B., and R. Wayne Johnson. "Status of silicon carbide (SiC) as a wide-bandgap semiconductor for high-temperature applications: A review." Solid-state electronics 39.10 (1996): 1409-1422.

[35] Fujita, Shizuo. "Wide-bandgap semiconductor materials: For their full bloom." Japanese journal of applied physics 54.3 (2015): 030101.

[36] Amano, Hiroshi. "Progress and prospect of the growth of wide-band-gap group III nitrides: Development of the growth method for single-crystal bulk GaN." Japanese Journal of Applied Physics 52.5R (2013): 050001.

[37] Sugiura, Takaya, et al. "High-temperature piezoresistance of silicon carbide and gallium nitride materials." IEEE Journal of the Electron Devices Society 10 (2022): 203-211.

[38] Willander, Magnus, et al. "Silicon carbide and diamond for high temperature device applications." Journal of Materials Science: Materials in Electronics 17 (2006): 1-25.

[39] Neudeck, Philip G., Robert S. Okojie, and Liang-Yu Chen. "High-temperature electronics-a role for wide bandgap semiconductors?" Proceedings of the IEEE 90.6 (2002): 1065-1076.

[40] Spetz, Anita Lloyd, and Susan Savage. "Advances in SiC field effect gas sensors." Silicon Carbide: Recent Major Advances, Springer-Verlag, Berlin (2004): 869-896.

[41] Behnel, N., T. Fuchs, and H. Seidel. "New SiC pressure-sensor-technologies for harsh environments." Mikrosystemtechnik Kongress 2009. Vol. 7. 2009.

[42] Wright, Nick G., Alton B. Horsfall, and Konstantin Vassilevski. "Prospects for SiC electronics and sensors." Materials today 11.1-2 (2008): 16-21.
[43] Wang, Li, et al. "Growth of 3C–SiC on 150-mm Si (100) substrates by alternating supply epitaxy at 1000 C." Thin solid films 519.19 (2011): 6443-6446.
[44] Maboudian, Roya, et al. "Advances in silicon carbide science and technology at the micro-and nanoscales." Journal of Vacuum Science & Technology A 31.5, 050805 (2013).
[45] He, Hualing, et al. "An ultralight self-powered fire alarm e-textile based on conductive aerogel fiber with repeatable temperature monitoring performance used in firefighting clothing." ACS nano 16.2 (2022): 2953-2967.
[46] Mehregany, Mehran, et al. "Silicon carbide MEMS for harsh environments." Proceedings of the IEEE 86.8 (1998): 1594-1609.
[47] Ha, Kyoung-Ho, et al. "Soft capacitive pressure sensors: trends, challenges, and perspectives." ACS nano 16.3 (2022): 3442-3448.
[48] Balavalad, Kirankumar B., and B. G. Sheeparamatti. "A critical review of MEMS capacitive pressure sensors." Sensors & Transducers 187.4 (2015): 120.
[49] Soo, Mun Teng, Kuan Yew Cheong, and Ahmad Fauzi Mohd Noor. "Advances of SiC-based MOS capacitor hydrogen sensors for harsh environment applications." Sensors and Actuators B: Chemical 151.1 (2010): 39-55.
[50] Roriz, Paulo, et al. "Review of fiber-optic pressure sensors for biomedical and biomechanical applications." Journal of biomedical optics 18.5 (2013): 050903-050903.
[51] Liang, Honglin, et al. "Review of optical fiber sensors for temperature, salinity, and pressure sensing and measurement in seawater." Sensors 22.14 (2022): 5363.
[52] Vorathin, E., et al. "Review of high sensitivity fibre-optic pressure sensors for low pressure sensing." Optics & Laser Technology 121 (2020): 105841.
[53] Kumar, S. Santosh, and B. D. Pant. "Design principles and considerations for the 'ideal'silicon piezoresistive pressure sensor: a focused review." Microsystem technologies 20 (2014): 1213-1247.
[54] Phan, Hoang-Phuong, et al. "The piezoresistive effect of SiC for MEMS sensors at high temperatures: A review." Journal of Microelectromechanical systems 24.6 (2015): 1663-1677.
[55] Javed, Yaser, Mohtashim Mansoor, and Irtiza Ali Shah. "A review of principles of MEMS pressure sensing with its aerospace applications." Sensor Review 39.5 (2019): 652-664.
[56] Chen, Yankun, et al. "Materials and sensing mechanisms for high-temperature pressure sensors: a review." IEEE Sensors Journal 23.22 (2023): 26910-26924.
[57] Han, X., Huang, M., Han, et al. "Advances in high-performance MEMS pressure sensors: design, fabrication, and packaging." Microsystems & Nanoengineering 9.1 (2023): 156.
[58] Chaussende, Didier, and Noboru Ohtani. "Silicon carbide." Single crystals of electronic materials (2019): 129-179.
[59] Kimoto, Tsunenobu, and James A. Cooper. "Physical properties of silicon carbide." (2014): 11-38.
[60] Kimoto T.; Cooper, J. A. Fundamentals of Silicon Carbide Technology: Groth, Claracerization, Devices and Aplication; John Wiley & Sons: Hoboken, NJ,USA, 2014.
[61] W.H. Backes, F. C. de Nooij, P. A. Bobbert, W. van Haevingen, Kronig-Penney-like description for band gap variation in SiC polytypes, PhysicaB217 (1996)207-211.
[62] Ma, Shujun, et al. A resonant method for determining the residual stress and elastic modulus of a thin

film. Applied Physics Letters 103.3 (2013).
[63] Cheng, Guangming, et al. "Mechanical properties of silicon carbide nanowires: effect of size-dependent defect density." Nano letters 14.2 (2014): 754-758.
[64] Jiang, Liudi, and Rebecca Cheung. A review of silicon carbide development in MEMS applications. International Journal of Computational Materials Science and Surface Engineering 2.3-4 (2009): 227-242.
[65] Harris, Gary Lynn, ed. Properties of silicon carbide. No. 13. Iet, 1995.
[66] Mehregany, Mehran, et al. "Silicon carbide MEMS for harsh environments." Proceedings of the IEEE 86.8 (1998): 1594-1609.
[67] N.G. Wright, A.B. Horsfall, K. Vassilevski, Prospects for Sic electronics and sensors, Materials Today 11 (2008) 16-21.
[68] Cheung, Rebecca. "Introduction to silicon carbide (SiC) microelectromechanical systems (MEMS)." Silicon carbide microelectromechanical systems for harsh environments. 2006. 1-17.
[69] Shur, Michael, and Sergey L. Rumyantsev. SiC materials and devices. Vol. 1. world scientific, 2006.
[70] Ching, W. Y., et al. "The electronic structure and spectroscopic properties of 3C, 2H, 4H, 6H, 15R and 21R polymorphs of SiC." Materials Science and Engineering: A 422.1-2 (2006): 147-156.
[71] Codreanu, C., et al. "Comparison of 3C–SiC, 6H–SiC and 4H–SiC MESFETs performances." Materials Science in Semiconductor Processing 3.1-2 (2000): 137-142.
[72] Wright, Nick G., Alton B. Horsfall, and Konstantin Vassilevski. "Prospects for SiC electronics and sensors." Materials today 11.1-2 (2008): 16-21.
[73] Mehregany, Mehran, et al. "Silicon carbide MEMS for harsh environments." Proceedings of the IEEE 86.8 (1998): 1594-1609.
[74] Wijesundara, Muthu, and Robert Azevedo. Silicon carbide microsystems for harsh environments. Vol. 22. Springer Science & Business Media, 2011.
[75] Cimalla, V., Joerg Pezoldt, and Oliver Ambacher. "Group III nitride and SiC based MEMS and NEMS: materials properties, technology and applications." Journal of Physics D: Applied Physics 40.20 (2007): S19.
[76] Soltys, L. M., et al. "Synthesis and Properties of Silicon Carbide." Physics and Chemistry of Solid State 24.1 (2023): 5-16.
[77] Feng, Zhe Chuan, ed. SiC power materials: devices and applications. Vol. 73. Springer Science & Business Media, 2004.
[78]Abderrazak, Houyem, and E. S. B. H. Hmida. "Silicon carbide: synthesis and properties." Properties and applications of Silicon Carbide (2011): 361-388.
[79] Augustine, G., V. Balakrishna, and C. D. Brandt. "Growth and characterization of high-purity SiC single crystals." Journal of crystal growth 211.1-4 (2000): 339-342.
[80] Semmelroth, K., N. Schulze, and G. Pensl. "Growth of SiC polytypes by the physical vapour transport technique." Journal of Physics: Condensed Matter 16.17 (2004): S1597.
[81] Straubinger, T. L., et al. "Aluminum p-type doping of silicon carbide crystals using a modified physical vapor transport growth method." Journal of crystal growth 240.1-2 (2002): 117-123.
[82] Wellmann, Peter, et al. "SiC single crystal growth by a modified physical vapor transport technique." Journal of Crystal Growth 275.1-2 (2005): e555-e560.
[83] Barth, Sven, et al. "Synthesis and applications of one-dimensional semiconductors." Progress in

Materials Science 55.6 (2010): 563-627.
[84] Kavecký, Štefan, et al. "Silicon carbide powder synthesis by chemical vapour deposition from silane/acetylene reaction system." Journal of the European Ceramic Society 20.12 (2000): 1939-1946.
[85] Fu, Qian-Gang, et al. "Synthesis of silicon carbide nanowires by CVD without using a metallic catalyst." Materials Chemistry and Physics 100.1 (2006): 108-111.
[86] From Saccharose, Carbide Powders. "A Comparison of Sol-Gel Derived Silicon." Ceramics International 21 (1995): 271-276.
[87] Raman, V. B. O. P., O. P. Bahl, and U. Dhawan. "Synthesis of silicon carbide through the sol-gel process from different precursors." Journal of materials science 30 (1995): 2686-2693.
[88] White, Douglas A., et al. "Preparation of silicon carbide from organosilicon gels: I. Synthesis and characterization of precursor gels." Advanced Ceramic Materials;(USA) 2.1 (1987).), pp: 45-52.
[89]White, Douglas A., Susan M. Oleff, and Joseph R. Fox. "Preparation of silicon carbide from organosilicon gels: II. Gel pyrolysis and SiC characterization." Advanced Ceramic Materials;(USA) 2.1 (1987), 53-59.
[90] Klein, L. C., and G. J. Garvey. "Kinetics of the sol/gel transition." Journal of Non-Crystalline Solids 38 (1980): 45-50.
[91] Brinker, C. J., et al. "Sol-gel transition in simple silicates." Journal of Non-Crystalline Solids 48.1 (1982): 47-64.
[92]Julbe, Anne, et al. "Effect of boric acid addition in colloidal sol-gel derived SiC precursors." Materials research bulletin 25.5 (1990): 601-609.
[93] Wolf, S.; Tauber, R. Silicon Processing for the VLSI Era Volume 1, 2nd ed.; Lattice Press: Sunset Beach, CA, USA, 2000.
[94] Zorman, Christian A., Xiao-An Fu, and Mehran Mehregany. "Deposition techniques for SiC MEMS." Silicon Carbide Microelectromechanical Systems For Harsh Environments (2006): 18-45.
[95] Yang, Nianjun, and Xin Jiang. "Cubic silicon carbide: Growth, properties, and electrochemical applications." Novel Carbon Materials and Composites: Synthesis, Properties and Applications (2019): 1-33.
[96] Wijesundara, Muthu BJ, et al. "SiC materials and processing technology." Silicon carbide microsystems for harsh environments (2011): 33-95.
[97] Pessoa, R. S., et al. "Low-pressure deposition techniques of silicon carbide thin films: An overview." Materials Science Research Journal 7.4 (2013): 329.
[98]Stoffel, A., et al. "LPCVD against PECVD for micromechanical applications." Journal of Micromechanics and Microengineering 6.1 (1996): 1.
[99] Tilli, Markku, et al., eds. Handbook of silicon based MEMS materials and technologies. Elsevier, (2020): 133–213.
[100] Zorman, Christian A., et al. "Epitaxial growth of 3C–SiC films on 4 in. diam (100) silicon wafers by atmospheric pressure chemical vapor deposition." Journal of Applied Physics 78.8 (1995): 5136-5138.
[101] Otani, Mana, et al. "Quartz Crystal Microbalances for Evaluating Gas Motion Differences between Dichlorosilane and Trichlorosilane in Ambient Hydrogen in a Slim Vertical Cold Wall Chemical Vapor Deposition Reactor." Advances in Chemical Engineering and Science 10.3 (2020): 190-200.
[102] Kim, Kang San, and Gwiy Sang Chung. "Characteristics of Porous 3C-SiC Thin Films Formed with Nitrogen Doping Concentrations." Materials Science Forum. Vol. 645. Trans Tech Publications Ltd,

(2010): 645–648, 391–394.

[103] Schillinger, Kai, Stefan Janz, and Stefan Reber. "Atmospheric pressure chemical vapour deposition of 3C-SiC for silicon thin-film solar cells on various substrates." Journal of nanoscience and nanotechnology 11.9 (2011): 8108-8113.

[104] Papasouliotis, George D., and Stratis V. Sotirchos. "Experimental study of atmospheric pressure chemical vapor deposition of silicon carbide from methyltrichlorosilane." Journal of Materials Research 14 (1999): 3397-3409.

[105] Funato, Yuichi, et al. "Fundamental evaluation of gas-phase elementary reaction models for silicon carbide chemical vapor deposition." ECS Journal of Solid State Science and Technology 6.7 (2017): P399.

[106] Mehregany, M., et al. "Silicon carbide for microelectromechanical systems." International materials reviews 45.3 (2000): 85-108.

[107] Lien, Wei-Cheng, et al. "Growth of epitaxial 3C-SiC films on Si (100) via low temperature SiC buffer layer." Crystal growth & design 10.1 (2010): 36-39.

[108] Liu, Fang, et al. "Residual stress characterization of polycrystalline 3C-SiC films on Si (100) deposited from methylsilane." Journal of Applied Physics 106.1 (2009). 013505.

[109] Liu, C. W., and J. C. Sturm. "Low temperature chemical vapor deposition growth of β-SiC on (100) Si using methylsilane and device characteristics." Journal of applied physics 82.9 (1997): 4558-4565.

[110] Arango-Ospina, Marcela, et al. "Silicon oxycarbide based materials for biomedical applications." Applied Materials Today 18 (2020): 100482.

[111] Fraga, Mariana, and Rodrigo Pessoa. "Progresses in Synthesis and Application of SiC Films: From CVD to ALD and from MEMS to NEMS." Micromachines 11.9 (2020): 799.

[112]Fraga, Mariana Amorim, et al. "Recent developments on silicon carbide thin films for piezoresistive sensors applications." Silicon Carbide—Materials, Processing and Applications in Electronic Devices; Mukherjee, M., Ed (2011): 369-388.

[113]Martinu, Ludvik, and Daniel Poitras. "Plasma deposition of optical films and coatings: A review." Journal of Vacuum Science & Technology A: Vacuum, Surfaces, and Films 18.6 (2000): 2619-2645.

[114] Pelegrini, Marcus V., Gustavo P. Rehder, and Inés Pereyra. "a-SiC: H films deposited by PECVD for MEMS applications." physica status solidi c 7.3-4 (2010): 786-789.

[115] KlM, Chang Kyo. "Deposition of SiC thin films by PECVD." Journal of the Korean Physical Society 34 (1999): 557.

[116] Sun, Guo Sheng, et al. "Heavily doped polycrystalline 3C-SiC growth on SiO2/Si (100) substrates for resonator applications." Materials science forum. Vol. 556. Trans Tech Publications Ltd, 2007. 556–557,179–182.

[117] Nagasawa, H., and Y-I. Yamaguchi. "Atomic level epitaxy of 3C-SiC by low pressure vapour deposition with alternating gas supply." Thin Solid Films 225.1-2 (1993): 230-234.

[118] Nagasawa, Hiroyuki, and Yoh-ichi Yamaguchi. "Mechanisms of SiC growth by alternate supply of SiH_2Cl_2 and C_2H_2." Applied surface science 82 (1994): 405-409.

[119] Hara, Shiro, et al. "Self-limiting growth on the β-SiC (001) surface." Surface science 273.3 (1992): 437-441.

[120] Johnson, Richard W., Adam Hultqvist, and Stacey F. Bent. "A brief review of atomic layer deposition:

from fundamentals to applications." Materials today 17.5 (2014): 236-246.
[121] Sumakeris, J. J., et al. "Layer-by-layer growth of SiC at low temperatures." Thin Solid Films 225.1-2 (1993): 219-224.
[122] Sadayuki, Eiichi Sadayuki Eiichi, Shigeru Imai Shigeru Imai, and Masakiyo Matsumura Masakiyo Matsumura. "Sub-atomic layer growth of SiC at low temperatures." Japanese journal of applied physics 34.11R (1995): 6166.
[123] Fuyuki, Takashi, Tatsuo Yoshinobu, and Hiroyuki Matsunami. "Atomic layer epitaxy controlled by surface superstructures in SiC." Thin Solid Films 225.1-2 (1993): 225-229.
[124] Fuyuki, Takashi, et al. "Atomic layer epitaxy of cubic SiC by gas source MBE using surface superstructure." Journal of Crystal Growth 95.1-4 (1989): 461-463.
[125] Hara, Shiro, et al. "Microscopic mechanisms of accurate layer-by-layer growth of β-SiC." Thin Solid Films 225.1-2 (1993): 240-243.
[126] Lei, Y. M., et al. "Compositional and structural studies of DC magnetron sputtered SiC films on Si (111)." Thin Solid Films 365.1 (2000): 53-57.
[127] Jeong, Taehee, et al. "Thermal characterization of SiC amorphous thin films." International Journal of Thermophysics 33 (2012): 1000-1012.
[128] Tang, Huidong, et al. "Surface morphology of α-SIC coatings deposited by RF magnetron sputtering." Surface and Coatings Technology 197.2-3 (2005): 161-167.
[129] Rajab, S. M., et al. "Effect of the thermal annealing on the electrical and physical properties of SiC thin films produced by RF magnetron sputtering." Thin Solid Films 515.1 (2006): 170-175.
[130] Mattox, Donald M. "Physical sputtering and sputter deposition (sputtering)." Handbook of physical vapor deposition (PVD) processing (2010): 237-286.
[131] Windischmann, Henry. "Intrinsic stress in sputter-deposited thin films." Critical Reviews in Solid State and Material Sciences 17.6 (1992): 547-596.
[132] Beheim, Glenn M., and Laura J. Evans. "Deep reactive ion etching for bulk micromachining of silicon carbide." MEMS. CRC Press, 2005.
[133] Lévy-Clément, Claude, Abdelghani Lagoubi, and Micha Tomkiewicz. "Morphology of Porous n-Type Silicon Obtained by Photoelectrochemical Etching: I. Correlations with Material and Etching Parameters." Journal of the Electrochemical Society 141.4 (1994): 958.
[134] Carrabba, Michael M., et al. "Photoelectrochemical fabrication of gratings in SiC." Presented at the Fall Meeting of the Electrochemical Society. 1989.
[135] Shor, Joseph S., and Anthony D. Kurtz. "Photoelectrochemical etching of 6 H-SiC." Journal of the Electrochemical Society 141.3 (1994): 778.
[136] Shor, J. S., et al. "Dopant-selective etch stops in 6H and 3C SiC." Journal of applied physics 81.3 (1997): 1546-1551.
[137] Shor, Joseph S., Leala Bemis, and Anthony D. Kurtz. "Characterization of monolithic n-type 6H-SiC piezoresistive sensing elements." IEEE Transactions on Electron Devices 41.5 (1994): 661-665.
[138] Ned A A, Okojie R S, Kurtz A D. "6H-SiC pressure sensor operation at 600℃." IEEE, 1998: 257-260.
[139] Gleria, M., and R. Memming. "Charge transfer processes at large band gap semiconductor electrodes: reactions at SiC-electrodes." Journal of Electroanalytical Chemistry and Interfacial Electrochemistry 65.1 (1975): 163-175.
[140] Pan, W-S., and A. J. Steckl. "Reactive ion etching of SiC thin films by mixtures of fluorinated gases

and oxygen." Journal of the Electrochemical Society 137.1 (1990): 212.
[141] Yih, P. H., V. Saxena, and A. J. Steckl. "A review of SiC reactive ion etching in fluorinated plasmas." physica status solidi (b) 202.1 (1997): 605-642.
[142] Szczęsny, Artur, et al. "Reactive ion etching of novel materials—GaN and SiC." Vacuum 70.2-3 (2003): 249-254.
[143] Reichert, W., et al. "Fabrication of smooth β-SiC surfaces by reactive ion etching using a graphite electrode." Materials Science and Engineering: B 46.1-3 (1997): 190-194.
[144] Jiang, Liudi, et al. "Fabrication of SiC microelectromechanical systems using one-step dry etching." Journal of Vacuum Science & Technology B: Microelectronics and Nanometer Structures Processing, Measurement, and Phenomena 21.6 (2003): 2998-3001.
[145] Tanaka, S., et al. "Deep reactive ion etching of silicon carbide." Journal of Vacuum Science & Technology B: Microelectronics and Nanometer Structures Processing, Measurement, and Phenomena 19.6 (2001): 2173-2176.
[146] Chabert, Pascal, et al. "High rate etching of 4H–SiC using a SF 6/O 2 helicon plasma." Applied Physics Letters 76.16 (2000): 2310-2312.
[147] Cho, H., et al. "Ultradeep, low-damage dry etching of SiC." Applied Physics Letters 76.6 (2000): 739-741.
[148] Okojie, Robert S., Carl W. Chang, and Laura J. Evans. "Reducing DRIE-induced trench effects in SiC pressure sensors using FEA prediction." Journal of microelectromechanical systems 20.5 (2011): 1174-1183.
[149] Yih, P. H., V. Saxena, and A. J. Steckl. "A review of SiC reactive ion etching in fluorinated plasmas." physica status solidi (b) 202.1 (1997): 605-642.
[150] Wang, Lukang, et al. "Femtosecond laser micromachining in combination with ICP etching for 4H–SiC pressure sensor membranes." Ceramics International 47.5 (2021): 6397-6408.
[151] Zhao, You, et al. "Femtosecond laser processing assisted SiC high-temperature pressure sensor fabrication and performance test." Micromachines 14.3 (2023): 587.
[152] Li, Jian, et al. "Ultrasonic vibration mill-grinding of single-crystal silicon carbide for pressure sensor diaphragms." Ceramics International 44.3 (2018): 3107-3112.
[153] Dong, Yuanyuan, et al. "Femtosecond-pulsed laser micromachining of a 4H–SiC wafer for MEMS pressure sensor diaphragms and via holes." Journal of Micromechanics and Microengineering 18.3 (2008): 035022.
[154] Simsek, Emrah, et al. "High-pressure deflection behavior of laser micromachined bulk 6H-SiC MEMS sensor diaphragms." Sensors and Actuators A: Physical 162.1 (2010): 29-35.
[155] Song, Juan, et al. "Surface birefringence of self-assembly periodic nanostructures induced on 6H-SiC surface by femtosecond laser." Applied Surface Science 363 (2016): 664-669.
[156] Ahn, Minhyung, et al. "Low damage electrical modification of 4H-SiC via ultrafast laser irradiation." Journal of Applied Physics 123.14 (2018): 145106.
[157] Wang, Lukang, et al. "Experimental investigation on ablation of 4H-SiC by infrared femtosecond laser." Micromachines 13.8 (2022): 1291.
[158] Wang, Lukang, et al. "Development of laser-micromachined 4H-SiC MEMS piezoresistive pressure sensors for corrosive environments." IEEE Transactions on Electron Devices 69.4 (2022): 2009-2014.
[159] Wu, Chen, et al. "Crystal cleavage, periodic nanostructure and surface modification of SiC ablated by

femtosecond laser in different media." Surface and Coatings Technology 424 (2021): 127652.
[160] Iwatani, Naoki, Hong Duc Doan, and Kazuyoshi Fushinobu. "Optimization of near-infrared laser drilling of silicon carbide under water." International Journal of Heat and Mass Transfer 71 (2014): 515-520.
[161] Zhang, J., et al. "Precise microfabrication of wide band gap semiconductors (SiC and GaN) by VUV–UV multiwavelength laser ablation." Applied surface science 127 (1998): 793-799.
[162] Zoppel, Sandra, et al. "Laser micro machining of 3C–SiC single crystals." Microelectronic Engineering 83.4-9 (2006): 1400-1402.
[163] Zhao, You, Yu-Long Zhao, and Lu-Kang Wang. "Application of femtosecond laser micromachining in silicon carbide deep etching for fabricating sensitive diaphragm of high temperature pressure sensor." Sensors and Actuators A: Physical 309 (2020): 112017.
[164] Li, Xinxin, et al. "Machine learning methods for prediction analyses of 4H–SiC microfabrication via femtosecond laser processing." Journal of Materials Research and Technology 18 (2022): 2152-2165.
[165] Zhang, Ru, et al. "Micromachining of 4H-SiC using femtosecond laser." Ceramics International 44.15 (2018): 17775-17783.
[166] Wang, Chengwu, et al. "SEM, AFM and TEM studies for repeated irradiation effect of femtosecond laser on 4H-SiC surface morphology at near threshold fluence." ECS Journal of Solid State Science and Technology 7.2 (2018): P29.
[167] Zehetner, J., et al. "Laser ablation for membrane processing of AlGaN/GaN-and micro structured ferroelectric thin film MEMS and SiC pressure sensors for extreme conditions." Smart Sensors, Actuators, and MEMS VII; and Cyber Physical Systems. Vol. 9517. SPIE, 2015.
[168] Zehetner, Johann, et al. "Manufacturing of membranes by laser ablation in SiC, sapphire, glass and ceramic for GaN/ferroelectric thin film MEMS and pressure sensors." Microsystem Technologies 22 (2016): 1883-1892.
[169] Marinova, Ts, et al. "Nickel based ohmic contacts on SiC." Materials Science and Engineering: B 46.1-3 (1997): 223-226.
[170] Okojie, Robert Sylvester. "Characterization and fabrication of a (6H)-SiC as a piezoresistive pressure sensor for high temperature applications." (1996). Dissertations. 997..
[171] Han, Sang Youn, et al. "Ohmic contact formation mechanism of Ni on n-type 4H–SiC." Applied Physics Letters 79.12 (2001): 1816-1818.
[172] Roccaforte, F., et al. "Improvement of high temperature stability of nickel contacts on n-type 6H–SiC." Applied surface science 184.1-4 (2001): 295-298.
[173] Macháč, Petr, Bohumil Barda, and Jaroslav Maixner. "Structural characterization of nickel–titanium film on silicon carbide." Applied Surface Science 254.6 (2008): 1691-1693.
[174] Lu, Jinling, Guang Xi, and Datong Qi. "Blade optimization of mixed-flow pump using inverse design method and neural network." an Jiaotong Daxue Xuebao(Journal of an Jiaotong University)(China) 38.3 (2004): 308-312.
[175] Wang, Zhongtao, Wei Liu, and Chunqing Wang. "Recent progress in ohmic contacts to silicon carbide for high-temperature applications." Journal of Electronic Materials 45 (2016): 267-284.
[176] Zhao, Lihuan, et al. "Manufacturability and reliability optimization for metallization of SiC piezoresistive pressure sensors." Journal of Materials Science: Materials in Electronics 32.13 (2021): 17637-17644.

[177] Li, Yongwei, et al. "Study on the stability of the electrical connection of high-temperature pressure sensor based on the piezoresistive effect of p-type SiC."Micromachines12.2 (2021): 216.
[178] Han, Sang Youn, et al. "Ohmic contact formation mechanism of Ni on n-type 4H–SiC." Applied Physics Letters 79.12 (2001): 1816-1818.
[179] Tanimoto, Satoshi, et al. "Toward a better understanding of Ni-based ohmic contacts on SiC." Materials Science Forum. Vol. 679. Trans Tech Publications Ltd, 2011.
[180] Cole, M. W., P. C. Joshi, and M. Ervin. "Fabrication and characterization of pulse laser deposited Ni 2 Si Ohmic contacts on n-SiC for high power and high temperature device applications." Journal of Applied Physics 89.8 (2001): 4413-4416.
[181] Kuchuk, A. V., et al. "Ni-based ohmic contacts to n-type 4H-SiC: The formation mechanism and thermal stability." Advances in Condensed Matter Physics 2016 (2016).
[182] Leech, Patrick W., et al. "Ohmic contacts to n-type 3C-SiC using Cr/Ni/Au and ni/cr/au metallizations." Microelectronic Engineering 215 (2019): 111016.
[183] Lu, Weijie, et al. "Catalytic graphitization and Ohmic contact formation on 4H–SiC." Journal of Applied Physics 93.9 (2003): 5397-5403.
[184] Luckowski, E. D., et al. "Improved ohmic contact to n-type 4H and 6H-SiC using nichrome." Journal of electronic materials 27 (1998): 330-334.
[185] S. Jin, X.A. Fu, M. Mehregany, Ohmic contacts on n-type polycrystalline silicon carbide with Ti/TaSi2/Pt, Solid-State Sensors, Actuators and Microsystems Conference, (2009) 1083-1086.
[186] Zeller, M. V., et al. "AES studies of the M/SiC interface with metal carbide formers." MRS Online Proceedings Library (OPL) 97 (1987): 283.
[187] Siad, M., M. Abdesslam, and A. C. Chami. "Role of carbon in the formation of ohmic contact in Ni/4HSiC and Ni/Ti/4HSiC." Applied Surface Science 258.18 (2012): 6819-6822.
[188] Sheng, Srihari Rajgopal, and Mehran Mehregany. "Characterization of poly-sic pressure sensors for high temperature and high pressure applications." Materials Science Forum. Vol. 717. Trans Tech Publications Ltd, 2012.
[189] He, Y, et al. "Ohmic contacts simultaneously formed on n-type and p-type 4H-SiC at low temperature." Journal of Alloys and Compounds 805 (2019): 999-1003.
[190] Gösele, U., and Q-Y. Tong. "Semiconductor wafer bonding." Annual review of materials science 28.1 (1998): 215-241.
[191] Akatsu, T., et al. "Wafer bonding of different III–V compound semiconductors by atomic hydrogen surface cleaning." Journal of Applied Physics 90.8 (2001): 3856-3862.
[192] Gösele, U., et al. "Wafer bonding for microsystems technologies." Sensors and Actuators A: Physical 74.1-3 (1999): 161-168.
[193] Christiansen, Silke H., Rajendra Singh, and Ulrich Gosele. "Wafer direct bonding: From advanced substrate engineering to future applications in micro/nanoelectronics." Proceedings of the IEEE 94.12 (2006): 2060-2106.
[194] Joly, J. P., et al. "New SiC on insulator wafers based on the smart-cut® approach and their potential a pplications." Progress in SOI Structures and Devices Operating at Extreme Conditions. Dordrecht: Springer Netherlands, 2002. 31-38.
[195] T. Liang,x. Wang,P.Jia,W. Zhang,C. Xue,J."Fabrication of bonded SiC structure with cavity based on direct bonding process for MEMS device applications." Microsystem Technologies 23 (2017): 225-229.

[196] Chung, Gwiy-Sang, and Roya Maboudian. "Bonding characteristics of 3C-SiC wafers with hydrofluoric acid for high-temperature MEMS applications." Sensors and Actuators A: Physical 119.2 (2005): 599-604.

[197] Van Veggel, A. A., et al. "Hydroxide catalysis bonding of silicon carbide." Journal of the European Ceramic Society 28.1 (2008): 303-310.

[198] Zhang, D., Y. Zhao, and Y. Jiang. "SiC-SiC Bonding Methods for Ultra High-Temperature Pressure Sensors. Nanotechnol." Precis. Eng 12 (2014): 258-262.

[199] Preston, Alix, and Guido Mueller. "Bonding SiC to SiC using a sodium silicate solution." International Journal of Applied Ceramic Technology 9.4 (2012): 764-771.

[200] Gottselig, Bernd, Erno Gyarmati, and Aristides Naoumidis. "Method and components for bonding a silicon carbide molded part to another such part or to a metallic part." U.S. Patent No. 4,961,529. 9 Oct. 1990.

[201] Ramm, Peter, James Jian-Qiang Lu, and Maaike MV Taklo, eds. Handbook of wafer bonding. John Wiley & Sons, 2012.

[202] Tong, Q-Y., et al. "Silicon carbide wafer bonding." Journal of The Electrochemical Society 142.1 (1995): 232.

[203] Grekhov, I. V., et al. "Direct bonding of silicon carbide wafers with a regular relief at the interface." Technical Physics Letters 32 (2006): 453-455.

[204] Grekhov, I. V., et al. "Direct bonding of silicon wafers with the concurrent formation of diffusion layers." Technical Physics 46 (2001): 690-695.

[205] Yushin, G. N., A. V. Kvit, and Z. Sitar. "Transmission electron microscopy studies of the bonded SiC-SiC interface." Journal of materials science 40.16 (2005): 4369-4371.

[206] Yushin, G. N., and Z. Sitar. "Influence of relative wafer rotation on the electrical properties of the bonded SiC/SiC interface." Applied physics letters 84.20 (2004): 3993-3995.

[207] Li, Jian, et al. "Direct bonding of silicon carbide with hydrofluoric acid treatment for high-temperature pressure sensors." Ceramics International 46.3 (2020): 3944-3948.

[208] Takagi, Hideki, Ryutaro Maeda, and Tadatomo Suga. "Wafer-scale spontaneous bonding of silicon wafers by argon-beam surface activation at room temperature." Sensors and Actuators A: Physical 105.1 (2003): 98-102.

[209] F.Mu, K. Iguchi,H. Nakazawa, et al. "A comparison study: Direct wafer bonding of SiC–SiC by standard surface-activated bonding and modified surface-activated bonding with Si-containing Ar ion beam." Applied Physics Express 9.8 (2016): 081302.

[210] Zuo-yun, Zhu, et al. "High temperature pressure sensor based on SIC films." Journal of Transducer Technology 20.2 (2001): 1.

[211] Barlian, A. Alvin, et al. "Semiconductor piezoresistance for microsystems." Proceedings of the IEEE 97.3 (2009): 513-552.

[212] Wejrzanowski, Tomasz, et al. "Design of SiC-doped piezoresistive pressure sensor for high-temperature applications." Sensors 21.18 (2021): 6066.

[213] Zhang, Jinbo, et al. "Surface acoustic wave platform integrated with ultraviolet activated rGO-SnS2 nanocomposites to achieve ppb-level dimethyl methylphosphonate detection at room-temperature." Talanta 282 (2025): 127063.

[214] Shor, Joseph S., David Goldstein, and Anthony D. Kurtz. "Characterization of n-type beta-SiC as a

piezoresistor." IEEE transactions on electron devices40.6 (1993): 1093-1099.
[215] Yasui, Kanji, et al. "Si-COI structure fabricated by catalytic chemical vapor deposition." Thin Solid Films 516.5 (2008): 644-647.
[216] Phan, Hoang-Phuong, et al. "Piezoresistive effect in p-type 3C-SiC at high temperatures characterized using Joule heating." Scientific reports 6.1 (2016): 28499.
[217] Phan, Hoang-Phuong, et al. "Piezoresistive effect of p-type single crystalline 3C-SiC thin film." IEEE Electron Device Letters 35.3 (2014): 399-401.
[218] Phan, Hoang-Phuong, et al. "Fundamental piezoresistive coefficients of p-type single crystalline 3C-SiC." Applied Physics Letters 104.11 (2014).
[219] Gupta, Aparna, and Chacko Jacob. "Selective epitaxy and lateral overgrowth of 3C-SiC on Si–A review." Progress in crystal growth and characterization of materials 51.1-3 (2005): 43-69.
[220] Ferro, Gabriel. "3C-SiC heteroepitaxial growth on silicon: the quest for holy grail." Critical Reviews in Solid State and Materials Sciences 40.1 (2015): 56-76.
[221] Ziermann, R., et al. "High temperature piezoresistive β-SiC-on-SOI pressure sensor with on chip SiC thermistor." Materials Science and Engineering: B 61 (1999): 576-578.
[222] Reichert, W., E. Obermeier, and J. Stoemenos. "β-SiC films on SOI substrates for high temperature applications." Diamond and Related Materials 6.10 (1997): 1448-1450.
[223] Reichert, W., et al. "Investigation of the effects of high temperature implantation and post implantation annealing on the electrical behavior of nitrogen implanted β-SiC films." Diamond and Related Materials 6.10 (1997): 1445-1447.
[224] Huang, X., & Zhang, X. "Investigating the advanced characteristics of SiC based piezoresistive pressure sensors." Materials Today Communications 25 (2020): 101493.
[225] Nguyen, Tuan-Khoa, et al. "Isotropic piezoresistance of p-type 4H-SiC in (0001) plane." Applied Physics Letters 113.1 (2018).
[226] Nguyen T K, Phan H P, Dinh T, et al. Highly sensitive 4H-SiC pressure sensor at cryogenic and elevated temperatures[J]. Materials & Design, 2018, 156: 441-445.
[227] Nguyen, Tuan-Khoa, et al. "High-temperature tolerance of the piezoresistive effect in p-4H-SiC for harsh environment sensing." Journal of Materials Chemistry C 6.32 (2018): 8613-8617.
[228] Akiyama, T., D. Briand, and N. F. De Rooij. "Design-dependent gauge factors of highly doped n-type 4H-SiC piezoresistors."Journal of Micromechanics and Microengineering 22.8 (2012): 085034.
[229] Shor, Joseph S., Leala Bemis, and Anthony D. Kurtz. "Characterization of monolithic n-type 6H-SiC piezoresistive sensing elements." IEEE Transactions on electron devices 41.5 (1994): 661-665.
[230] Okojie, Robert S., et al. "Characterization of highly doped n-and p-type 6H-SiC piezoresistors." IEEE transactions on electron devices 45.4 (1998): 785-790.
[231] Okojie, Robert S. "Stable 600 C silicon carbide MEMS pressure transducers." Sensors and Systems for Space Applications. Vol. 6555. SPIE, (2007): 230-240.
[232] Nguyen, Tuan-Khoa, et al. "Experimental investigation of piezoresistive effect in p-type 4H–SiC." IEEE Electron Device Letters 38.7 (2017): 955-958.
[233] Li, Yongwei, et al. "Quantitative analysis of piezoresistive characteristic based on a P-ty1pe 4H-SiC epitaxial layer."Micromachines10.10 (2019): 629.
[234] Meng Meng, et al." Nonlinear piezoresistive effect of 4H–SiC for applications of high temperature pressure sensors." Journal of Materials Science prepublish (2024):1-15.

[235] Wei, Guang, et al. "Temperature-dependent field emission properties of 3 C-SiC nanoneedles." The Journal of Physical Chemistry C 115.26 (2011): 13063-13068.

[236] Nakamura, Koichi, Toshiyuki Toriyama, and Susumu Sugiyama. "First-principles simulation on piezoresistivity in alpha and beta silicon carbide nanosheets." Japanese Journal of Applied Physics 50.6S (2011): 06GE05.

[237] Zhang, Huan, et al. "Synthesis and characterization of crystalline silicon carbide nanoribbons." Nanoscale research letters 5 (2010): 1264-1271.

[238] Naderi, N., and M. R. Hashim. "Nanocrystalline SiC sputtered on porous silicon substrate after annealing." Materials letters 97 (2013): 90-92.

[239] Wang, Li, et al. "Aluminum induced in situ crystallization of amorphous SiC." Applied physics letters 94.18 (2009).

[240] Verucchi, Roberto, et al. "Epitaxy of nanocrystalline silicon carbide on Si (111) at room temperature." Journal of the American Chemical Society 134.42 (2012): 17400-17403.

[241] Komura, Yusuke, et al. "Film properties of nanocrystalline 3C–SiC thin films deposited on glass substrates by hot-wire chemical vapor deposition using CH4 as a carbon source." Japanese journal of applied physics 46.1R (2007): 45

[242] Somogyi, Bálint, Viktor Zólyomi, and Adam Gali. "Near-infrared luminescent cubic silicon carbide nanocrystals for in vivo biomarker applications: an ab initio study." Nanoscale 4.24 (2012): 7720-7726.

[243] Eickhoff, M., et al. "Piezoresistive properties of single crystalline, polycrystalline, and nanocrystalline n-type 3C-SiC." Journal of applied physics 96.5 (2004): 2872-2877.

[244] Phan, Hoang-Phuong, et al. "The effect of strain on the electrical conductance of p-type nanocrystalline silicon carbide thin films." Journal of Materials Chemistry C 3.6 (2015): 1172-1176.

[245] Meng, Alan, et al. "Synthesis and field emission properties of silicon carbide nanobelts with a median ridge." CrystEngComm 14.20 (2012): 6755-6760.

[246] Fang, X., Hu, L., Ye, C., & Zhang, L. "One-dimensional inorganic semiconductor nanostructures: a new carrier for nanosensors." Pure and Applied Chemistry 82.11 (2010): 2185-2198.

[247] Li, Hejun, et al. "Large-scale synthesis, growth mechanism, and photoluminescence of 3C-SiC nanobelts." Materials letters 109 (2013): 275-278.

[248] Li, Xiaoxiao, et al. "Fabrication of N-doped 3C-SiC nanobelts with selected (110) top surface and their enhanced transverse piezoresistance behaviours." Ceramics International 44.15 (2018): 19021-19027.

[249] Li, Xiaoxiao, et al. "Giant piezoresistance in B-doped SiC nanobelts with a gauge factor of– 1800." ACS Applied Materials & Interfaces 12.42 (2020): 47848-47853.

[250] Shao, Ruiwen, et al. "Piezoresistance behaviors of ultra-strained SiC nanowires." Applied Physics Letters 101.23 (2012).

[251] Wei, Guang, et al. "Temperature-dependent field emission properties of 3 C-SiC nanoneedles." The Journal of Physical Chemistry C 115.26 (2011): 13063-13068.

[252] Gao F, Zheng J, Wang M, et al. "Piezoresistance behaviors of p-type 6 H-SiC nanowires." Chemical Communications 47.43 (2011): 11993-11995.

[253] Phan, Hoang-Phuong, et al. "The piezoresistive effect in top–down fabricated p-type 3C-SiC nanowires." IEEE Electron Device Letters 37.8 (2016): 1029-1032.

[254] Sasaki, K., et al. "High-temperature electrical properties of 3C-SiC epitaxial layers grown by chemical vapor deposition." Applied physics letters 45.1 (1984): 72-73.

[255] Bi J, Wei G, Wang L, et al. Highly sensitive piezoresistance behaviors of n-type 3C-SiC nanowires[J]. Journal of Materials Chemistry C, 2013, 1(30): 4514-4517.

[256] Li X, Chen S, Ying P, et al. A giant negative piezoresistance effect in 3C-SiC nanowires with B dopants[J]. Journal of Materials Chemistry C, 2016, 4(27): 6466-6472.

[257] Rurali, Riccardo, et al. "Theoretical evidence for the kick-out mechanism for B diffusion in SiC." Applied physics letters 81.16 (2002): 2989-2991.

[258] Yang, Yang, et al. "Enhanced field emission of p-type 3C-SiC nanowires with B dopants and sharp corners." Journal of Materials Chemistry C 2.23 (2014): 4515-4520.

[259] Li, Ya-Lin, Pei Gong, et al. "Comparative study on transport properties of N-, P-, and As-doped SiC nanowires: Calculated based on first principles." Chinese Physics B 29.3 (2020): 037304.

[260] Cheng, X., Wang, L., Gao, F., Yang, W., Du, Z., Chen, D. and Chen, S., 2019. The N and P co-doping-induced giant negative piezoresistance behaviors of SiC nanowires. Journal of Materials Chemistry C, 7(11), pp.3181-3189.

[261] Zeng, Hongjiang, et al. "In situ SEM electromechanical characterization of nanowire using an electrostatic tensile device." Journal of Physics D: Applied Physics 46.30 (2013): 305501.

[262] Bernal, Rodrigo A., et al. "In situ electron microscopy four-point electromechanical characterization of freestanding metallic and semiconducting nanowires." Small 10.4 (2014): 725-733.

[263] Zhang, Yuefei, et al. "Direct observation of super-plasticity of beta-SiC nanowires at low temperature." Advanced Functional Materials 17.17 (2007): 3435-3440.

[264] Cui, Junfeng, et al. "Unprecedented piezoresistance coefficient in strained silicon carbide." Nano Letters 19.9 (2019): 6569-6576.

[265] Pan, Caofeng, et al. "Piezotronic effect on the sensitivity and signal level of Schottky contacted proactive micro/nanowire nanosensors." Acs Nano 7.2 (2013): 1803-1810.

[266] Nguyen, Thanh, et al. "Giant piezoresistive effect by optoelectronic coupling in a heterojunction." Nature communications 10.1 (2019): 4139.

[267] Nguyen, Thanh, et al. "Opto-electronic coupling in semiconductors: towards ultrasensitive pressure sensing." Journal of Materials Chemistry C 8.14 (2020): 4713-4721.

[268] Zhang, Jinbo, et al. "Integration strategies of ternary hierarchical nanocomposite designs with activated ultraviolet lights and surface acoustic waves for enhancing NO2 sensing at room-temperature." Chemical Engineering Journal 482 (2024): 149067.

[269] Li, Xiaoxiao, et al. "Enhanced piezoresistive performance of 3C-SiC nanowires by coupling with ultraviolet illumination." Journal of Materials Chemistry C 7.43 (2019): 13384-13389.

[270] Wu, Renbing, et al. "Recent progress in synthesis, properties and potential applications of SiC nanomaterials." Progress in Materials Science 72 (2015): 1-60.

[271] Fraga, M. A., et al. "Fabrication and characterization of a SiC/SiO2/Si piezoresistive pressure sensor."Procedia Engineering (2010): 609-612.

[272]Ziermann, R., et al. "High temperature piezoresistive β-SiC-on-SOI pressure sensor with on chip SiC thermistor." Materials Science and Engineering: B 61 (1999): 576-578.

[273] Reichert, W., et al. "Investigation of the effects of high temperature implantation and post implantation annealing on the electrical behavior of nitrogen implanted β-SiC films." Diamond and Related Materials 6.10 (1997): 1445-1447.

[274] Reichert, W., E. Obermeier, and J. Stoemenos. "β-SiC films on SOI substrates for high temperature

applications." Diamond and Related Materials 6.10 (1997): 1448-1450.
[275] Ziermann, René, et al. "A high temperature pressure sensor with β-SiC piezoresistors on SOI substrates." Proceedings of International Solid State Sensors and Actuators Conference (Transducers' 97). Vol. 2. IEEE, 1997.
[276] Serre, C., et al. "SiCOI Structures. Technology and Characterization." Progress in SOI Structures and Devices Operating at Extreme Conditions (2002): 17-29.
[277] Via, Francesco La, et al. "New approaches and understandings in the growth of cubic silicon carbide." Materials 14.18 (2021): 5348.
[278] Masri, Pierre. "Silicon carbide and silicon carbide-based structures: The physics of epitaxy." Surface science reports 48.1-4 (2002): 1-51.
[279] Locke, C. W., et al. "SiC films and coatings: amorphous, polycrystalline, and single crystal forms." Silicon carbide biotechnology (2011): 17-61.
[280] La Via, F., et al. "From thin film to bulk 3C-SiC growth: Understanding the mechanism of defects reduction." Materials Science in Semiconductor Processing 78 (2018): 57-68.
[281] Eickhoff, M., et al. "A high temperature pressure sensor prepared by selective deposition of cubic silicon carbide on SOI substrates." Sensors and Actuators A: Physical 74.1-3 (1999): 56-59.
[282] Von Berg J, Ziermann R, Reichert W, et al. "Measurement of the cylinder pressure in combustion engines with a piezoresistive/spl beta/-SiC-on-SOI pressure senso. " IEEE, (1998): 245-249.
[283] Zhang, Jiangchun, et al. "Electrical, mechanical and metal contact properties of polycrystalline 3C-SiC films for MEMS in harsh environments." Surface and Coatings Technology 201.22-23 (2007): 8893-8898.
[284] Wu, Chien-Hung, Christian A. Zorman, and Mehran Mehregany. "Fabrication and testing of bulk micromachined silicon carbide piezoresistive pressure sensors for high temperature applications." IEEE Sensors Journal 6.2 (2006): 316-324.
[285] Tong, Braiden, et al. "Highly sensitive and robust 3C-SiC/Si pressure sensor with stress amplification structure." Materials & Design 224 (2022): 111297.
[286] Fedder, Gary K., et al. "Technologies for cofabricating MEMS and electronics." Proceedings of the IEEE 96.2 (2008): 306-322.
[287] Wu, C-H., et al. "Fabrication and testing of single crystalline 3C-SiC piezoresistive pressure sensors." Transducers’ 01 Eurosensors XV: The 11th International Conference on Solid-State Sensors and Actuators June 10–14, 2001 Munich, Germany. Springer Berlin Heidelberg, (2001): 514-517.
[288] Syväjärvi, Mikael, Rositsa Yakimova, and Erik Janzén. "Anisotropic etching of SiC."Journal of the Electrochemical Society147.9 (2000): 3519.
[289] Marsi, Noraini, et al. "The fabrication of back etching 3C-SiC-on-Si diaphragm employing KOH+ IPA in MEMS capacitive pressure sensor." Microsystem Technologies 21 (2015): 1651-1661.
[290] Fraga, M. A., et al. "Preliminary evaluation of the influence of the temperature on the performance of a piezoresistive pressure sensor based on a-SiC film." Microsystem technologies 17.3 (2011): 477-480.
[291] Huang, X., & Zhang, X. "Investigating the advanced characteristics of SiC based piezoresistive pressure sensors." Materials Today Communications 25 (2020): 101493.
[292] Wu, Chen, et al. "Exploring the nonlinear piezoresistive effect of 4H-SiC and development MEMS pressure sensors for extreme environments." Microsystems & Nanoengineering 9.1 (2023): 41.
[293] Okojie, R. S., et al. "α (6H)-SiC pressure sensors at 350 °C." International Electron Devices Meeting.

Technical Digest. IEEE, (1996): 525-528.

[294] Okojie, Robert, et al. "Characteristics of a hermetic 6H-SiC pressure sensor at 600°C." AIAA Space 2001 Conference and Exposition. 2001.

[295] Okojie, Robert S., et al. "4H-SiC piezoresistive pressure sensors at 800°C with observed sensitivity recovery." IEEE Electron Device Letters 36.2 (2014): 174-176.

[296] Okojie, Robert S., et al. "Demonstration of SiC pressure sensors at 750° C." Additional Papers and Presentations 2014.HITEC (2014): 000028-000032.

[297] Okojie, Robert S., et al. "High temperature dynamic pressure measurements using silicon carbide pressure sensors." Additional Papers and Presentations 2014.HITEC (2014): 000047-000052.

[298] Hattori, Junya, et al. "High-speed observation of pulse energy and pulse width dependences of damage generation in SiC during ultrashort pulse laser drilling." Applied Physics A 126 (2020): 1-6.

[299] Ito, Yusuke, et al. "Mechanisms of damage formation in glass in the process of femtosecond laser drilling." Applied Physics A 124 (2018): 1-8.

[300] Wu, Chen, et al. "Fabrication of 4H-SiC piezoresistive pressure sensor for high temperature using an integrated femtosecond laser-assisted plasma etching method." Ceramics International 49.18 (2023): 29467-29476.

[301] Wang, Lukang, et al. "Design and fabrication of bulk micromachined 4H-SiC piezoresistive pressure chips based on femtosecond laser technology." Micromachines 12.1 (2021): 56.

[302] Wang, Lukang, et al. "Development of laser-micromachined 4H-SiC MEMS piezoresistive pressure sensors for corrosive environments." IEEE Transactions on Electron Devices 69.4 (2022): 2009-2014.

[303] Okojie R S, Blaha C, Lukco D, et al. "Zero offset drift suppression in SiC pressure sensors at 600 °C." Sensors, 2010 IEEE. IEEE, (2010): 2269-2274.

[304] Hoogerwerf, Arno C., et al. "Silicon carbide pressure sensors for harsh environments."2019 20th International Conference on Solid-State Sensors, Actuators and Microsystems & Eurosensors. IEEE, 2019.

[305] Fang, Xudong, et al. "Development of a 4H-SiC Piezoresistive Pressure Sensor for High Temperature Applications."2019 IEEE 14th International Conference on Nano/Micro Engineered and Molecular Systems (NEMS). IEEE, 2019.

[306] Okojie, R. S., et al. "Improved reliability of SiC pressure sensors for long term high temperature applications."2011 16th International Solid-State Sensors, Actuators and Microsystems Conference. IEEE, 2011.

[307] Okojie, Robert S., et al. "High temperature SiC pressure sensors with low offset voltage shift."Sensors for Extreme Harsh Environments. Vol. 9113. SPIE, 2014.

[308] Fang, Ziyan, et al. "Pt thin-film resistance thermo detectors with stable interfaces for potential integration in SiC high-temperature pressure sensors." Microsystems & Nanoengineering 10.1 (2024): 133

[309]Fang, Xudong, et al. "A 350° C piezoresistive n-type 4H-SiC pressure sensor for hydraulic and pneumatic pressure tests." Journal of Micromechanics and Microengineering 30.5 (2020): 055009.

[310] Mackowiak, Piotr, et al. "Piezoresistive 4H-silicon carbide (SiC) pressure sensor." 2021 IEEE Sensors. IEEE, 2021.

[311] Burla, Ravi K., et al. "High-temperature testing of nickel wire bonds for SiC devices." Design, Characterization, and Packaging for MEMS and Microelectronics. Vol. 3893. SPIE, 1999.

[312] Wang, Lukang, et al. "High temperature characteristics of piezoresistive silicon carbide pressure sensors implemented by leadless packaging." 2023 IEEE SENSORS. IEEE, 2023.
[313] von Berg, Jochen, et al. "A piezoresistive low-pressure sensor fabricated using silicon-on-insulator (SOI) for harsh environment applications." Transducers' 01 Euro-sensors XV: The 11th International Conference on Solid-State Sensors and Actuators June 10–14, 2001 Munich, Germany. Springer Berlin Heidelberg, 2001.
[314] Khan, Muhib, et al. "Effect of high thermal stability mold material on the gold-aluminum bond reliability in epoxy encapsulated VLSI devices." 26th Annual Proceedings Reliability Physics Symposium 1988. IEEE, (1988):40-49.
[315] Okojie, Robert S., John C. DeLaat, and Joseph R. Saus. "SiC pressure sensor for detection of combustor thermoacoustic instabilities [aircraft engine applications]." The 13th International Conference on Solid-State Sensors, Actuators and Microsystems, 2005. Digest of Technical Papers. TRANSDUCERS'05. Vol. 1. IEEE, (2005):470-473.
[316] Okojie, Robert S., et al. "Reliability evaluation of direct chip attached silicon carbide pressure transducers." SENSORS, 2004 IEEE, (2004): 638-638.
[317]Masheeb, Fawzia, et al. "Leadless sensor packaging for high temperature applications." Technical Digest. MEMS 2002 IEEE International Conference. Fifteenth IEEE International Conference on Micro Electro Mechanical Systems (Cat. No. 02CH37266). IEEE, (2002): 392-395.
[318]Ned, Alexander A., et al. "Improved SiC leadless pressure sensors for high temperature, low and high pressure applications." catálogo Kulite Semiconductors Products (2004).
[319]Okojie, Robert S. "Stable 600 C silicon carbide MEMS pressure transducers." Sensors and Systems for Space Applications. Vol. 6555. SPIE, 2007.
[320] Okojie, R. S., et al. "Failure mechanisms in MEMS based silicon carbide high temperature pressure sensors." 2007 IEEE International Reliability Physics Symposium Proceedings. 45th Annual. IEEE, (2007): 429-432.
[321] Wieczorek G, Schellin B, Obermeier E, et al. "SiC based pressure sensor for high-temperature environments." SENSORS, 2007 IEEE. IEEE, (2007): 748-751
[322] Belwanshi, Vinod. "Analytical modeling to estimate the sensitivity of MEMS technology-based piezoresistive pressure sensor." Journal of Computational Electronics 20.1 (2021): 668-680.
[323] Belwanshi, Vinod, and Anita Topkar. "Quantitative analysis of MEMS piezoresistive pressure sensors based on wide band gap materials."IETE Journal of Research 68.1 (2022): 667-677.
[324] Tian, Baohua, et al. "Performance optimization of SiC piezoresistive pressure sensor through suitable piezoresistor design." Microsystem Technologies 27 (2021): 3083-3093.
[325] Chen Wu, Xudong Fang, Xin Guo, Libo Zhao, Bian Tian, " Zhuangde Jiang; Optimal design of SiC piezoresistive pressure sensor considering material anisotropy. " Rev. Sci. Instrum. 91 (1): 015004(2020).
[326] Kanekal, Dadasikandar, and Sumit Kumar Jindal. "Investigation of MEMS piezoresistive pressure sensor with a freely supported rectangular silicon carbide diaphragm as a primary sensing element for altitudinal applications."Silicon15.4 (2023): 1947-1959.
[327] Young, Darrin J., et al. "High-temperature single-crystal 3C-SiC capacitive pressure sensor." IEEE sensors Journal 4.4 (2004): 464-470.
[328] Suthar, Rahul Jitendrab, and Mohammad Shavezipur. "Temperature compensation in MEMS capacitive

pressure sensors for harsh environment applications." International Design Engineering Technical Conferences and Computers and Information in Engineering Conference. Vol. 51791. American Society of Mechanical Engineers, 2018.

[329] Suster, Michael, Wen H. Ko, and Darrin J. Young. "Optically-powered wireless transmitter for high-temperature MEMS sensing and communication." TRANSDUCERS'03. 12th International Conference on Solid-State Sensors, Actuators and Microsystems. Digest of Technical Papers (Cat. No. 03TH8664). Vol. 2. IEEE, 2003.

[330] Fonseca, Michael A., et al. "Wireless micromachined ceramic pressure sensor for high-temperature applications." Journal of microelectromechanical systems 11.4 (2002): 337-343.

[331] Ko, Wen H., and Qiang Wang. "Touch mode capacitive pressure sensors." Sensors and Actuators A: Physical 75.3 (1999): 242-251.

[332] Lv, Haojie, Huiyong Yu, and Guoqing Hu. "A SiC high-temperature pressure sensor operating in severe condition." Telkomnika 10.8 (2012).: 2247-2252.

[333] Timoshenko, Stephen, and Sergius Woinowsky-Krieger. "Theory of plates and shells." (1959).

[334] Eaton, William P., and James H. Smith. "Micromachined pressure sensors: review and recent developments." Smart Materials and Structures 6.5 (1997): 530-539.

[335] Eswaran, P., and S. Malarvizhi. "MEMS capacitive pressure sensors: a review on recent development and prospective." Int. J. Eng. Technol 5.3 (2013): 2734-2746.

[336] Marsi, Noraini, et al. "The mechanical and electrical characteristics of a 3C-SiC for mems capacitive pressure sensor diaphragm." Middle east journal of scientific Research (2013).

[337] Marsi, N., et al. "The Mechanical and Electrical Effects of MEMS Capacitive Pressure Sensor Based 3C-SiC for Extreme Temperature." Journal of engineering 2014.1 (2014): 715167.

[338] Pozzi, Michele, et al. "Design, fabrication, and testing of 3C-SiC sensors for high temperature applications." Device and Process Technologies for Microelectronics, MEMS, Photonics, and Nanotechnology IV. Vol. 6800. SPIE, 2008.

[339] Du J, Young D J, Zorman CA, et al. "Single crystal SiC capacitive pressure sensor at 400℃." IEEE International Electron Devices Meeting 2003. (2003): 783-786

[340] Du, Jiangang, and Christian A. Zorman. "A polycry stalline SiC-on-Si architecture for capacitive pressure sensing applications beyond 400℃: Process development and device performance." Journal of Materials Research28.1 (2013): 120-128.

[341] Du, Jiangang, et al. "Poly-SiC capacitive pressure sensors made by wafer bonding." SENSORS, 2005 IEEE. IEEE, (2005): 4.

[342] Marsi, Noraini, et al. "A MEMS packaged capacitive pressure sensor employing 3C-SiC with operating temperature of 500℃." Microsystem Technologies21 (2015): 9-20.

[343] Marsi, Noraini, et al. "High reliability of MEMS packaged capacitive pressure sensor employing 3C-SiC for high temperature." Energy Procedia68 (2015): 471-479.

[344] Sheng, Srihari Rajgopal, and Mehran Mehregany. "Silicon carbide pressure sensor for high temperature and high pressure applications: Influence of substrate material on performance."2011 16th International Solid-State Sensors, Actuators and Microsystems Conference. IEEE, 2011.

[345] Marsi, N., et al. "The capacitance and temperature effects of the SiC-and Si-based MEMS pressure sensor."Journal of Physics: Conference Series. Vol. 431. No. 1. IOP Publishing, 2013.

[346] X. Ding, L. Tong, W. He, J. Hsu， W. H. Ko， "Touch mode silicon capacitive pressure sensors," Win

ter Annual Meeting of the American Society of Mechanical Engineers, Conference Dallas, USA, pp.111-117,1990.

[347] Jindal,SK, Sumit Kumar, Ankush Mahajan, and Sanjeev Kumar Raghuwanshi. "A complete analytical model for clamped edge circular diaphragm non-touch and touch mode capacitive pressure sensor." Microsystem Technologies 22 (2016): 1143-1150.

[348] Hezarjaribi, Y., et al. "Capacitive pressure sensors based on MEMS, operating in harsh environments." 2008 IEEE International Conference on Semiconductor Electronics. IEEE, 2008.

[349] Meng, Guangqing, and Wen H. Ko. "Modeling of circular diaphragm and spreadsheet solution programming for touch mode capacitive sensors." Sensors and Actuators A: physical 75.1 (1999): 45-52.

[350] Ko, Wen H., and Qiang Wang. "Touch mode capacitive pressure sensors." Sensors and Actuators A: Physical 75.3 (1999): 242-251.

[351] Chen, Li, and Mehran Mehregany. "A silicon carbide capacitive pressure sensor for high temperature and harsh environment applications." TRANSDUCERS 2007-2007 International Solid-State Sensors, Actuators and Microsystems Conference (pp.2597-2600). IEEE, 2007.

[352] Chen, Li, and Mehran Mehregany. "A silicon carbide capacitive pressure sensor for in-cylinder pressure measurement." Sensors and Actuators A: Physical 145 (2008): 2-8.

[353] Pakula, L. S., et al. "Fabrication of a CMOS compatible pressure sensor for harsh environments." Journal of Micromechanics and Microengineering 14.11 (2004): 1478-1483.

[354] Tang, Wei, et al. "Complementary metal-oxide semiconductor-compatible silicon carbide pressure sensors based on bulk micromachining." Micro & Nano Letters 6.4 (2011): 265-268.

[355] Meng, B., et al. "An optimized fabrication of high yield CMOS-compatible silicon carbide capacitive pressure sensors." 2012 7th IEEE International Conference on Nano/Micro Engineered and Molecular Systems (NEMS). IEEE, 2012.

[356] Meng, Bo, et al. "Silicon carbide capacitive pressure sensors with arrayed sensing membranes." The 8th Annual IEEE International Conference on Nano/Micro Engineered and Molecular Systems. IEEE, 2013.

[357] Lian, Zhang, et al."Conditioning circuit for precise SiC capacitive pressure sensors."2011 6th IEEE Conference on Industrial Electronics and Applications. IEEE, 2011.

[358] Beker, Levent, et al. "A silicon carbide differential output pressure sensor by concentrically matched capacitance." 2017 IEEE 30th International Conference on Micro Electro Mechanical Systems (MEMS). IEEE, 2017.

[359] Beker, Levent, et al. "Modeling, fabrication, and characterization of SiC concentrically matched differential capacitance output pressure sensors." Sensors and Actuators A: Physical 273 (2018): 293-302.

[360] Wang, Qiang, and Wen H. Ko. "Modeling of touch mode capacitive sensors and diaphragms." Sensors and Actuators A: Physical 75.3 (1999): 230-241.

[361] Lv, Haojie, Kaiyuan Lai, and Yonglan Qi. "Analysis of a micro pressure-sensor employing SiC-AlN-SiC structure." Australian Journal of Mechanical Engineering20.1 (2022): 194-200.

[362] Lv, Haojie, Qiang Guo, and Guoqing Hu. "A touch mode capacitive pressure sensor with long linear range and high sensitivity." 2008 3rd IEEE International Conference on Nano/Micro Engineered and Molecular Systems. IEEE, 2008.

[363] Jindal, Sumit Kumar, M. Aditya Varma, and Deepali Thukral. "Study of MEMS touch-mode capacitive pressure sensor utilizing flexible sic circular diaphragm: Robust design, theoretical modeling, numerical simulation and performance comparison." Journal of Circuits, Systems and Computers 28.12 (2019): 1950206.

[364] Jindal, Sumit Kumar, M. Aditya Varma, and Deepali Thukral. "Comprehensive assessment of MEMS double touch mode capacitive pressure sensor on utilization of SiC film as primary sensing element: Mathematical modelling and numerical simulation." Microelectronics Journal 73 (2018): 30-36.

[365] Jindal, SK, and Sanjeev Kumar Raghuwanshi. "Capacitance and sensitivity calculation of double touch mode capacitive pressure sensor: theoretical modelling and simulation." Microsystem Technologies 23 (2017): 135-142.

[366] Lv, Hao Jie, et al. "A study of a silicon carbide capacitive pressure sensor applied in harsh environment."Applied Mechanics and Materials 241 (2013): 984-987.

[367] Lv, Hao Jie, Tao Geng, and Guo Qing Hu. "Study of high-temperature mems pressure sensor based on sic-aln structure." Key Engineering Materials 562 (2013): 471-476.

[368] Meredith, Roger D., et al. "High temperature capacitive pressure sensor employing a SiC based ring oscillator." Materials Science Forum. Vol. 717. Trans Tech Publications Ltd, (2012): 1215-1218.

[369] Scardelletti, Maximilian C., et al. "Demonstration of a packaged capacitive pressure sensor system suitable for jet turbofan engine health monitoring." 2016 IEEE 66th Electronic Components and Technology Conference (ECTC). IEEE, (2016): 877-888.

[370] Scardelletti, Maximilian C., and Christian A. Zorman. "Packaged capacitive pressure sensor system for aircraft engine health monitoring." 2016 IEEE SENSORS. IEEE, (2016): 1.

[371] Scardelletti, Maximilian C., et al. "Novel High Temperature Capacitive Pressure Sensor Utilizing SiC Integrated Circuit Twin Ring Oscillators." 2017 IEEE 67th Electronic Components and Technology Conference (ECTC). IEEE, 2017.: 197-205.

[372] Murphy, Kent A., et al. "Quadrature phase-shifted, extrinsic Fabry–Perot optical fiber sensors." Optics letters 16.4 (1991): 273-275.

[373] Chen, Ye, et al. "Recent Progress in MEMS Fiber-Optic Fabry–Perot Pressure Sensors." Sensors 24.4 (2024): 1079.

[374]Di Giovanni, M. (1982). Flat and Corrugated Diaphragm Design Handbook (1st ed.). Routledge.

[375] Lee, Byoungho. "Review of the present status of optical fiber sensors." Optical fiber technology 9.2 (2003): 57-79.

[376] Riza, Nabeel A., Muzzamil Arain, and Frank Perez. "6-H single-crystal silicon carbide thermo-optic coefficient measurements for ultrahigh temperatures up to 1273 K in the telecommunications infrared band." Journal of applied physics 98.10 (2005).

[377] Pulliam, Wade J., et al. "Micromachined SiC fiber optic pressure sensors for high-temperature aerospace applications." Industrial Sensing Systems. Vol. 4202. SPIE, 2000.

[378]Jiang, Yonggang, et al. "Fabrication of all-SiC fiber-optic pressure sensors for high-temperature applications." Sensors 16.10 (2016): 1660.

[379] Sheng, Tianyu, et al. "Fabrication and Measurement of SiC Fiber-optic Fabry-Perot Pressure Sensors for High-temperature Applications." China Mechanical Engineering 33.15 (2022): 1803.

[380] Liang, Ting, et al. "All-SiC fiber-optic sensor based on direct wafer bonding for high temperature pressure sensing." Photonic Sensors 12 (2022): 130-139.

[381]Pulliam, Wade J. et al. “High-temperature high-bandwidth fiber optic MEMS pressure-sensor technology for turbine engine component testing.” SPIE Optics East (2002).
[382] Riza, Nabeel A., Farzan Ghauri, and Frank Perez. "Silicon carbide-based remote wireless optical pressure sensor." IEEE Photonics Technology Letters19.7 (2007): 504-506.
[383] Riza, Nabeel A., Farzan Ghauri, and Frank Perez. "Hybrid sensors using laser targeting." Optical Sensing II. Vol. 6189. SPIE, 2006.
[384] Riza, Nabeel A., Farzan Ghauri, and Frank Perez. "Wireless pressure sensor using laser targeting of silicon carbide." Optical Engineering 46.1 (2007): 014401-014401.
[385] Riza, Nabeel A., A. Arain, and Frank Perez. "Harsh environments minimally invasive optical sensor using free-space targeted single-crystal silicon carbide." IEEE Sensors Journal 6.3 (2006): 672-685.
[386] Chakravarty, A., N. R. Quick, and A. Kar. "Decoupling of silicon carbide optical sensor response for temperature and pressure measurements." Journal of Applied Physics 102.7 (2007): 073111.
[387] Dakshinamurthy, Surendramohan, Nathaniel R. Quick, and Aravinda Kar. "SiC-based optical interferometry at high pressures and temperatures for pressure and chemical sensing." Journal of Applied Physics 99.9 (2006).
[388] Zhang, Dongzhi, Guoqing Hu, and Gengxin Zeng. "Modeling and simulation of high temperature optical-fiber SiC-based pressure sensor."2009 International Conference on Mechatronics and Automation. IEEE, 2009.
[389] Xue Tiange, et al." Study on high temperature reliability of electrical interconnection material of SiC pressure sensor." Journal of Materials Science: Materials in Electronics 35.10(2024)
[390] Middelburg, Luke M., et al. "Toward a self-sensing piezoresistive pressure sensor for all-SiC monolithic integration." IEEE Sensors Journal 20.19 (2020): 11265-11274.
[391] Phan, Hoang-Phuong, et al. "Highly sensitive pressure sensors employing 3C-SiC nanowires fabricated on a free standing structure." Materials & Design156 (2018): 16-21.